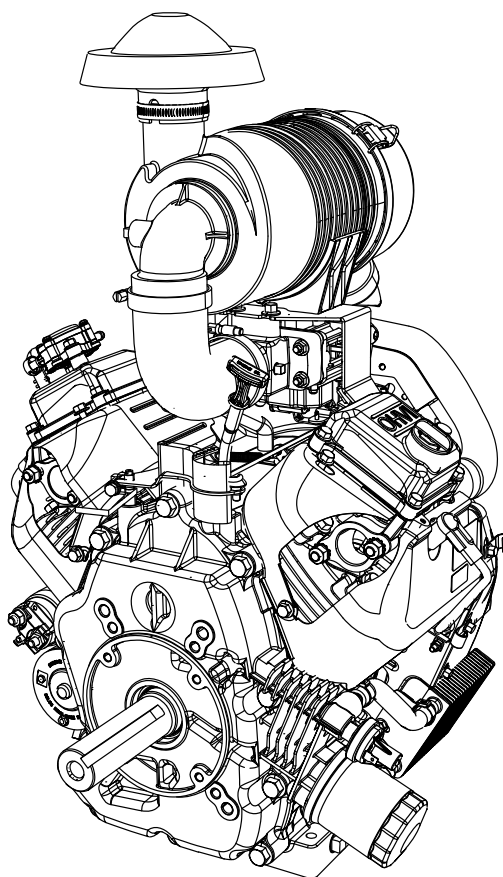
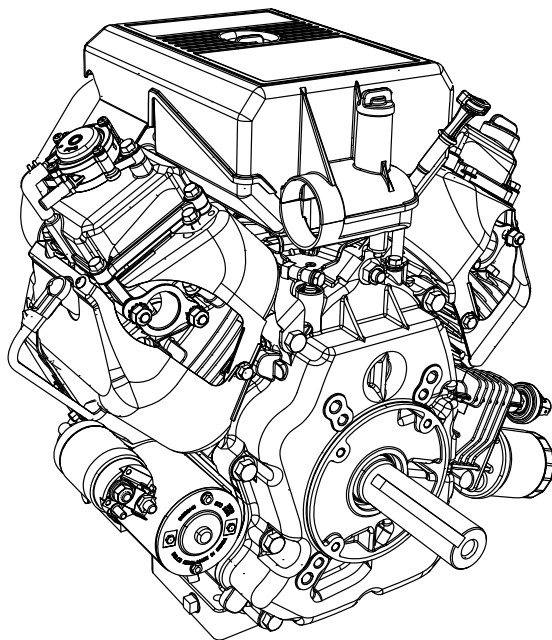


GENERAC®

OHVI® V-Twin Engine

Service Manual



MODELS:

GTH760/990/1000

GTV760/990/1000

SAVE THIS MANUAL FOR FUTURE REFERENCE

 WARNING

CANCER AND REPRODUCTIVE HARM

www.P65Warnings.ca.gov

(000393a)

The information in this manual is accurate based on products produced at the time of publication. The manufacturer reserves the right to make technical updates, corrections, and product revisions at any time without notice.

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Foreword

This manual has been written and published by Generac Power Systems, Inc. to aid our dealers' mechanics, company service personnel, and general consumers when servicing the products described herein.

It is assumed these personnel are familiar with servicing procedures for these products, or like or similar products, manufactured, and marketed by Generac Power Systems, Inc. It is also assumed they have been trained in recommended servicing procedures for these products, which includes the use of mechanics' hand tools and any special tools which might be required.

Proper service and repair is important to safe, economical, and reliable operation of the products described herein. Troubleshooting, testing, service, and repair procedures recommended by Generac Power Systems, Inc. and described in this manual are effective methods of performing such operations. Some operations or procedures may require specialized equipment. Such equipment should be used when and as recommended.

We could not possibly know of and advise the service trade of all conceivable procedures or methods by which a service might be performed, nor of any possible hazards and/or results of each procedure or method. We have not undertaken any such wide evaluation. Therefore, anyone who uses a procedure or method not recommended by the manufacturer must first satisfy himself/herself that neither his/her safety, nor the product's safety, will be endangered by the service or operating procedure selected.

All information, illustrations, and specifications contained in this manual are based on the latest product information available at the time of publication. However, Generac Power Systems, Inc. reserves the right to change, alter, or otherwise improve the product at any time without prior notice.

Some components or assemblies of products described in this manual may not be considered repairable. Disassembly, repair, and reassembly of such components may not be included in this manual.

Engines described herein may be used to power a wide variety of products. Service and repair instructions relating to any such products are not covered in this manual. For information pertaining to use of these engines with other products, refer to any owner's or service manuals pertaining to said products.

Special Tools

Many procedures depicted in this manual require special tools. Some required tools are available as Generac parts and are listed as such in this manual.

WARNING!

Generac Power Systems does not approve or authorize the use of these engines on All Terrain Vehicles (ATVs), go-carts, motorbikes, aircraft products, personal watercraft, or vehicles intended for use in competitive events. The use of this product in any other than its intended application will void the warranty! Use of these engines in such applications could result in property damage, serious injury (including paralysis), or even death.

Service Information

Service on this engine within and after the warranty period can be performed by any Independent Authorized Service Dealer (IASD). Service technicians are factory trained and capable of handling all service needs.

When contacting an IASD about parts and service, always supply the complete model number and serial number of your unit as given on its data plate decal. See [Figure 2-1](#) for the location of the decal.

The warranty for this engine is included in the documents packaged with the generator.

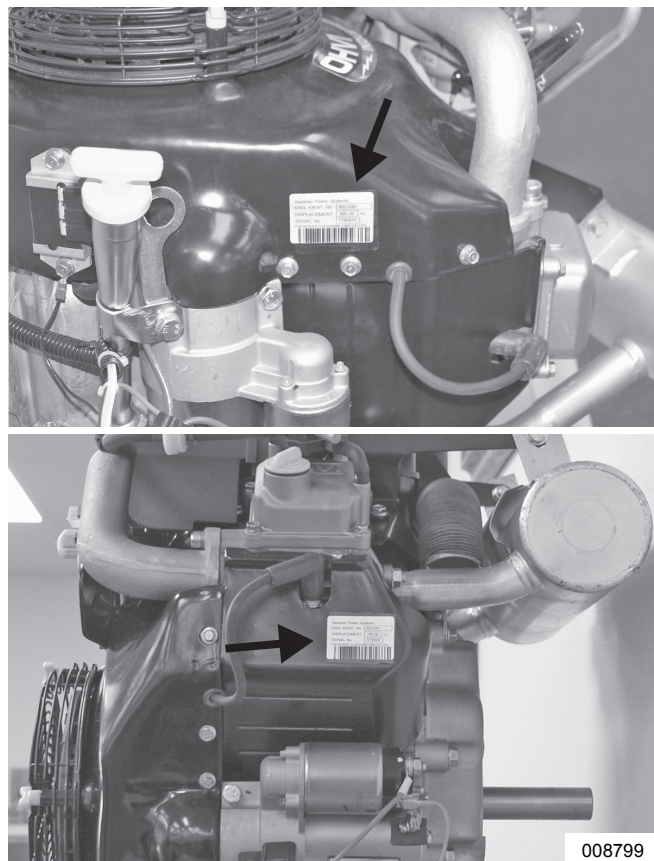


Figure 2-1. Data Plate Decals

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Section 1: Rules for Safe Operation

Notice to Users of this Manual

Study these **Rules for Safe Operation** carefully before operating or servicing this equipment. Become familiar with the equipment Owner's Manual and with the engine. The engine can operate safely, efficiently and reliably only if it is properly operated and maintained. Many accidents are caused by failing to follow simple and fundamental rules or precautions.

Read this manual thoroughly and understand all of the instructions, cautions, and warnings before using this equipment.

If you do not understand any portion of this manual, contact Generac or your nearest Generac Independent Authorized Service Dealer (IASD) for starting, operating and servicing procedures. Save these instructions for future reference. ALWAYS have this service manual with you when contacting Technical Support.

The information in this manual is accurate based on products produced at the time of publication. The manufacturer reserves the right to make technical updates, corrections, and product revisions at any time without notice.

Safety Rules

The manufacturer cannot anticipate every possible circumstance which might involve a hazard. The warnings in this manual, and on tags and decals affixed to the unit are, therefore, not all inclusive. If using a procedure, work method or operating technique which the manufacturer does not specifically recommend, verify it is safe for others. Also verify the procedure, work method or operating technique utilized does not render the equipment unsafe.

Safety Symbols and Meanings

Throughout this publication, and on tags and decals affixed to the engine, DANGER, WARNING, CAUTION, and NOTE blocks are used to alert personnel to special instructions about a particular operation which may be hazardous if performed incorrectly or carelessly. Observe them carefully. Their definitions are as follows:

DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

(000001)

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

(000002)

CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

(000003)

NOTE: Notes contain additional information important to a procedure and will be found within the regular text of this manual.

These safety alerts cannot eliminate the hazards they indicate. Common sense and strict compliance with the special instructions while performing the action or service are essential to preventing accidents.

General Safety Information

NOTE: Improper treatment of this engine can damage it and shorten its life.

- Contact an IASD if you have questions about intended use.
- NEVER operate engines with broken or missing parts, or without protective housing or covers.
- DO NOT bypass any safety device on this machine.

Intended Use

This engine was designed and manufactured for specific applications. Do not attempt to modify the equipment or use it for any application for which it was not designed.

Generac Power Systems does not approve or authorize the use of these engines on All Terrain Vehicles (ATV's), go-carts, motorbikes, aircraft products, personal watercraft, or vehicles intended for use in competitive events. The use of this product in any other than its intended application will void the warranty! Use of these engines in such applications could result in property damage, serious injury (including paralysis), or even death.

Replacement Parts

This equipment is designed to be used with Generac authorized parts ONLY. If equipment is used with parts which DO NOT comply with minimum specifications, user assumes all risks and liabilities. During any maintenance procedure, replacement fasteners must have the same measurements and strength as the removed fasteners. Metric bolts and nuts have numbers indicating their strength. Customary bolts use radial lines to indicate strength while most customary nuts do not have strength markings. Mismatched or incorrect fasteners can cause damage, malfunction and possible injury.

Exhaust Hazards



⚠ DANGER

Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury. (000103)



⚠ WARNING

Asphyxiation. Always use a battery operated carbon monoxide alarm indoors and installed according to the manufacturer's instructions. Failure to do so could result in death or serious injury. (000178a)

- Carbon monoxide produced during use can kill in minutes. NEVER use indoors, in a vehicle, or in other sheltered areas, EVEN IF doors and windows are open. Only use OUTSIDE and far away from windows, doors, and vents.
- Breathing carbon monoxide can cause headache, fatigue, dizziness, vomiting, confusion, seizures, nausea, fainting, or death. If you start experiencing any of these symptoms, IMMEDIATELY move to fresh air. Seek medical attention.
- Keep exhaust gas from entering a confined area through windows, doors, ventilation intakes, or other openings. Warn any occupants inside about the symptoms of carbon monoxide so they know to move to fresh air if they begin to feel ill.
- NEVER start or run engine indoors, in a vehicle, or in an enclosed area, even if windows and doors are open.
- Use a respirator or mask whenever there is a possibility of inhaling harmful gas or vapors.
- Read all instructions with respirator or mask so you are certain the respirator or mask will provide the necessary protection against inhaling harmful gas or vapors. If you begin to experience any of the CO symptoms listed above, IMMEDIATELY move to fresh air.

Electrical Safety



⚠ DANGER

Electrocution. Water contact with a power source, if not avoided, will result in death or serious injury.

(000104)

- Never work on this engine or handle any electrical device while standing in water, while barefoot, or while hands or feet are wet. DANGEROUS ELECTRIC SHOCK will result.
- Never operate the engine (a) in the rain; (b) in any enclosed compartment; (c) if the engine speed changes; (d) if the engine sparks; (e) if flame or smoke is observed while the engine is running.

Fire and Fuel Hazards



⚠ DANGER

Explosion and Fire. Fuel and vapors are extremely flammable and explosive. Add fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury.

(000105)



⚠ DANGER

Explosion and Fire. Fuel and vapors are extremely flammable and explosive. Store fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury.

(000143)



⚠ DANGER

Explosion and Fire. Do not fill fuel tank past full line. Allow for fuel expansion. Overfilling may cause fuel to spill onto engine causing fire or explosion, which will result in death or serious injury.

(000214)

- Avoid spilling gasoline on a hot engine. Comply with all of the laws regulating storage and handling of gasoline.
- Store gasoline and other fuels only in containers designed and approved for the storage of such materials.
- Store engine and fuel containers away from furnaces, stoves, water heaters, clothes dryers, or other appliances with pilot lights or other ignition source because they can ignite fuel vapors.
- Add gasoline in a clean, well-ventilated area. Wipe up any spilled gasoline immediately. If gasoline has been spilled, let it dry completely before starting the engine.

- Stop the engine and let it cool at least two minutes before removing fuel cap. Loosen cap slowly to relieve pressure in tank.
- Fill or drain fuel tank outdoors.
- DO NOT light a cigarette or smoke.
- It is a violation of California Public Resource Code, Section 4442, to use or operate the engine on any forest-covered, brush-covered, or grass-covered land unless the exhaust system is equipped with a spark arrester, as defined in Section 4442, maintained in effective working order. Other states or federal jurisdictions may have similar laws. Contact the original equipment manufacturer, retailer, or IASD to obtain a spark arrester designed for the exhaust system installed on this engine.

Hearing Protection Hazard



⚠ WARNING

Hearing Loss. Hearing protection is recommended when using this machine. Failure to wear hearing protection could result in permanent hearing loss.

(000107)

Moving Parts Hazard



⚠ WARNING

Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury.

(000115)

- Tie up long hair and remove jewelry.

Before Starting Equipment

- Verify spark plug, muffler, fuel cap, and air cleaner are in place.
- DO NOT crank engine with spark plug removed.
- Thoroughly inspect the engine for loose or damaged parts before each use. Do not use the engine until adjustments or repairs are made.
- Check the oil level in the engine before each use.

Kickback Hazard



⚠️ WARNING

Recoil Hazard. Recoil could retract unexpectedly. Kickback could result in death or serious injury.

(000183)

- When starting engine, pull cord slowly until resistance is felt and then pull rapidly to avoid kickback.

When Operating Equipment

⚠️ WARNING

Equipment damage. Operating engine with obstructed air cooling slots could result in death, serious injury, and unit damage.

(000255a)



⚠️ WARNING

Fire hazard. Do not obstruct cooling and ventilating airflow around the engine. Inadequate ventilation could result in fire hazard, possible equipment damage, death or serious injury. (000256)

⚠️ WARNING

Personal injury and equipment damage. Do not tamper with engine governed speed. Operating engine too fast or slow increases the risk of injury or permanent engine damage.

(000254)

- DO NOT tip engine or equipment at angle which causes fuel to spill.
- Reflective exhaust heat may damage the fuel tank, causing fire. Keep at least 5 ft (1.3 m) of clearance on all sides of the engine for adequate cooling, maintenance, and servicing.
- DO NOT stop the engine by moving the choke lever to the CHOKE position.

Hot Surfaces Hazard



⚠️ WARNING

Hot Surfaces. When operating machine, do not touch hot surfaces. Keep machine away from combustibles during use. Hot surfaces could result in severe burns or fire.

(000108)

- DO NOT touch hot parts and AVOID hot exhaust gases.
- Allow equipment to cool before touching.

When Transporting or Repairing Equipment

- Transport/repair with fuel tank EMPTY.
- Disconnect spark plug wire.

Spark Hazards

⚠️ WARNING

Accidental start-up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.

(000141)

When Testing For Engine Spark

- Use approved spark plug tester.
- DO NOT check for spark with spark plug removed.

Battery Hazards



⚠️ WARNING

Explosion. Do not dispose of batteries in a fire. Batteries are explosive. Electrolyte solution can cause burns and blindness. If electrolyte contacts skin or eyes, flush with water and seek immediate medical attention.

(000162)



⚠️ WARNING

Explosion. Batteries emit explosive gases while charging. Keep fire and spark away. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000137a)



⚠️ WARNING

Electrical shock. Disconnect battery ground terminal before working on battery or battery wires. Failure to do so could result in death or serious injury.

(000164)



⚠️ WARNING

Risk of burns. Batteries contain sulfuric acid and can cause severe chemical burns. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000138a)

⚠️ WARNING

Environmental Hazard. Always recycle batteries at an official recycling center in accordance with all local laws and regulations. Failure to do so could result in environmental damage, death or serious injury.

(000228)

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit the Battery Council International website at: <http://batterycouncil.org>.

General Maintenance

- Inspect the engine periodically. Repair or replace all damaged or defective parts immediately.
- Inspect fuel system frequently for leaks or damage. Repair or replace any damaged or leaking component immediately. Never attempt to change, alter or modify the engine fuel system in any way which might affect safety or compliance with applicable codes and standards.

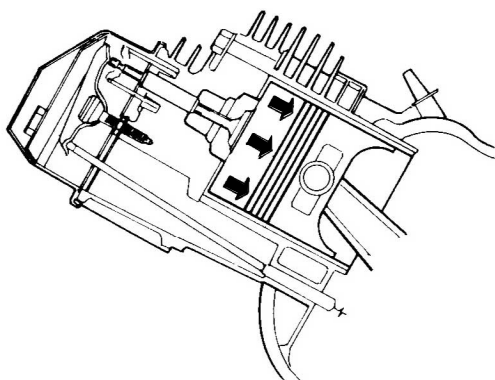
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Section 2: Four-Cycle Engine Theory

Introduction

If the engine is to run properly, four events must occur in proper sequence and at the correct time. These events are (A) intake, (B) compression, (C) ignition and power, and (D) exhaust.

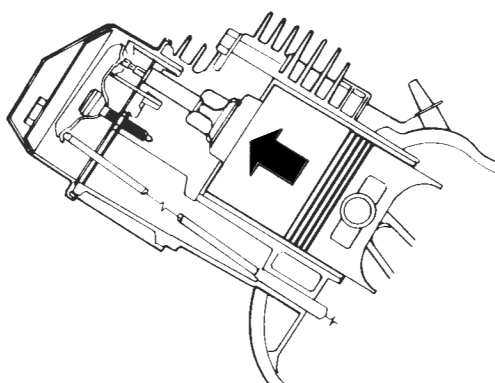
(A) Intake



001684

The piston is traveling from top dead center (TDC) to bottom dead center (BDC). The cam has opened the intake valve. The piston's downward movement in the cylinder creates a partial vacuum in the cylinder. Air at atmospheric pressure is drawn into the cylinder through the carburetor and is mixed with fuel in the carburetor. The fuel-air mixture flows through the open intake valve into the cylinder. When the piston reaches BDC, the intake stroke is over.

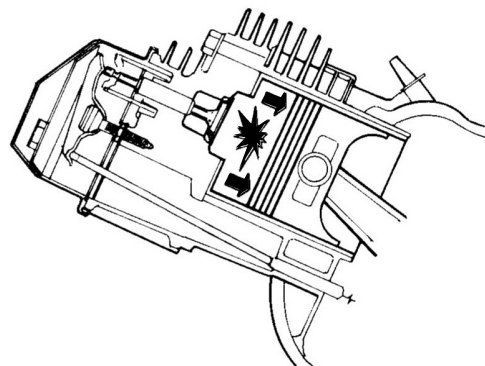
(B) Compression



001685

As the piston reaches bottom dead center (BDC), both the intake and exhaust valves are closed. The piston moves upward toward TDC and the fuel-air mixture is compressed. Just before the piston reaches TDC, ignition occurs.

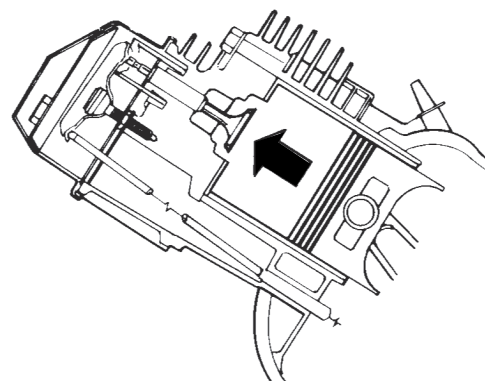
(C) Power



001686

By the time the piston reaches TDC, combustion is already in progress. The intake and exhaust valves remain closed as the expanding gases of combustion force the piston downward.

(D) Exhaust



001687

The exhaust stroke begins when the piston has reached BDC and has started its upward movement. The intake valve is closed. The exhaust valve is open to let gases escape.

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Section 3: General Information

Maintenance Schedule

	Every 8 Hours or Daily	Every 50 Hours or Yearly	Every 100 Hours or Yearly	Every 500 Hours or Yearly
Check Oil Level	•			
Change Oil	Note 1		Note 1	
Change Oil Filter	Note 1		Note 1	
Service Air Filter				Note 2
Fuel Filter				•
Replace or Clean Spark Plug			•	
Clean Spark Arrestor Screen			•	
Check Valve Clearance		Note 3	Note 3	

NOTE 1: Change oil and filter after first 25 hours of operation and then every 100 hours thereafter. Change sooner when operating under a heavy load or in a dusty or dirty environment or in high ambient temperatures.

NOTE 2: Clean more often when operating in dirty or dusty conditions.

NOTE 3: Check valve clearance and adjust if necessary after first 25 hours of operation and every 100 hours thereafter.

Fuel And Oil Recommendations

Gasoline

Use clean, fresh lead-free gasoline with a minimum of 85 octane. Lead-free gasoline results in fewer combustion deposits and longer valve life.

NOTE: A fuel additive such as Sta-Bil® fuel stabilizer, or an equivalent, will prevent gum deposits from forming in the engine fuel system.

NOTE: Some fuels, called oxygenated or reformulated gasolines, are gasolines blended with alcohols or ethers. Excessive amounts of these blends can damage the fuel system or cause performance problems. Do not use gasoline which contains Methanol. If any undesirable operating symptoms occur, use gasoline with a lower percentage of alcohol or ether. Recommended percentage is 10% or less.

Gasoline should be purchased in small quantities—not more than a 30 day supply. Fresh gasoline minimizes gum deposits, and also ensures fuel volatility tailored for the season in which the engine will be operated.

Lubrication

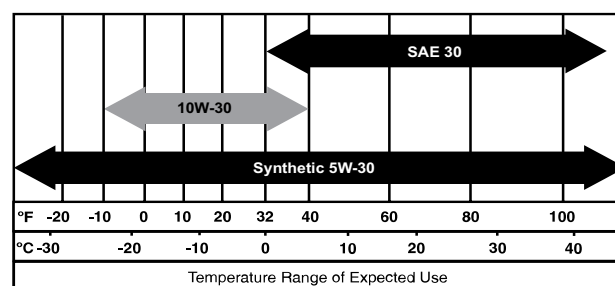
Oil has four purposes—it cools, cleans, seals, and lubricates. During normal operation, small particles of metal from the cylinder walls, pistons, bearings, and combustion deposits gradually contaminate oil. Dust particles from the air also contaminate oil, forming an abrasive mixture which causes wear to all of the internal moving parts of the engine if oil is not changed regularly. Fresh oil also assists in cooling. Old oil gradually becomes thick and loses its cooling ability and lubricating qualities.

Recommended Oil

Proper crankcase oil type and oil weight is extremely important. Check oil before each use and change oil regularly (see [Figure 3-2](#) through [Figure 3-6](#)). Incorrect or dirty oil can cause premature engine wear and failure.

Use only high quality detergent oil rated with API service classification SN or higher. Recommended oil weights include the following:

- During summer months: SAE 30. An acceptable substitute is SAE 10W-30. After first oil change, synthetic oil is acceptable.
- During winter months: SAE 5W-30 or Synthetic 5W-30. DO NOT USE SAE 10W-40. After first oil change, synthetic oil is acceptable.



000399

Figure 3-1. Recommended Oil

Check and Change Oil

⚠ WARNING

Skin irritation. Avoid prolonged or repeated contact with used motor oil. Used motor oil has been shown to cause skin cancer in laboratory animals. Thoroughly wash exposed areas with soap and water.

(000210)

See **Specifications** for crankcase oil capacities. Use no special additives. Verify the unit is level when filling with oil.

Change oil and filter after first eight hours of operation. Change oil and filter every 100 hours of operation thereafter. Change oil more often if engine is operated in dirty or dusty conditions, or if engine is operated under heavy loads or in high ambient air temperatures.

IMPORTANT: Do not overfill. Excessive oil will foam, smoke, and eventually damage internal engine parts.

Check Engine Oil Level

See **Figure 3-2** and **Figure 3-3**. Regularly check oil level on the dipstick (A) and maintain engine oil level.

- On vertical engines, add oil through the dipstick opening.
- See **Figure 3-4**. On horizontal engines, add oil through the oil fill port (B).

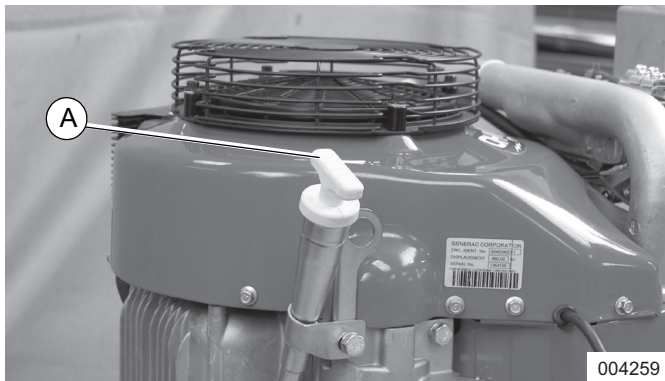


Figure 3-2. Oil Fill / Check—Vertical Engine

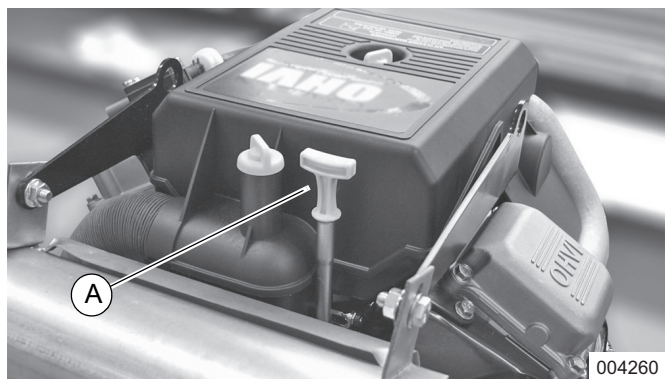


Figure 3-3. Oil Check—Horizontal Engine

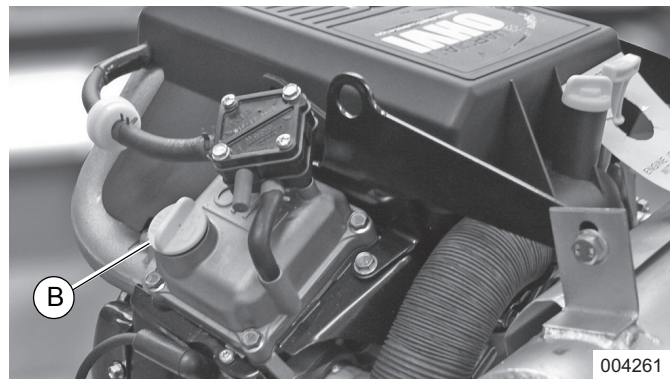


Figure 3-4. Oil Fill—Horizontal Engine

Replace Engine Oil

1. See **Figure 3-5**. Remove oil drain plug (C) and drain oil while engine is still warm.

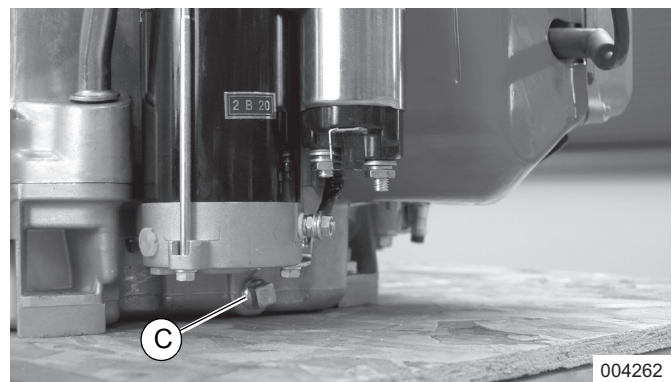


Figure 3-5. Oil Drain

2. See **Figure 3-2** through **Figure 3-4**. Remove dipstick (A) or oil fill cap (B) and refill slowly with new oil of proper service classification and viscosity grade. Refill to bottom of threads in oil fill opening.

Change Oil Filter

Replace oil filter every 100 hours.

1. See **Figure 3-6**. Remove used oil filter (D) and replace oil drain cap.



Figure 3-6. Oil Filter

2. Before installing a new filter, lightly oil filter gasket with fresh clean engine oil.
3. Screw filter on by hand until gasket contacts filter adapter. Tighten 3/4 to one full turn farther.
4. Start and run engine at idle for 30 seconds. Stop engine. Check oil level and add if required.
5. Start engine and check for oil leaks.

Clean Intake Screen

See **Figure 3-7**. Grass particles, chaff, or dirt can clog the air cooling system, especially after prolonged service in cutting dry grass or when operating in extremely dusty or dirty conditions. Continued operation with a clogged cooling system can cause severe overheating and possible engine damage. Therefore, clean the intake screen (E) and oil cooler fins (F) after each use.

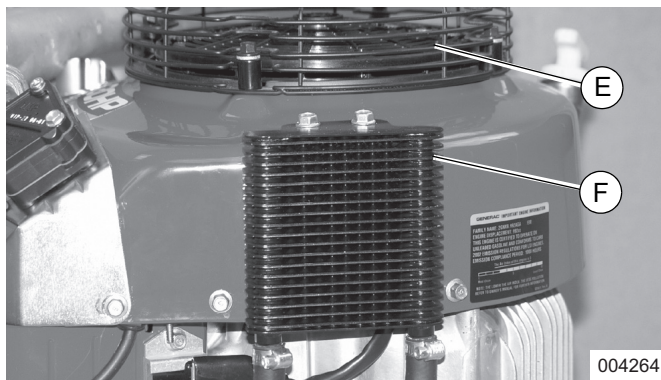
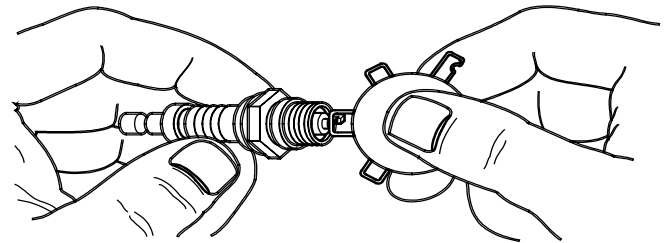


Figure 3-7. Intake Screen and Oil Cooler Fins

Replace Spark Plugs

Replace spark plugs every 100 hours of operation or every season, whichever occurs first. Replace spark plug if electrode is burned away, or if porcelain is cracked. Refer to generator owner's manual for spark plug replacement specification.

See **Figure 3-8**. Check spark plug gap and set gap to 0.040 in (1.02 mm) for all models. Tighten spark plug to 168 in-lb (1.09 Nm) torque.



000211

Figure 3-8. Check Spark Plug Gap

Maintain Air Cleaner



WARNING

Risk of fire. Never operate engine without the air cleaner installed. Operating engine without the air cleaner could result in death or serious injury.

(000249)

A properly serviced air cleaner protects internal engine parts from dirt and dust particles in the air. If air cleaner instructions are not carefully followed, dirt and dust normally collected in the air cleaner will instead be drawn into the engine. Worn cylinder bores and piston rings allow these particles to enter the crankcase and contaminate the oil, forming an abrasive mixture which will quickly wear internal moving parts.

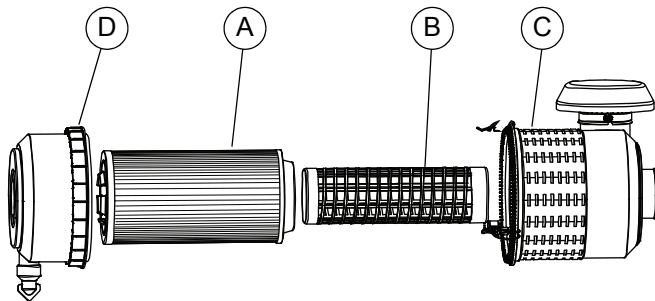
Examine the air cleaner on every engine brought in for service or repair. If the air cleaner shows signs of neglect, show it to the customer before cleaning. Instruct the customer on proper care techniques to assure long engine life.

NOTE: Replace worn or damaged air cleaner gaskets and mounting gaskets to prevent dirt and dust from entering the engine due to improper sealing. Replace bent air cleaner mounting bracket if necessary.

NOTE: Air cleaner assemblies on some equipment may have been supplied by the equipment manufacturer. See the equipment manufacturer's owner's manual for service information specific to that product.

Service Canister Air Cleaners

See [Figure 3-9](#). Canister-type air cleaners consist of a primary air filter (A) and secondary air filter (B) contained inside a filter housing (C) with removable cover (D).



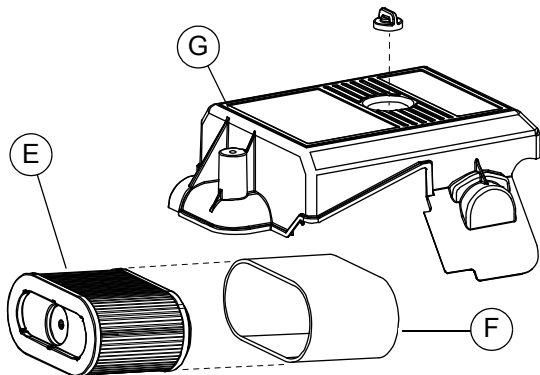
004344

Figure 3-9. Canister Air Cleaner Components

Clean air filter elements, filter housing, and removable cover with low-pressure compressed air every 50 hours or every season, whichever occurs first. Replace air filter elements every 500 hours or if damaged.

Service Dual Element Air Cleaners

See [Figure 3-10](#). Dual element air cleaners consist of a paper air filter element (E) wrapped with a foam pre-cleaner (F). The filter element and foam pre-cleaner are contained inside a cover (G).



004295

Figure 3-10. Dual Element Air Cleaner Components

Remove and service foam pre-cleaner every 25 hours or every season, whichever occurs first. Soak pre-cleaner in hot water and wash with a mild detergent (for example, dish soap) to dissolve grease. Rinse thoroughly, squeeze out excess water, and allow to dry before installing.

Use low-pressure compressed air to clean paper air filter element. Replace paper air filter element every 500 hours or if damaged.

Troubleshooting

Most complaints concerning engine operation can be classified as one or a combination of the following:

- Will not start
- Hard starting
- Lack of power
- Runs rough
- Vibration
- Overheating
- High oil consumption

NOTE: What appears to be an engine malfunction may be a fault of the powered equipment rather than the engine. If equipment is suspect, see [Equipment Affecting Engine Operation](#).

Systematic Check

If the engine will not start and the cause of malfunction is not readily apparent, perform a systematic check in the following order:

1. Fuel
2. Ignition
3. Compression

These checks, performed in a systematic manner, can usually be done in a matter of minutes. It is the quickest and surest method of determining the cause of failure. The basic check procedure is the same for all engine models. Any variation, by model, will be described under the subject heading.

Check Fuel

Fuel pressure on LP and NG generator engines can be checked using a pressure test kit for LP and NG systems. For gasoline engines, check the following:

- Are tanks full?
- Is fuel stale?
- Is tank vent open?
- Is fuel shutoff valve open?
- Is fuel pump working?
- Is fuel solenoid working?

Check Ignition

See [Ignition](#) for testing procedures.

Check Compression

The following procedure is a simple and accurate test of cylinder compression.

1. Remove both spark plugs and insert a compression gauge into either cylinder (one cylinder at a time).
2. Open throttle to Wide Open Throttle (WOT) position.
3. Turn engine over with engine starter until there is no further increase in pressure. Record this reading.
4. Repeat procedure on other cylinder and record that reading.

The difference between both cylinders should not exceed 25%. More than 25% indicates loss of compression in the cylinder with lower pressure.

To calculate the difference in percentage, divide the difference in PSI (kPa) by the higher pressure reading. Refer to the examples below.

	Cylinder 1	Cylinder 2	Difference
Engine #1	165 psi (1137.6 kPa)	160 psi (1103.1 kPa)	5 psi (34.5 kPa)
Engine #2	175 psi (1206.6 kPa)	155 psi (1068.7 kPa)	20 psi (137.9 kPa)
Engine #3	160 psi (1103.1 kPa)	112 psi (772.2 kPa)	48 psi (330.9 kPa)

Engine #1: $(5 \text{ psi} \div 165 \text{ psi}) [34.5 \text{ kPa} \div 1103.1 \text{ kPa}]$
 $= 0.03 = 3\% \text{ difference}$

Engine #2: $(20 \text{ psi} \div 175 \text{ psi}) [137.9 \text{ kPa} \div 1206.6 \text{ kPa}]$
 $= 0.11 = 11\% \text{ difference}$

Engine #3: $(48 \text{ psi} \div 160 \text{ psi}) [330.9 \text{ kPa} \div 1103.1 \text{ kPa}]$
 $= 0.30 = 30\% \text{ difference}$

Using this method of calculation, it is obvious Cylinder 2 in Engine #3 has lost significant compression.

If compression is poor, look for one or more of the following causes:

- Loose cylinder head bolts
- Failed cylinder head gasket
- Burned / worn valves or valve seats
- Insufficient valve clearance
- Warped cylinder head
- Warped valve stem or worn valve guides
- Worn or broken piston ring(s)
- Worn or damaged cylinder bore
- Broken connecting rod

Model	Normal Compression
GTH/GTV-760	130–160 psi (896.3–1310.0 kPa)
GTH/GTV-990/1000	160–190 psi (1103.1–1310.0 kPa)

Cylinder Leakdown Test

A cylinder leakdown tester may be used to test sealing capability of the compression components of each cylinder and quickly identify the problem component.

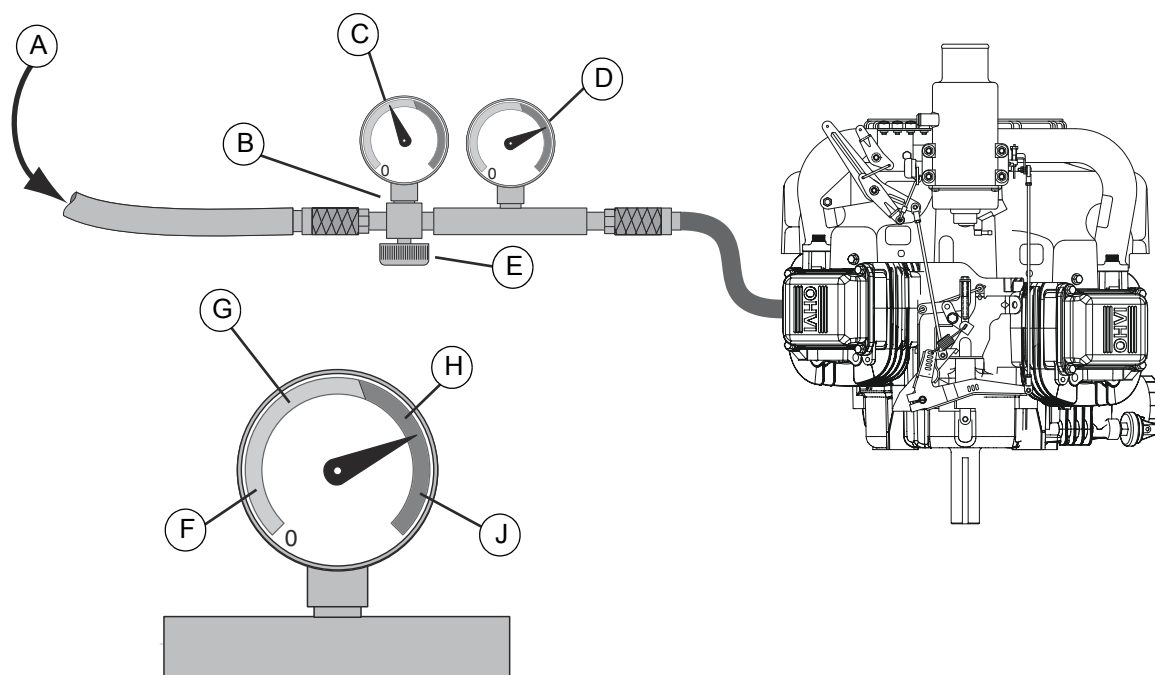
A cylinder leakdown test indicates cylinder condition. It will assist in troubleshooting leaking valves or rings.

See [Figure 3-11](#). There are two gauges on the tester. One measures incoming air pressure, and the other measures the percent of cylinder leakage.

Testing Procedure

1. Align piston at TDC.
2. Install tester in the spark plug hole.
3. Pressurize cylinder to 90 psi (621 kPa).
4. Observe cylinder leakage and identify location of air leak.

NOTE: If leaking into intake or exhaust port, check valve clearance, valve face, and seat condition.



004296

Figure 3-11. Cylinder Leakdown Tester

A	Compressed air inlet
B	Air pressure regulator
C	Inlet gauge pressure set point
D	Outlet gauge pressure
E	Regular adjustment knob
F	Outlet gauge
G	Red range indicates unacceptable air leakage
H	Needle indicates minimal air leakage
J	Green range indicates acceptable air leakage

Results

- 0-10%—Cylinder compression is acceptable
- 10-30%—There may be a leak
- 30-100%—Cylinder requires repair

Factors Affecting Cylinder

- Carburetion
- Crankcase vacuum
- Ignition timing
- A partially sheared flywheel key will affect ignition timing and engine performance.
- Spark plug
 - A fouled spark plug may indicate carburetor is out of adjustment.
- Leak in spark plug wire
- Head gasket
- Intake manifold
- Valves
- Rings
- Piston
- Cylinder Bore

NOTE: A twin cylinder engine will run well on one cylinder as long as power required for the application does not exceed the power produced by the cylinder.

Equipment Affecting Engine Operation

Frequently, what appears to be a problem with engine operation, such as hard starting or vibration, may be the fault of the equipment being powered rather than the engine itself. Listed are the most common effects of equipment problems—and what to look for as the most common cause.

Hard Starting or Will Not Start

- Loose drive mechanism—A loose drive mechanism (for example, a belt), can cause a backlash effect which will counteract engine cranking effort.
- Starting under load—Observe if unit is disengaged when engine starts; or, if engaged, unit should not have a heavy starting load.
- Check remote control assembly for proper adjustment.
- Check interlock system for shorted wires, loose or corroded connections, or defective modules or switches.

Engine Will Not Stop

- Check equipment ignition stop switch.
- Check for loose or disconnected equipment stop switch wire.
- Check ground wire harness on engine.
- See [Ignition](#) for test procedure.

Vibration

- Unit load out of balance—remove and balance.
- Mounting bolts loose—tighten.

Power Loss

- Bind or drag in unit. If possible, disengage engine and operate unit manually to feel for any binding action.
- Unit load has excess drag.

Noise

- Engine coupling or pulley. An oversize or worn coupling can result in knocking, usually under acceleration. Check for fit or tightness.
- Equipment needs lubrication.

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Section 4: Ignition

Overview

Generac GTH/GTV-999 OHVI V-Twin engines use a magneto ignition, which is an ignition coil with a self-contained transistor module (no moving parts). Two magneto ignition coils are used, with a flywheel containing a permanent magnet.

NOTE: The magneto ignition system requires a minimum of 250 rpm to produce a consistent spark.

Ignition Check

If the engine runs but misses during operation, first check to determine if the ignition system is functioning properly.

1. See [Figure 4-7](#). Install a spark tester (B) between the spark plug lead (A) and each spark plug (C) as shown. Test each plug separately if only one spark tester is available.

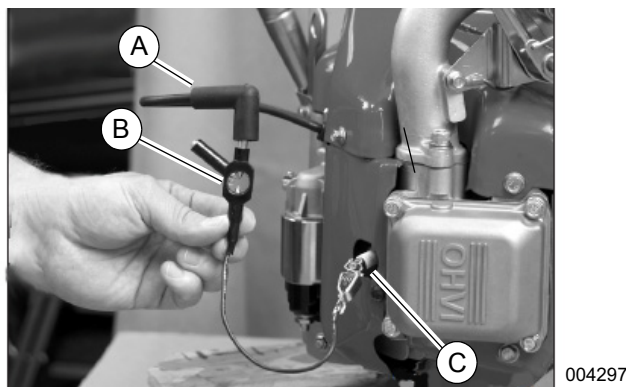


Figure 4-1. Spark Tester on Engine (Typical)

2. Start engine and verify if a steady spark is visible in the window of the tester. A lack of spark or miss will be readily apparent when the engine is running, indicating further investigation of the ignition system is required.
3. If spark is steady for both cylinders but engine still misses or will not run, check for a fouled spark plug. Also, verify fuel is present in the carburetor and power is being supplied to the fuel shut-off solenoid.

Ignition Coil Test

Test ignition coil if the ignition check indicates one or both of the coils are not delivering a strong consistent spark.

1. Remove ground wire from ignition ground stud on the back side of the blower housing (oil filter side).
2. Repeat ignition check procedure to determine if ignition now has spark.
3. If spark is present, check wiring harness for broken insulation or pinched wires. These faults may cause a ground contact in the circuit between the grounding stud and the low oil pressure switch or engine kill switch.
4. If there is no spark present, then proceed with [Coil Diode Test](#).

NOTE: Verify engine has oil and there is no continuity between the two spade lugs on the low oil pressure switch while cranking or during operation. Continuity between the spade lugs indicates low or nonexistent oil pressure, or a faulty low oil pressure switch. The low oil pressure switch will ground the coils if <10 psig (69 kPa) is present at the switch; stopping the engine.

Coil Diode Test

See [Figure 4-2](#). A digital multimeter (DMM) with a diode test position is required to test the diode in the coils. With the DMM in the diode test position, the display indicates the forward voltage drop across the diode.

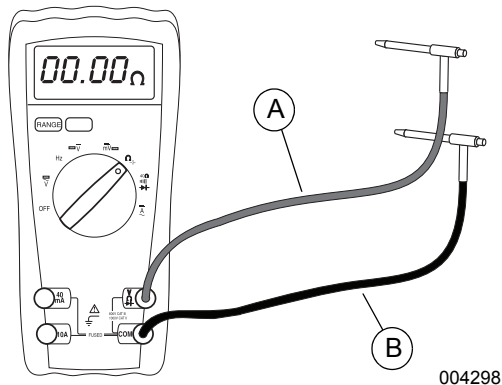
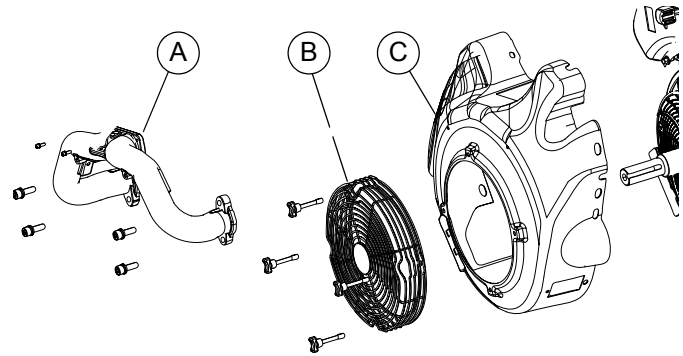


Figure 4-2. Digital Multimeter (DMM)

1. Insert RED test lead (A) into **VΩ→+** receptacle on DMM.
2. Insert BLACK test lead (B) into DMM “COM” receptacle.
3. Rotate selector to **→+ IIII** (Diode Test) position.
4. Disconnect wires from coil ground terminal on back side of blower housing (oil filter side).
5. Insert RED test lead clip onto coil ground terminal on back side of the blower housing (oil filter side)
6. Touch BLACK test lead probe to crankcase ensuring a good contact.
 - a. If DMM beeps once, voltage drop is less than 0.7 volts. This indicates diode is functioning properly.
 - b. If DMM produces a continuous tone, at least one diode is faulty (shorted) and further investigation is required.
 - c. If DMM displays OL, circuit is incomplete (open). At least one diode is faulty and further investigation is required.

7. See [Figure 4-3](#). Remove intake manifold (A), rotating screen (B), and blower housing (C) from engine.



004299

Figure 4-3. Remove Intake Manifold

8. See [Figure 4-4](#). Inspect coil grounding wires leading from each coil to grounding stud in the blower backing plate on left side of engine as viewed from flywheel side. Look for signs of broken insulation, pinching, or other damage allowing the wire to ground out the coil in operation. If any damage is identified, replace wire and check for spark.



008801

Figure 4-4. Inspect Coil Grounding Wires

9. Inspect grounding stud in blower backing plate to verify it has not worn through the nylon grommet; allowing stud to ground to backing plate. If contact is identified, replace stud/grommet assembly and check for spark.

Continued on next page.

10. Check cylinder #1 coil as follows:
 - a. Remove grounding wire from spade terminal on coil.
 - b. Using DMM configured as in [Coil Diode Test](#), connect RED test lead clip to grounding spade on coil.
 - c. Touch BLACK test lead probe to crankcase or metal laminate on the coil ensuring a good contact.
 - If DMM beeps once, voltage drop is less than 0.7 volts. This indicates the diode is functioning properly.
 - If DMM produces a continuous tone, diode is faulty (shorted) and coil must be replaced.
 - If DMM displays OL, diode is faulty (shorted) and coil must be replaced.
11. Repeat step 9 for cylinder #2 coil.

Coil Resistance Check

Follow the procedure below to check coil resistance.

1. Set DMM to measure resistance.
2. Touch RED test lead probe to metal in spark plug boot.
3. Touch BLACK lead to crankcase or metal laminate on coil. Resistance reading should be within 7k–14k Ohms.
4. Remove wire from ignition ground stud on blower housing, and connect BLACK lead to ignition ground stud and RED lead to crankcase or metal laminate of coil. Resistance reading should be between 2M–2.5M Ohms.
5. Connect RED lead to ignition ground stud and BLACK lead to metal in spark plug boot. Again, resistance reading should be between 2M–2.5M Ohms.
6. With ignition ground stud wire still disconnected, place RED lead on ignition ground stud and BLACK lead on crankcase or metal laminate of the coil. Resistance should read O/L or infinity (∞).
7. Place BLACK lead on ignition ground stud and RED lead on metal inside plug wire. Resistance should read O/L or infinity.

Additional Ignition Test

With engine cranking at a minimum of 250 rpm, 3–5 volts AC should be measured at the ignition ground stud. This voltage is the sum total of the individual voltage for each coil. Therefore, a reading of 1.5–2.5 volts AC indicates a single coil failure or loose wiring.

20 Amp Regulated Alternator

General Alternator Design

The 20 Amp regulated alternator system on this engine provides AC current through two output leads to the regulator-rectifier. The regulator-rectifier converts the AC current to DC, and regulates the current to the battery. Charging rate varies with engine rpm and temperature.

Alternator Output Test

1. Temporarily disconnect stator wire harness from regulator-rectifier.
2. Using DMM, insert RED test lead into **V Ω →+** receptacle on DMM.
3. Insert BLACK test lead into COM receptacle on DMM.
4. Rotate selector to AC Volt ($\overline{\sim}$) position.

NOTE: Attach DMM test leads to AC output terminals (white wires) before starting engine. If stator is grounded (faulty), and DMM test leads contact center DC output pin, arcing may occur which may damage wiring.

5. See [Figure 4-5](#). Attach RED and BLACK test lead probes to AC output terminals (white wires) as shown. DMM test clip leads may be attached to either AC output terminal.



Figure 4-5. Test AC Output

6. With engine running at 3600 rpm, output should be no less than 26 volts—20 Amp system. If no or low output is found, check for bare wires or other obvious defects.
7. Replace stator if shorted leads are not visible.

DC Output Charging Wire Test

Proceed as follows to test the DC output charging wire circuit. If a problem exists in this circuit, it can be corrected before testing the regulator-rectifier.

1. Verify key switch for engine/equipment is in the OFF position.
2. Verify stator wire is disconnected from regulator-rectifier.
3. If necessary, set DMM as for **Alternator Output Test**.
4. Rotate DMM selector to DC volts ($\overline{\text{V}}$) position.
5. See **Figure 4-6**. Attach RED test lead probe to DC output wire terminal (Red wire).



004302

Figure 4-6. Test DC Output Charging Wire

6. Attach BLACK test lead probe to negative battery terminal.
7. Turn engine/equipment key switch to ON. DMM should display battery voltage.
8. If DMM does not display battery voltage, check for a blown fuse or broken/shorted wire(s).

Regulator-Rectifier Test

Proceed as follows to test the voltage rectifier-regulator circuit. This circuit maintains battery voltage.

1. Using DMM, test battery voltage while engine is NOT running.
2. With charging system properly connected, start engine.
3. Bring engine up to normal operating speed and test battery voltage again.

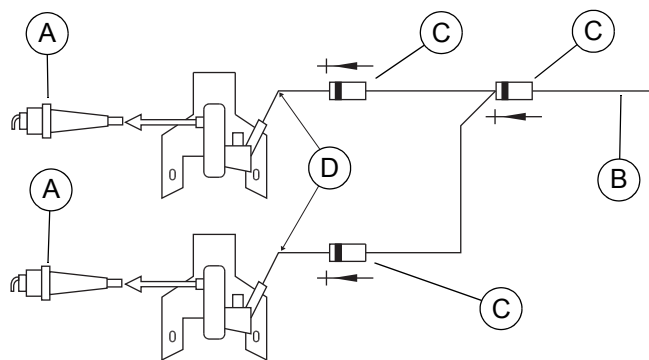
If battery voltage while engine is running is greater than when engine is stopped, charging system is working properly.

NOTE: When engine is running, battery voltage should be at least 13 V.

Engine Wiring Harness

See **Figure 4-7**. The engine wiring harness supplies power to spark plugs (A). The harness consists of an ignition ground wire (B) with a diode (C) for each ignition coil and a separate wire for the carburetor solenoid. The engine ground wire harness (D) is connected to the wiring harness provided by the equipment manufacturer.

NOTE: Models built after 2007 have diodes built into ignition coils or spark plug leads. Check the wire harness. If there are no diodes in the harness, test the coil ground terminal.



004303

Figure 4-7. Engine Wiring Harness

Test Ground Wires

Use a DMM in Diode test position to test ground wires.

In Diode test position, the DMM displays forward voltage drop across diode(s). If voltage drop is less than 0.7 volts, DMM will "beep" once and display voltage drop. A continuous tone indicates continuity (a shorted diode) An incomplete circuit (open diode) displays as "OL."

1. See [Figure 4-8](#). Insert RED test lead (A) into **VΩ→+** receptacle in DMM.

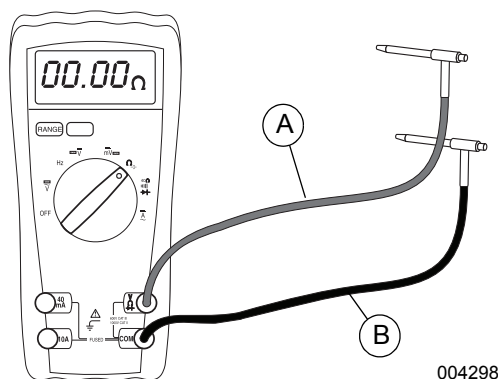


Figure 4-8. Ground Wire Test Setup

2. Insert BLACK test lead (B) into DMM "COM" receptacle.
3. Rotate selector to **→+ IIII** (Diode Test) position.
4. See [Figure 4-9](#). Insert RED test lead clip into connector (C). Leave attached for remainder of test.

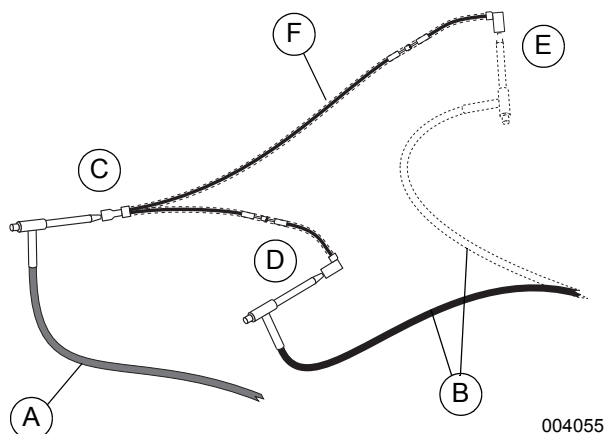


Figure 4-9. Testing Ground Wire

5. Touch BLACK test lead probe to terminal (D).
 - a. If DMM "beeps" once, diode is functioning normally.
 - b. If DMM makes a continuous tone, diode is defective (shorted). Replace ground harness.
 - c. If DMM displays "OL", the diode is defective (open). Replace ground harness.
6. Now repeat test for terminal (E). Results must be the same.
7. Replace wiring if defective.

8. If wiring tests normally, proceed to section [Remove Ignition Coil](#) to replace defective coil.

Table 4-1. Diode Failure Diagnosis

Switch On	Turned Off	Cause
Engine runs on 1 cylinder	Engine shuts off	1 closed diode
Engine runs (both cylinders)	Only one cylinder	1 open diode
Engine won't run (no spark)	—	2 closed diodes
Engine runs (both cylinders)	Engine won't shut off	2 open diodes

Wire Harness Check

If ignition does not have spark, disconnect ground wire from ignition ground stud on engine and check again. If ignition now has spark, check wire harness for pinched wires, broken insulation, or (if equipped) faulty diode. If there is no spark, proceed to [Remove Ignition Coil](#) section.

Removing and Installing Wire Harness

1. Remove spark plug leads.
2. Remove intake manifold and cover intake ports with a shop towel.
3. Remove rotating screen and blower housing.
4. Disconnect stop switch wire(s) at ignition coil(s).
5. Reverse order of above to install new wire harness.

Remove Ignition Coil

Ignition coil condition can be accurately diagnosed using a spark tester (Generac P/N 0C5969) as described in section [Troubleshooting](#).

1. Disconnect spark plug leads.
2. Remove intake manifold and cover intake ports with a shop towel.
3. Remove rotating screen and blower housing.
4. Disconnect ground wire(s) at ignition coil(s).
5. See [Figure 4-10](#). Remove ignition coil screws and remove ignition coil(s).

NOTE: Flywheel does not need to be removed to service ignition except to check the flywheel key.

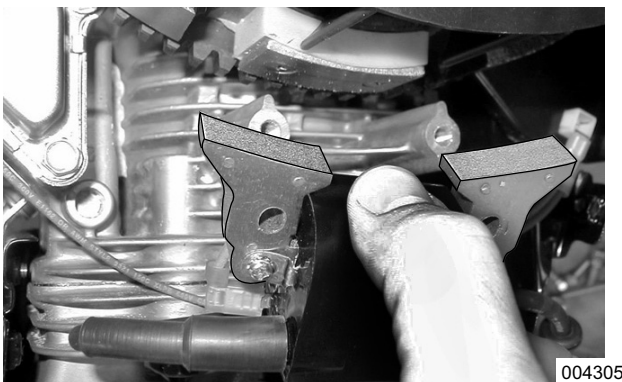


Figure 4-10. Remove Ignition Coil

Install Ignition Coil

1. Turn flywheel so magnet is facing away from ignition coil.
2. Install ground wire onto tab terminal on ignition coil.

NOTE: Verify wires are routed over ignition coil mounting posts and away from flywheel.

3. See [Figure 4-11](#). Assemble ignition coil on engine, noting position of ground wire routing (A).

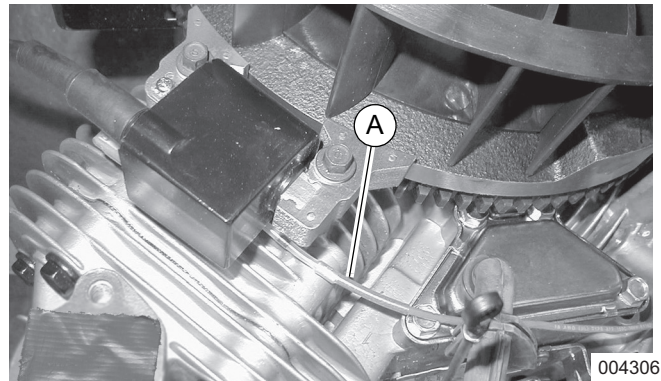


Figure 4-11. Install Ignition Coil

NOTE: Mounting holes in ignition coil are slotted. Push ignition coil away from flywheel as far as possible and tighten screws.

4. Repeat for second ignition coil.
5. Proceed to [Adjust Ignition Coil Air Gap](#).

Adjust Ignition Coil Air Gap

1. Rotate flywheel until magnet is aligned with ignition coil laminations.
2. See [Figure 4-12](#). Place a 0.008–0.012 in (0.20–0.30 mm) thick, non-magnetic shim (A) between magnet and ignition coil laminations.

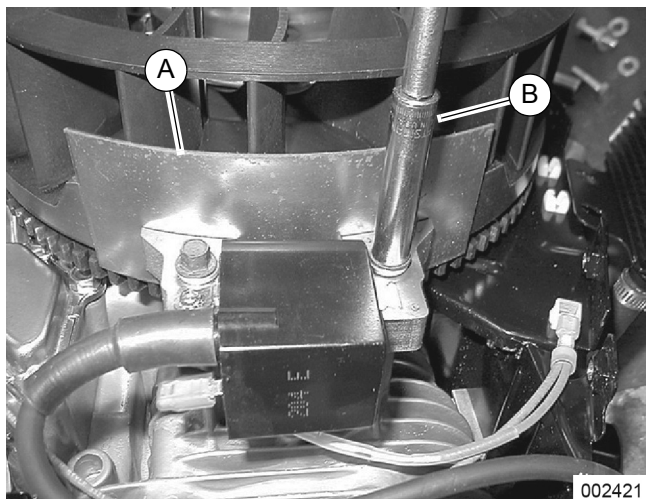


Figure 4-12. Adjusting Air Gap

3. Using a socket wrench (B), loosen mounting screws so magnet will pull ignition coil against the shim.
4. Tighten screws to 40 in-lb (4.75 Nm) torque.
5. Rotate flywheel to remove shim.
6. Repeat for second ignition coil.

NOTE: See [Figure 4-14](#). Route ignition coil ground wire (B) under breather tube and away from flywheel.

Flywheel

The flywheel is a weighted disc which helps keep the crankshaft rotating between power strokes.

Remove Flywheel

1. Remove two screws attaching fan and fan retainer to flywheel.
2. Remove fan retainer and fan.
3. See [Figure 4-13](#). Loosen flywheel nut (A) until it is flush with end of crankshaft threads

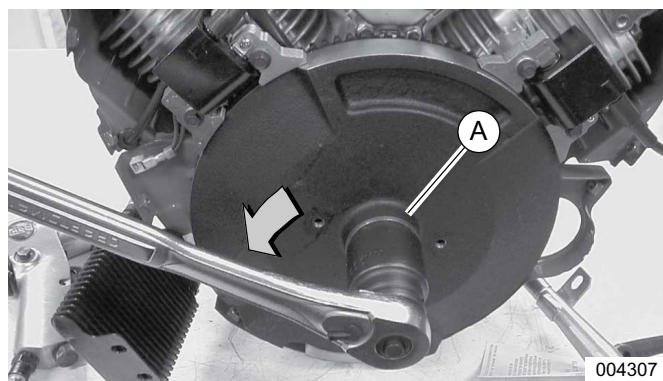


Figure 4-13. Removing Flywheel Nut

4. Install flywheel puller.
5. See [Figure 4-14](#). Tighten puller screws equally until flywheel loosens.

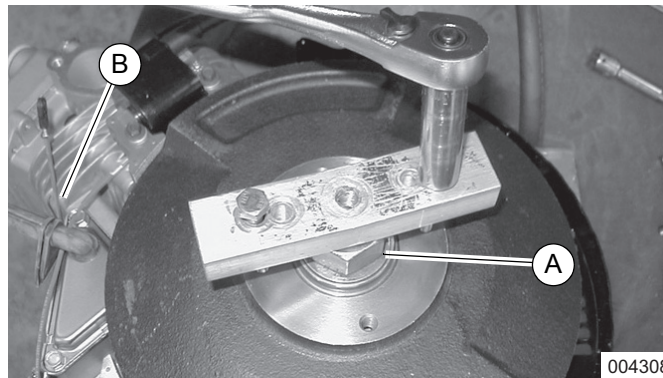


Figure 4-14. Removing Flywheel

6. Remove puller, flywheel nut (A), washer and flywheel. ("B" identifies ignition coil ground wire.)

IMPORTANT NOTES:

- Do not overtighten flywheel puller bolts. Overtightening bolts may damage lighting coil.
- DO NOT strike flywheel with a hard object or a metal tool. Doing so may cause flywheel to shatter in operation. Always use approved flywheel removal tools.

Inspect Flywheel Key, Keyways, Flywheel, and Crankshaft

1. See [Figure 4-14](#). Check flywheel key and keyway (A) for damage.

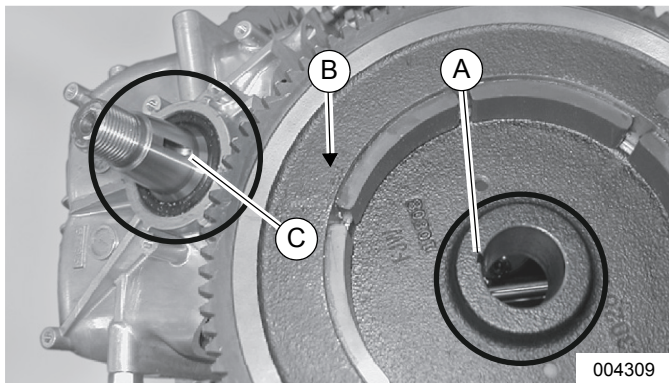


Figure 4-15. Check Flywheel and Crankshaft

2. Check flywheel (B) for cracks or keyway damage.
3. Check crankshaft keyways and taper (C) for damage. Replace crankshaft if damaged.

Install Flywheel

Clean flywheel and crankshaft taper before installing flywheel. Remove all oil, dirt, and grease.

1. Insert flywheel key into crankshaft.
2. Align keyways and assemble flywheel to crankshaft.
3. Install washer and flywheel nut.
4. See [Figure 4-16](#). Tighten flywheel nut to 150 ft lb (204.0 Nm) torque.

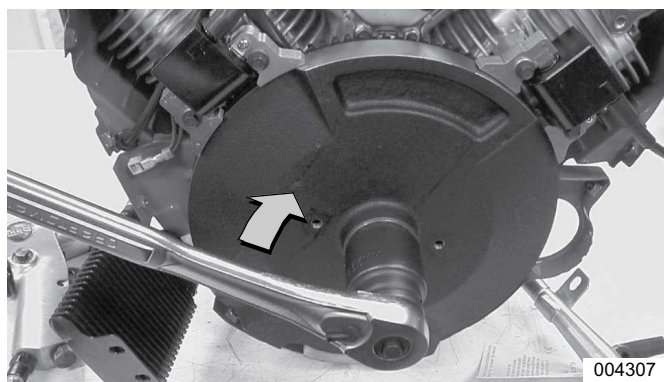


Figure 4-16. Tighten Flywheel Nut

5. See [Figure 4-17](#). Assemble fan and retainer on flywheel.



Figure 4-17. Install Flywheel Fan

6. Tighten screws to 182 in-lb (21.7 Nm) torque.

Spark Plug

Spark plug(s) should be cleaned and re-gapped periodically. Plug(s) should be replaced every 100 hours of operation, or once annually, whichever comes first.

See [Figure 4-18](#). Set spark plug gap to 0.040 in (1.02 mm).

NOTE: For proper spark plug replacement, refer to the owner's manual for the specific product. Always use original GENERAC® replacement parts.

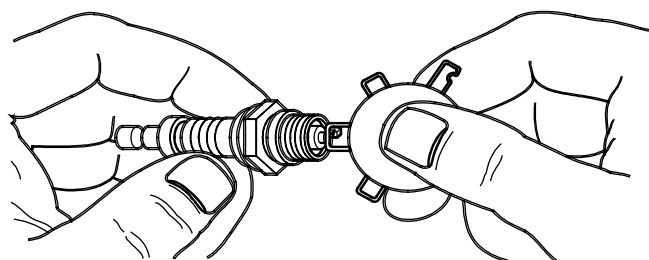


Figure 4-18. Check Spark Plug Gap

Engine Start Switch

The engine start switch opens and closes the electrical circuit delivering voltage to the spark plug. It is impossible to generate a spark when the engine is cranked if engine start switch is in STOP position.

Move engine start switch to RUN position before starting the engine. Always move switch from RUN to STOP when engine is not in use.

Replace engine start switch if it is loose or visibly damaged.

Section 5: Carburetion and Fuel System

Carburetion

Overview

Proper engine performance depends on the carburetion system. Clean, fresh fuel and a well maintained air cleaner are extremely important to engine operation, engine reliability, and power.

All Generac gasoline powered OHVI V-Twin engines feature a two-barrel type carburetor. Some LP engines use a single barrel carburetor.

Common Carburetor Problems

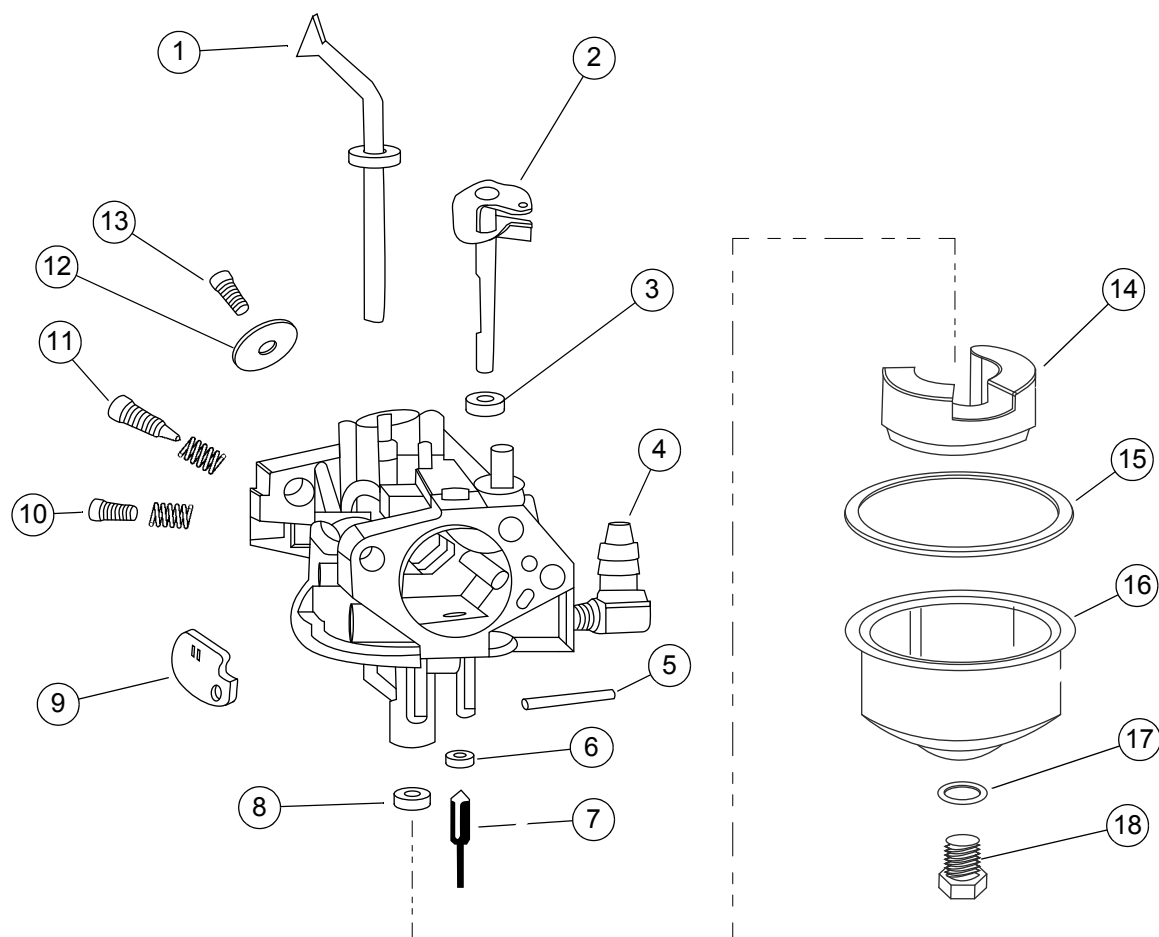
Most carburetor problems are related to the use of stale, gummy fuel and dirt ingestion. Check for evidence of these conditions before servicing any carburetor.

Gasoline left in the fuel system for long periods can form gum or varnish deposits adversely affecting carburetor operation.

NOTE: A reliable fuel stabilizer (available from an IASD) will minimize the formation of gum deposits during storage. Add stabilizer to the gasoline in the fuel tank or storage container. Follow the mix ratio recommended on the stabilizer container. After adding stabilizer to the engine fuel tank, run the engine for about 10 minutes to draw stabilized fuel through the carburetor.

Service Tip

Carefully inspect carburetor choke and throttle shafts for wear. Grasp each shaft and attempt to move it horizontally and vertically. Maximum allowable end play for choke and throttle shafts is 0.004 in (0.1 mm). Replace if excessive play is evident.



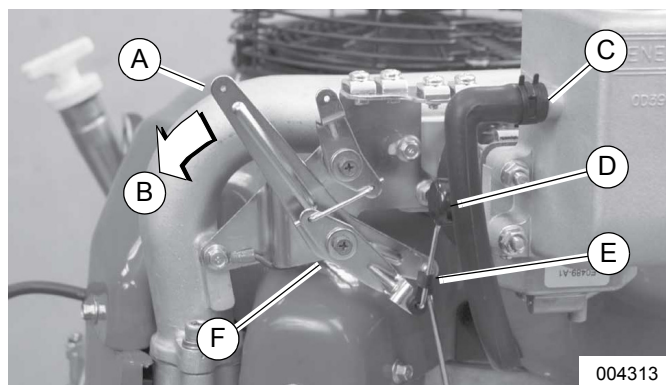
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Figure 5-1. Exploded View of Typical Carburetor (for reference only)

Item	Quantity	Description	Item	Quantity	Description
1	1	Choke lever and shaft	10	1	Low idle speed adjusting screw and spring
2	1	Throttle lever and shaft	11	1	Low idle fuel adjusting screw and spring
3	1	Throttle lever dust seal	12	1	Throttle plate
4	1	Fuel inlet fitting	13	1	Throttle plate screw
5	1	Float shaft	14	1	Float
6	1	Fuel inlet seat	15	1	Bowl gasket
7	1	Fuel inlet needle	16	1	Float bowl
8	1	Main jet	17	1	Bowl retaining screw gasket
9	1	Choke plate	18	1	Bowl retaining screw

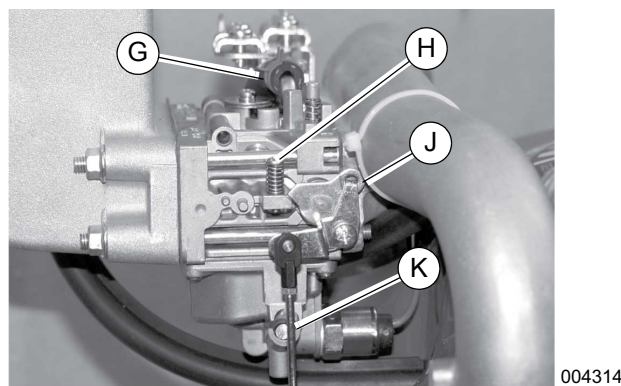
Carburetor Removal— Vertical Shaft Engines

1. See [Figure 5-2](#). Detach choke link (D) from bell crank (F) and remove link from carburetor.

**Figure 5-2. Vertical Shaft Carburetor**

A	Speed control lever	D	Choke link
B	Slow	E	Clip
C	Breather tube	F	Bell crank

2. Disconnect fuel shutoff solenoid by unplugging power wire.
3. Disconnect breather tube (C) and EVAP hose (if equipped).
4. Separate throttle link ball joint from carburetor by rotating ball socket.
5. See [Figure 5-3](#). Disconnect fuel line clamp and fuel line (G).

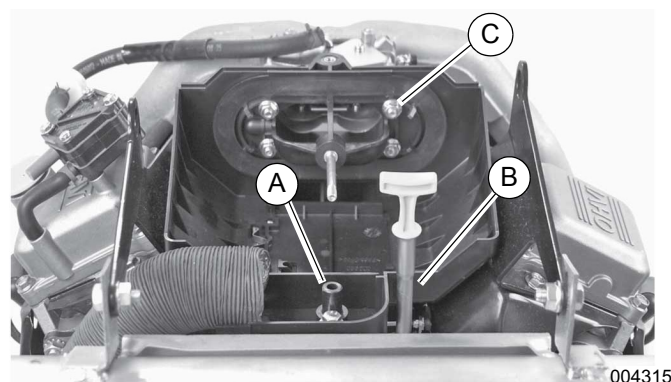
**Figure 5-3. Vertical Shaft Carburetor**

G	Fuel line	J	Throttle lever
H	Idle stop screw	K	Speed control lever

6. Remove four nuts holding carburetor and plenum to intake manifold.
7. Remove plenum, carburetor, and gaskets from manifold, and discard gaskets.

Carburetor Removal— Horizontal Shaft Engines

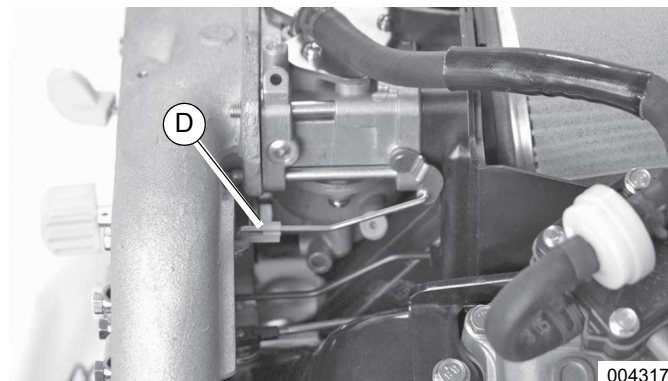
1. Remove air cleaner cover and air cleaner.
2. Unscrew and remove yellow plastic knob from Summer/Winter intake control (if equipped).
3. See [Figure 5-4](#). Remove dipstick tube retainer bolt (A) and tube assembly (B).

**Figure 5-4. Horizontal Shaft Carburetor**

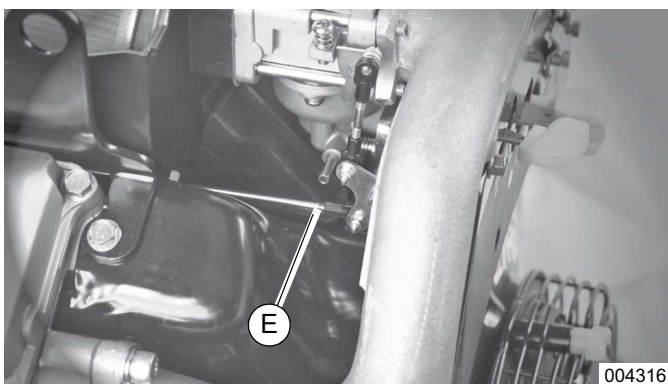
A	Dipstick tube retainer bolt	C	Air cleaner base nuts (4)
B	Dipstick tube and hold-down assembly	—	—

4. Remove four base nuts (C) and one bolt retaining air cleaner base, breather tube, and EVAP hose if equipped; remove base.
5. Remove fuel line clamp and fuel line from top of carburetor.

6. See **Figure 5-5**. Remove choke link (D), throttle link (E), and fuel shut-off solenoid wire from carburetor.



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Figure 5-5. Horizontal Shaft Carburetor

D	Choke link	E	Throttle link
---	------------	---	---------------

7. Slide carburetor off mounting studs.

Cleaning and Inspecting the Carburetor

See **Figure 5-6**.

1. Separate out all nonmetallic parts. Then, clean metallic parts with solvent or commercial carburetor cleaner. **DO NOT PERMIT NONMETALLIC PARTS TO CONTACT SOLVENT OR CLEANER.**
2. Inspect throttle shaft (2). Replace if worn or damaged.
3. Inspect choke shaft (if equipped). Replace if worn or damaged.
4. Inspect float bowl (16). Float bowl must be free of dirt and corrosion.
5. Inspect float (14). Replace if damaged.
6. After all metallic parts have been cleaned, blow out carburetor passages with compressed air.



WARNING

Vision Loss. Eye goggles are required to be worn when using this machine. Failure to wear eye goggles could result in permanent vision loss.

(000101)

When working with compressed air and solvents or cleaners, wear appropriate eye protection to avoid personal injury.

NOTE: Generac recognizes that there are spray type carburetor cleaners on the market, but does not recommend their use. Also, Generac does not recommend soaking a carburetor in cleaning solutions because they may damage the carburetor.

See **Figure 5-6**. Carburetor float bowl (16) may be removed for cleaning. Leave all jets and inlets (7, 8) in place while cleaning carburetor. Use a carburetor cleaning solution which will not damage rubber, neoprene, or plastic parts.

Note: Left and right main jets are different sizes.

Checking Fuel Shutoff Solenoid

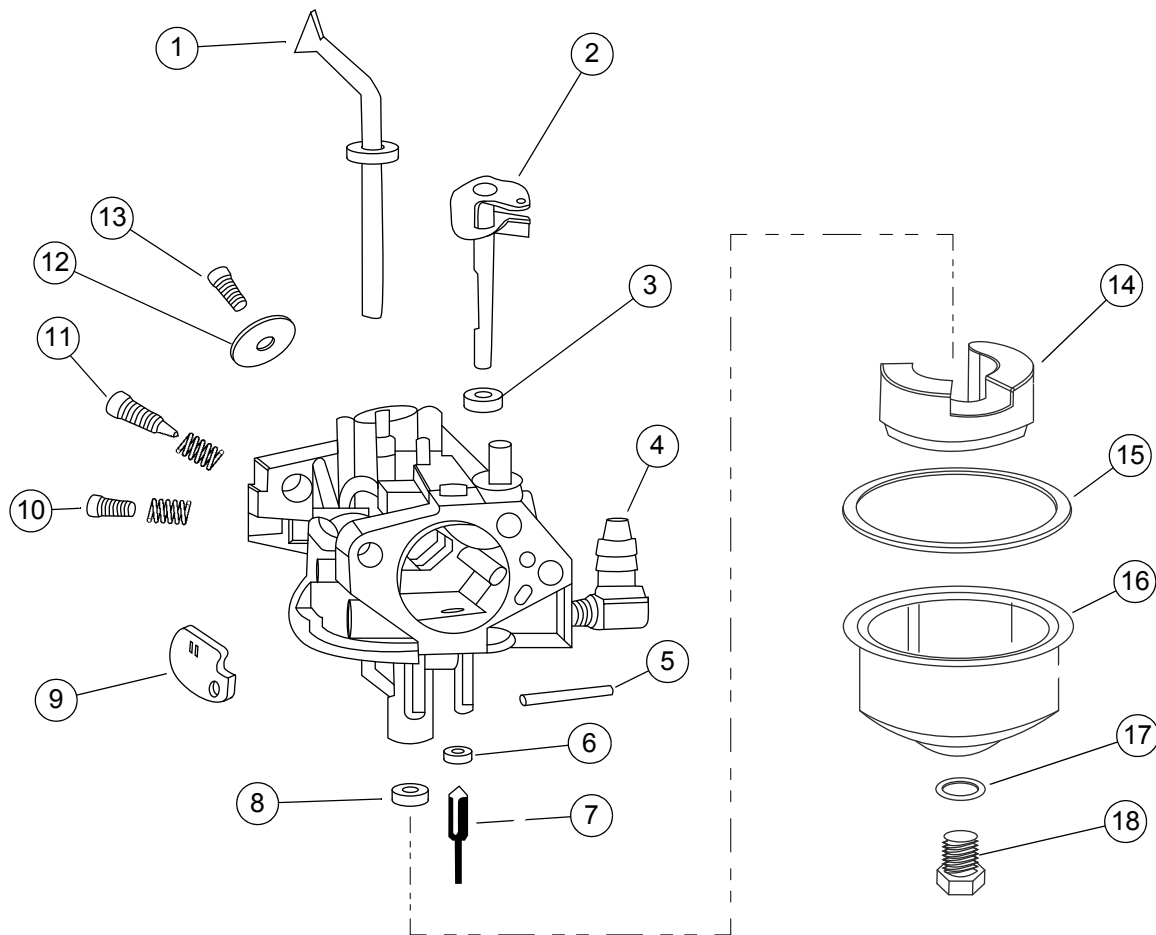
The fuel shutoff solenoid is an electrical device which opens and closes the valve supplying fuel to the carburetor.

Check solenoid operation by applying 12 volts to the terminals. A "click" indicates solenoid is operating normally. Remove solenoid from float bowl and install a replacement if it does not appear to be working properly. Replace entire carburetor if any other defects are found.

NOTE: Do not lose spring inside fuel shutoff solenoid when removing plunger.

See **Figure 5-6**. Proceed as follows to disassemble carburetor:

1. Remove float bowl retaining screw (18) and retaining screw gasket (17). Remove float bowl (16).
2. Remove float shaft (5). Remove float (14) and fuel inlet fitting (4).
3. Remove idle speed adjusting screw (10) with spring.
4. Rotate throttle plate (12) to closed position and remove throttle plate screw (13).
5. If equipped, rotate choke plate (9) to closed position and remove choke retainer spring.
6. If equipped, remove choke lever and shaft (1) along with choke shaft seal.
7. Remove throttle lever/shaft (2) and throttle lever seal (3).



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Figure 5-6. Exploded View of Typical Carburetor (for reference only)

Item	Quantity	Description	Item	Quantity	Description
1	1	Choke lever and shaft	10	1	Idle speed adjusting screw and spring
2	1	Throttle lever and shaft	11	1	Idle fuel adjusting screw and spring
3	1	Throttle lever seal	12	1	Throttle plate
4	1	Fuel inlet fitting	13	1	Throttle plate screw
5	1	Float shaft	14	1	Float
6	1	Fuel inlet seat	15	1	Bowl gasket
7	1	Fuel inlet needle	16	1	Float bowl
8	1	Main jet	17	1	Bowl retaining screw gasket
9	1	Choke plate	18	1	Bowl retaining screw

Carburetor Assembly

See [Figure 5-6](#). Assemble carburetor in reverse order of disassembly. Note the following:

- 1. Blow metallic parts dry with compressed air.
- 2. A notch on the fuel inlet valve mates with the float.
- 3. Float setting is fixed and nonadjustable.
- 4. Verify float bowl is properly aligned with bowl gasket and carburetor body before tightening float bowl nut.

Carburetor Installation—Vertical Shaft

- 1. See [Figure 5-7](#). Slide new carburetor gasket (D) and carburetor onto studs.

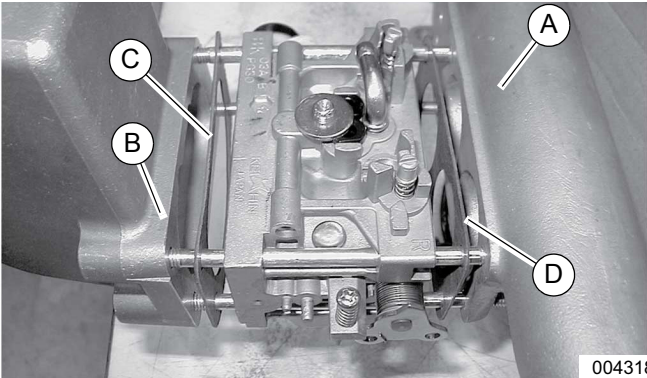


Figure 5-7. Carburetor Installation—Vertical Shaft

A	Manifold	C	Plenum gasket
B	Plenum	D	Carburetor gasket

- 2. Slide new plenum gasket (C) and plenum (B) onto studs.
- 3. Install four nuts and tighten to 48 in-lb (5.4 Nm) torque.
- 4. Connect fuel line and clamp.
- 5. Install throttle link on carburetor ball joint.
- 6. If equipped, connect breather tube and EVAP hose.
- 7. Connect fuel shutoff solenoid.
- 8. Install choke link on carburetor and clip link to bell crank.
- 9. Perform a static governor adjustment. (See [Static Governor Adjustment](#) in Section 6.)
- 10. Start engine and allow to warm up for five minutes.

- 11. See [Figure 5-8](#). Set throttle lever (G) to slow speed position and hold throttle lever against idle stop screw (F).

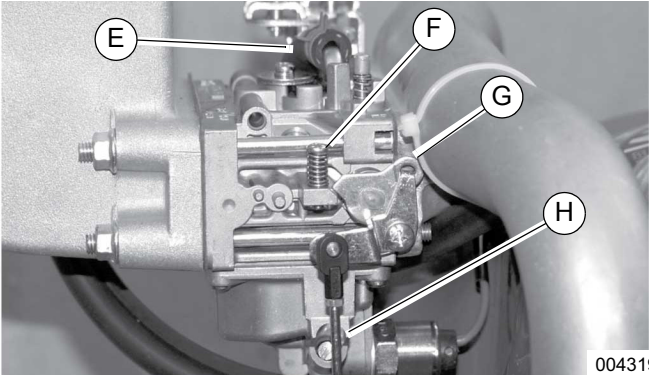


Figure 5-8. Adjust Idle Stop Screw

E	Fuel line	G	Throttle lever
F	Idle stop screw	H	Throttle link

- 12. Adjust idle top screw to maintain ~1800 rpm idle.
- 13. Slowly move speed control lever to high speed position.

NOTE: See [Figure 5-9](#). High speed screw (K) may need adjusting to reach speed control stop without exceeding 3800 RPM.

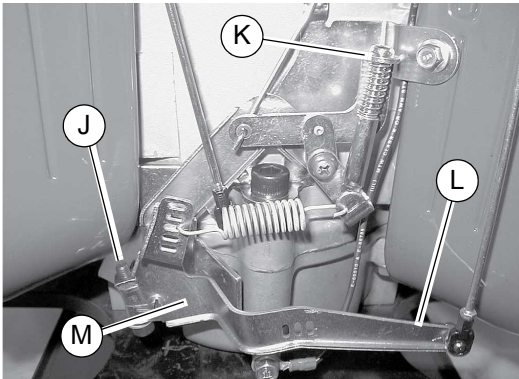


Figure 5-9. Speed Adjustment

J	Clinching screw	L	Governor lever
K	High speed screw	M	Governor arm

- 14. With speed control in high speed position, adjust high speed screw to obtain desired engine speed. Do not exceed 3800 rpm.

Carburetor Installation—Horizontal Shaft

1. Slide new carburetor gasket and carburetor onto studs.
2. Slide new plenum gasket and plenum onto studs.
3. Install air cleaner base, breather tube and EVAP hose (if equipped), bolt and nuts and tighten to 48 **in-lb** (5.4 Nm) torque.
4. Attach choke, throttle links, and solenoid wire to carburetor.
5. Install fuel line and fuel line clamp.
6. Install oil fill tube and hold-down bolt. Tighten to 40 **in-lb** (4.7 Nm) torque.
7. Install air cleaner and air cleaner cover.
8. Install yellow knob for Summer/Winter control.
9. Perform a static governor adjustment. (See [Static Governor Adjustment](#) in Section 6.)
10. Start engine and allow to warm up for five minutes.
11. See [Figure 5-10](#). Set throttle lever (G) to slow speed position and hold throttle lever against idle stop screw (F).

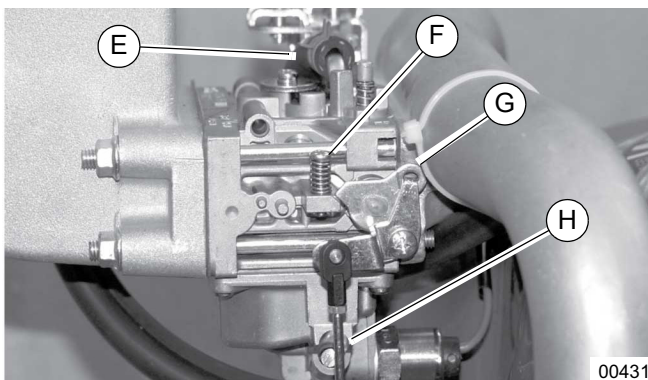


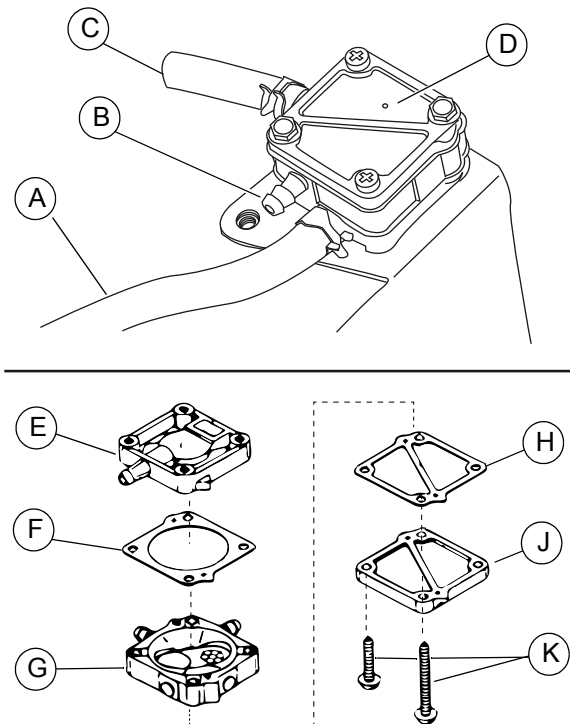
Figure 5-10. Adjust Idle Stop Screw

E	Fuel line	G	Throttle lever
F	Idle stop screw	H	Throttle link

12. Adjust idle stop screw to maintain ~1800 rpm idle.
13. Slowly move speed control lever to high speed position.
14. With speed control in high speed position, adjust high speed screw to obtain desired engine speed. Do not exceed 3800 rpm.

Fuel Pump

See [Figure 5-11](#). The fuel pump supplied with the engine is a pulse type pump. It uses crankcase vacuum pulses drawn from the valve cover to pump the fuel. It is capable of priming at 12 in (30.5 cm) maximum lift, with a maximum outlet pressure of 1.5 psi (10.3 kPa).



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Figure 5-11. Fuel Pump

A	Pulse line
B	Fuel from tank
C	To carburetor inlet
D	Vent hole
E	Lower spring assembly with spring return
F	Lower diaphragm
G	Check valve assembly (check valves 1 and 2)
H	Upper diaphragm
J	Upper vent assembly
K	Mounting hardware

Check vent hole (D) on top of the pump for obstructions if fuel pump is not working properly.

If a fuel pump other than the factory recommended pump is used, fuel line pressure at carburetor inlet (C) must not exceed 3 psi (20.7 kPa). Pressures above 3 psi (20.7 kPa) may produce an overly rich carburetor mixture leading to engine damage.

As crankcase vacuum increases, lower diaphragm (F) pulls down against spring return (E) and allows fuel to flow through check valve 2 (G). As crankcase pressure increases, the spring return pushes up against lower diaphragm, allowing fuel to flow through check valve 2 (G) and out pulse line (A).

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Section 6: Mechanical Governor

Disassemble

1. Drain oil from engine.
2. Remove any rust, nicks, or burrs from crankshaft.
3. Remove four oil cooler screws.
4. Disconnect wiring from oil pressure switch.
5. Remove governor lever from shaft.
6. Separate ball joint on winging arm.
7. Remove all crankcase bolts and slide crankcase cover off.
8. Discard crankcase gasket and oil passage o-ring.

Both the spool and the flyweights must move freely for the governor to work properly. Check for wear on spool and flyweights. If wear is noticed, replace governor gear assembly (gear and flyweights), spool, and governor arm. Lubricate all moving parts when assembling.

Governor Removal And Installation

1. See [Figure 6-1](#). Leverage gear assembly (A) and governor spool (B) off of governor shaft. Note the location of the oil passage O-ring (C).

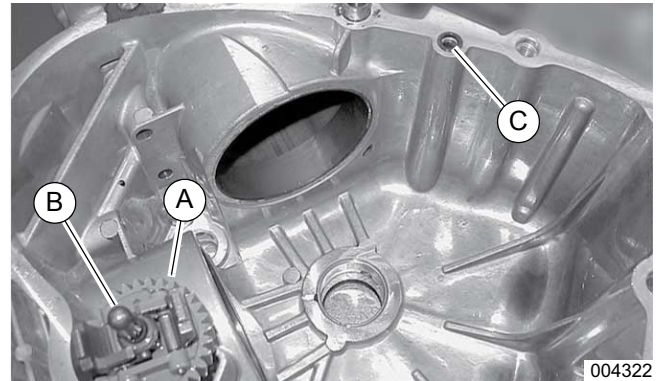


Figure 6-1. Governor Gear Assembly and Spool

2. Remove any remaining plastic from notch in governor shaft.
3. See [Figure 6-2](#). Verify all governor bearing parts (bearing E and both roller bearing plates D) are on the shaft, and it moves smoothly.
4. Slide new gear assembly and spool onto governor shaft. Slide until gear engages notch in governor shaft.

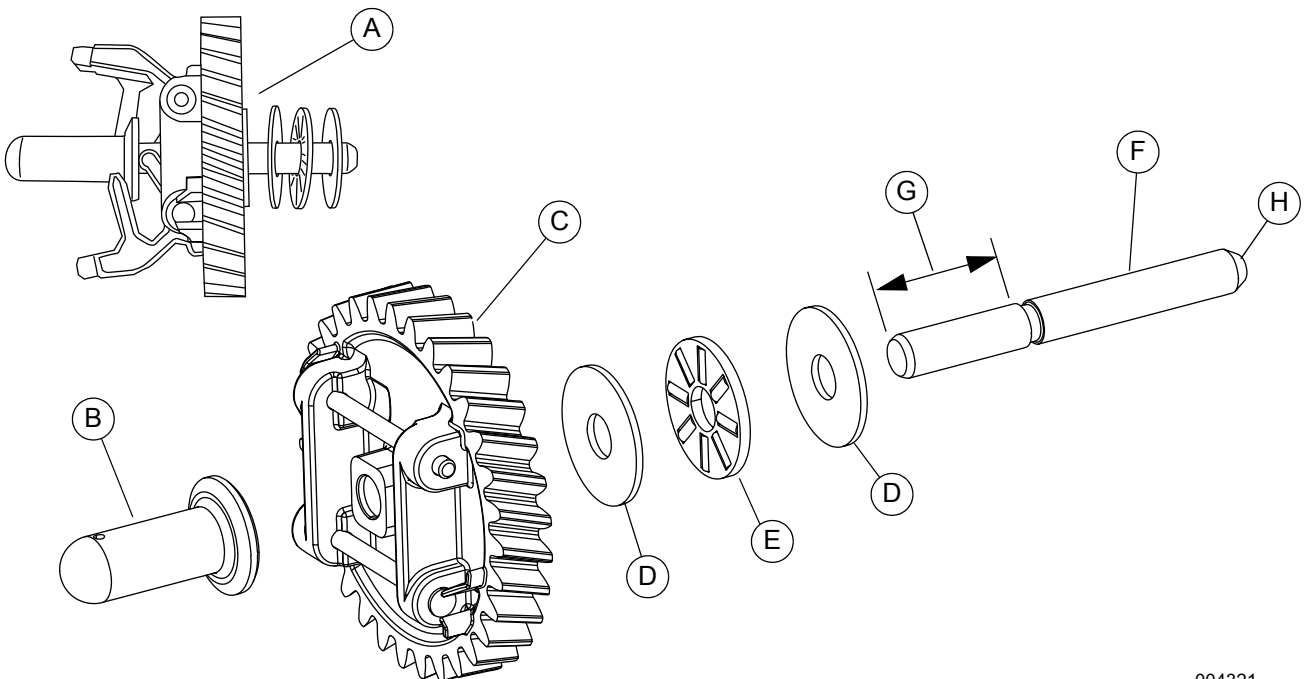


Figure 6-2. Governor Assembly and Components

A	Governor assembly (Spool must engage under flyweights)	D	Roller bearing plate	G	Exposed shaft distance: 1.456 in. (37 mm)
B	Spool	E	Roller bearing (cupped side faces away from gear)	H	Press this end into block
C	Governor gear and flyweight assembly	F	Governor shaft	—	—

Governor Arm

See [Figure 6-3](#). If governor arm (J) does not move freely, or if arm feels loose in bushings, it may need replacing. Change governor arm, governor gear assembly (gear and flyweights), spool, and bushings as needed if wear is noticed. Lubricate all moving parts when assembling.

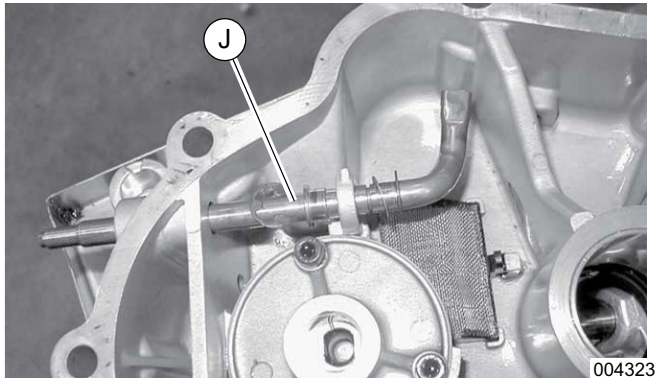


Figure 6-3. Governor Arm Assembly

Disassemble Governor Arm

Proceed as follows to disassemble the governor arm:

1. Remove e-clips.
2. Slide arm down and out of bushings.
3. Replace worn parts.

NOTE: Lower bushing is a slip fit, and upper bushing is pressed in.

Assemble Governor Arm

Proceed as follows to assemble governor arm:

1. Slide thrust washer halfway onto new governor arm.
2. Insert governor arm in lower bushing holder, and slide halfway in.
3. Install lower e-clip on arm, and slide thrust washer down to meet e-clip.
4. Slide lower bushing halfway onto governor arm.
5. Slide governor arm into thrust washer until tight.
6. Slide lower bushing down and into holder, then install upper e-clip.

Assemble Crankcase Cover

Proceed as follows to assemble crankcase cover:

1. Clean old gasket material from crankcase and cover mating surfaces.
2. See [Figure 6-1](#). Verify new oil passage o-ring (C) is in place.
3. Install new gasket the crankcase.

NOTE: On vertical shaft engines, align camshaft drive and oil pump gerotor.

4. Slide crankcase cover onto crankcase.

NOTE: Hold governor arm in counter-clockwise position while installing.

5. See [Figure 6-4](#). Hand-tighten all crankcase bolts, and then tighten to 35 ft-lb (47.5 Nm), following the proper torque sequence.

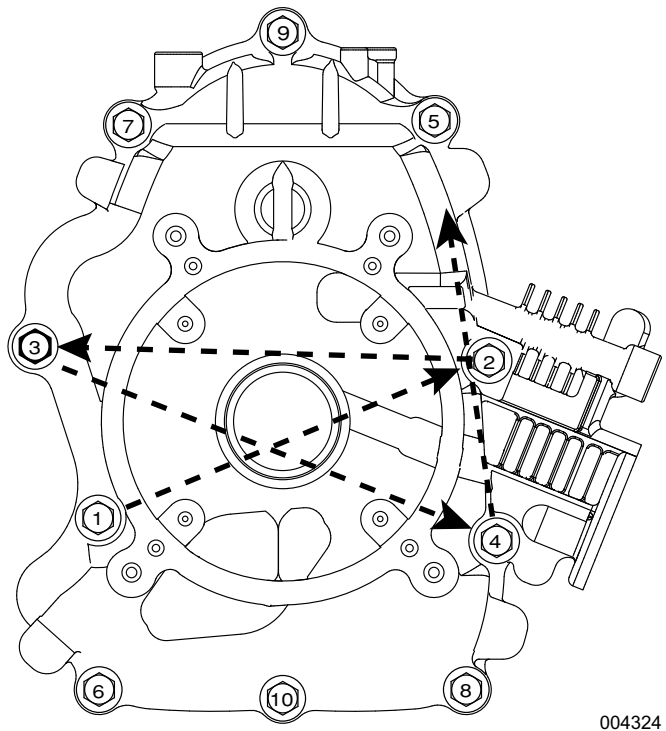


Figure 6-4. Crankcase Bolt Torque Sequence

6. Connect ball joint on the swing arm.
7. Place governor lever on governor arm. Place governor spring in the fourth hole out.
8. Perform a static governor adjustment as described in [Static Governor Adjustment](#).
9. Connect wires to oil pressure switch.
10. Attach oil cooler to blower housing.

Static Governor Adjustment

Determine which version of governor linkage is installed on engine. Refer to [Figure 6-5](#) (earlier version) or [Figure 6-6](#) (later version).

1. Loosen clinching screw on governor lever.
2. Rotate governor arm clockwise and hold governor lever in WOT (Wide Open Throttle) position.
3. While holding this position, tighten clinching screw to 100 in-lb (11.3 Nm) torque.
4. Verify throttle travels from WOT to IDLE. If it doesn't, the governor must be readjusted.

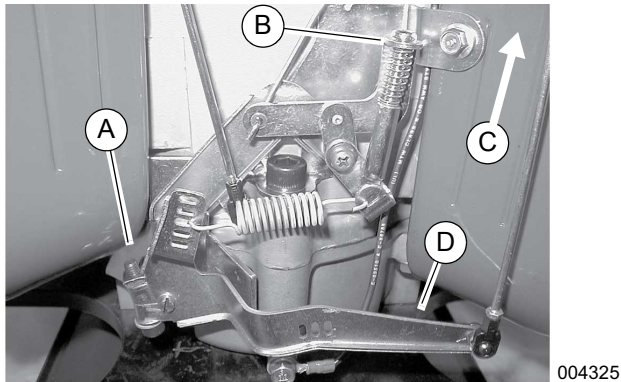


Figure 6-5. Static Governor Adjustment

A	Clinching screw	C	Idle
B	High speed screw	D	Governor lever

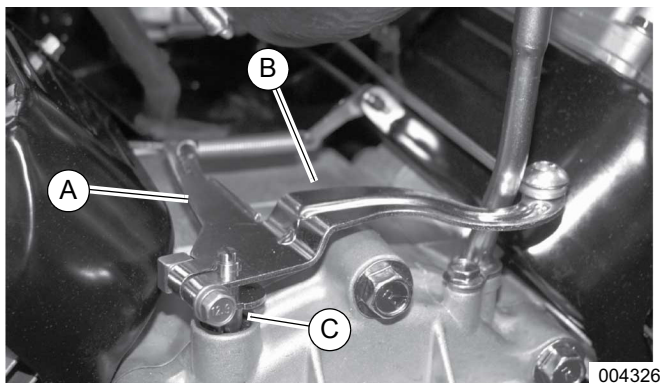


Figure 6-6. Static Governor Adjustment (Later Version)

A	Governor arm	C	Clinching screw
B	Governor lever	—	—

Dynamic Governor Adjustment

1. Start engine and allow to warm up for five minutes.
2. See [Figure 6-7](#). Set speed control lever to SLOW speed position and hold throttle lever against idle stop screw (A).

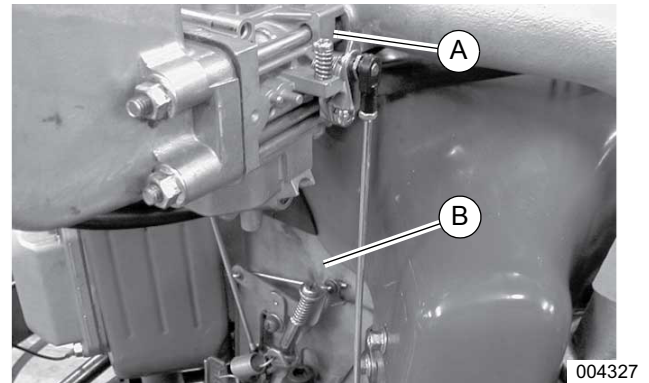


Figure 6-7. Dynamic Governor Adjustment (Earlier Version)

A	Idle stop screw	B	High speed screw
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3. Adjust idle stop screw to maintain 1800 rpm idle.
4. Slowly move speed control lever to high-speed position. Do not exceed 3800 rpm (see [Figure 6-7](#) or [Figure 6-10](#) depending on governor version).

NOTE: On the later governor version, insert a key key or similar object into the hole in the control panel behind the speed control lever. This acts as a temporary stop during adjustment.

NOTE: High-speed screw (B) may need adjusting to reach speed control stop without exceeding 3800 rpm.

5. With speed control in the high-speed position, adjust high-speed screw to obtain desired engine speed (see [Figure 6-7](#) or [Figure 6-10](#) depending on governor version).

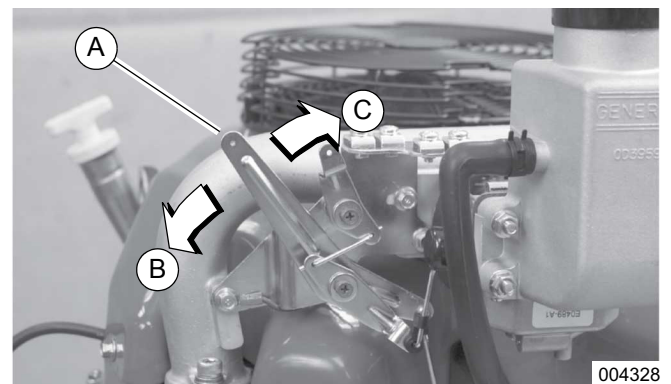
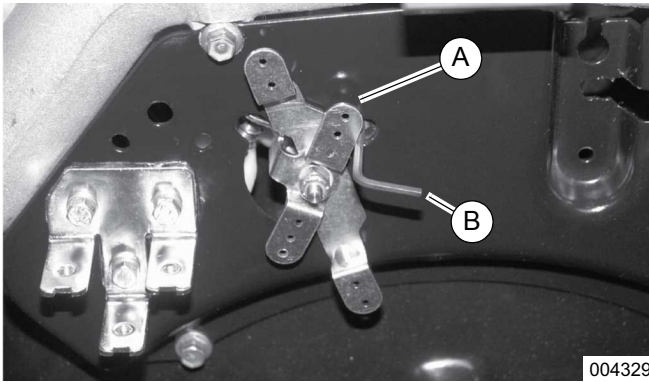


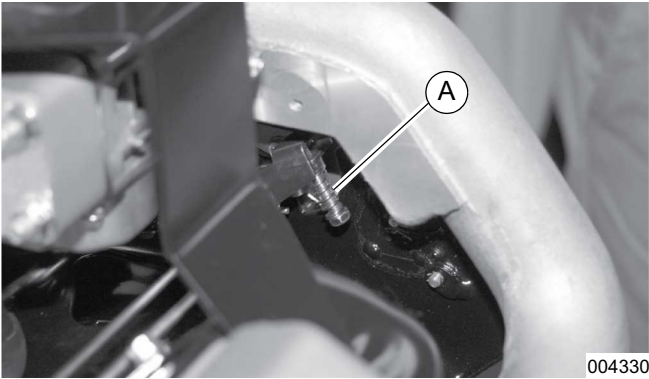
Figure 6-8. Dynamic Governor Adjustment (Vertical Shaft)

A	Speed control lever	C	Fast
B	Slow	—	—



**Figure 6-9. Dynamic Governor Adjustment
(Horizontal Shaft)**

A	Speed control lever	B	Hex key
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**Figure 6-10. Dynamic Governor Adjustment
(Horizontal Shaft)**

A	High speed screw	—	—
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Section 7: Cylinder Head and Valves

General Information

Compression testing information and procedures are described in Section 4 under [Troubleshooting](#).

See [Figure 7-1](#). Cylinders are numbered as shown.

NOTE: Cylinder #1 is closest to the flywheel.

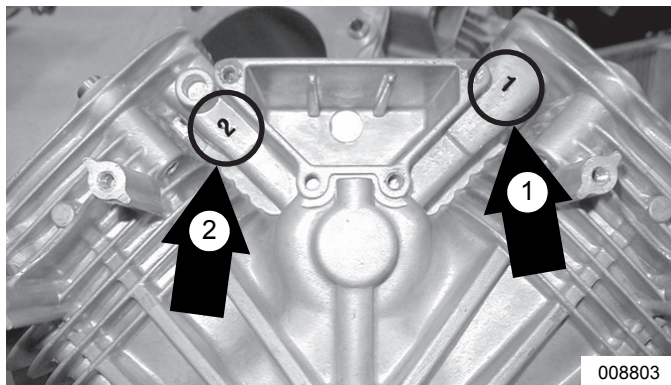


Figure 7-1. Cylinder Numbers

Remove Cylinder Heads

Proceed as follows to remove the cylinder heads:

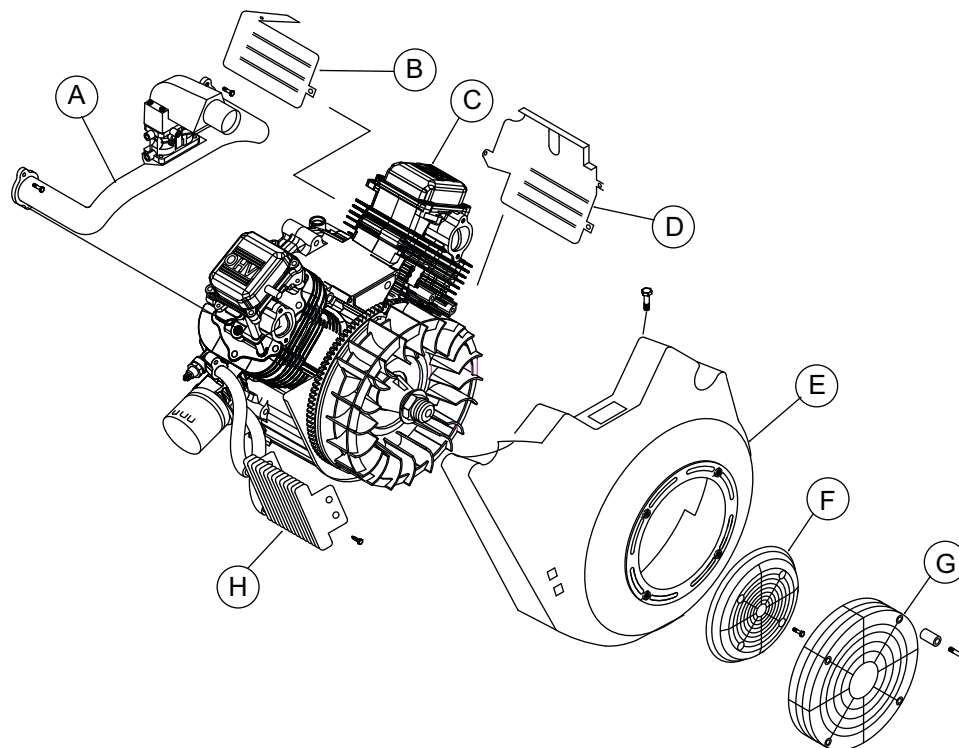
1. Remove exhaust system from engine. Disconnect choke and throttle control cables.
2. Remove spark plugs.
3. See [Figure 7-2](#). Remove parts as depicted.
4. Discard gaskets and valve cover seals.

Remove Rocker Arms

Proceed as follows to remove the rocker arms:

1. See [Figure 7-3](#). Unlock jam nuts and remove two ball studs and rocker arm assemblies.
2. Remove push rods and identify each.

NOTE: Push rods develop a wear pattern. Moving them to different locations can increase this wear, leading to more frequent adjustments and/or loss of performance.



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Figure 7-2. Exploded View

A	Intake manifold assembly	D	Lower wrapper	G	Finger guard
B	Upper wrapper	E	Blower housing	H	Oil cooler
C	Valve cover	F	Rotating screen	—	—

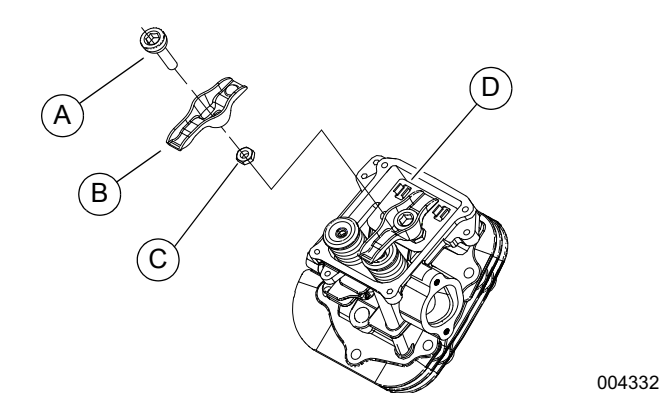


Figure 7-3. Remove Rocker Arm Assemblies

A	Ball stud	C	Jam nut
B	Rocker arm	D	Push rod guide plate (Tabs must face upward)

3. Remove push rod guide plate.
4. See [Figure 7-4](#). Remove six head bolts and cylinder head. Discard gasket.

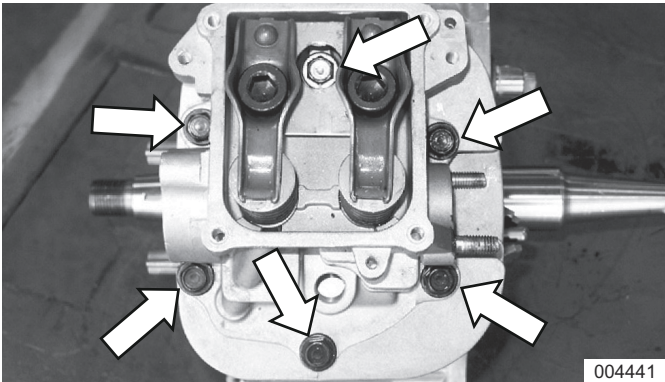


Figure 7-4. Remove Cylinder Head

5. Repeat steps 1-4 for other cylinder head.

Disassemble Cylinder Head

Proceed as follows to disassemble the cylinder head:

1. Place a shop rag or short section of rubber fuel line beneath valves inside combustion chamber to hold valve in place while compressing spring.
2. See [Figure 7-5](#). Hold down valve spring retainer with a valve spring compressor (A). Remove the following:
 - a. Valve spring keepers (B)
 - b. Valve spring retainer (C)
 - c. Valve spring (D)
 - d. Intake and exhaust valves

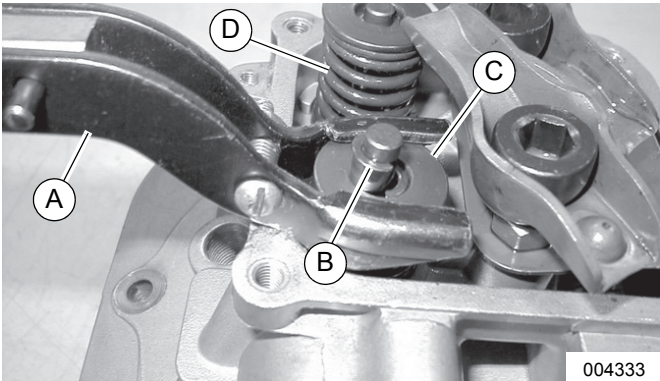


Figure 7-5. Remove Retainers

A	Valve spring compressor	C	Valve spring retainer
B	Valve spring keeper	D	Valve spring

3. See [Figure 7-6](#). Remove and discard two valve stem seals (E).

NOTE: Valve stem seals are used only on intake side.

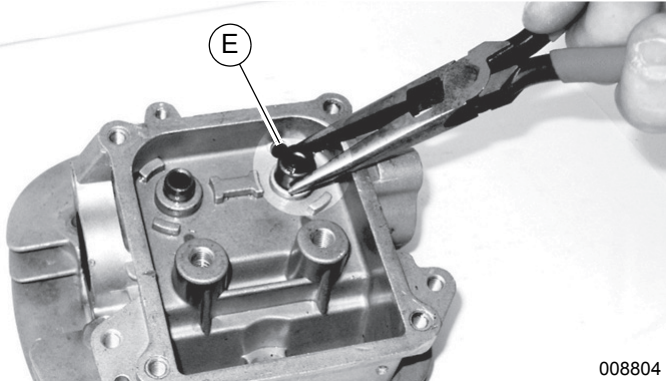


Figure 7-6. Remove Valve Stem Seals

Inspect And Repair

1. See [Figure 7-7](#). Check condition of cylinder head. Verify all gasket material is removed from surfaces before checking.
 - a. Inspect cylinder head for cracks or damage.
 - b. Use a surface plate or straightedge and check cylinder head mounting surface for distortion.

If mounting surfaces are distorted more than 0.004 in (0.1 mm), the cylinder head must be replaced.

NOTE: Resurfacing cylinder head mounting surfaces is not recommended.

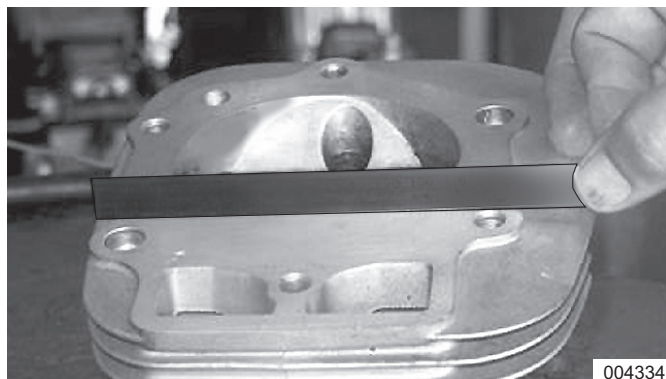


Figure 7-7. Check Cylinder Head Distortion

2. See [Figure 7-8](#). Clean valve guides and measure inner diameter using a split ball bore gauge. Replace head if either valve guide measures 0.278 in (7.06 mm) or more.

NOTE: Valve guides are not replaceable.

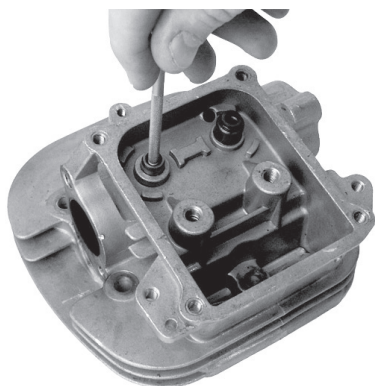


Figure 7-8. Measure Valve Guide Inner Diameter

Reface Valves and Seats

See [Figure 7-9](#). Valve seats may be reconditioned with a valve seat cutter. If valve seat is wider than dimension A (1/16–3/64 in. [0.8–1.2 mm]), use a narrowing cutter to verify contact area of valve seat is centered on face of valve.

1. Use a 60° cutter to narrow seat from bottom and a 15° cutter to narrow seat from top.

NOTE: If valve seat is loose or cracked, replace cylinder head.

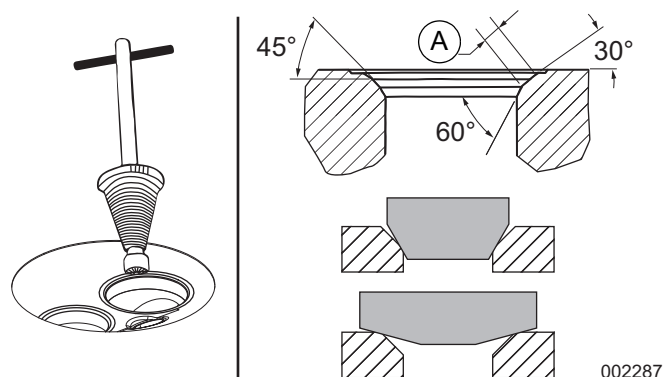


Figure 7-9. Reface Valve Seats

2. See [Figure 7-10](#). Resurface valve faces to 45°.

NOTE: In most instances, it is more economical to replace valves rather than reface them.

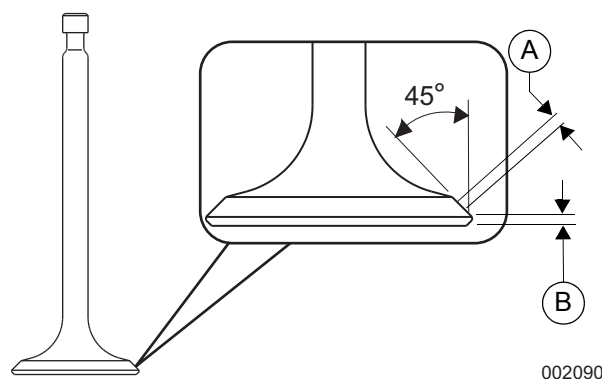
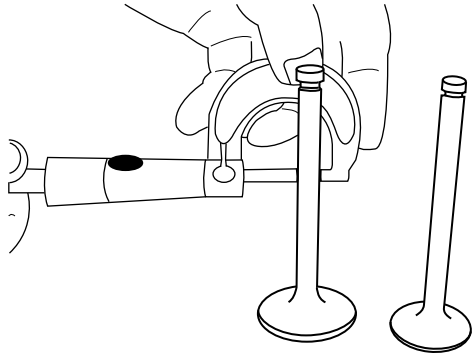


Figure 7-10. Valve Dimensions

A	3/64–1/16 in. (1.2–1.6 mm) Seating area centered on valve
B	Minimum 1/32 in. (0.79375 mm)

3. See **Figure 7-11**. Measure valve stem diameter at 1.600 in. (40 mm) from end of valve.

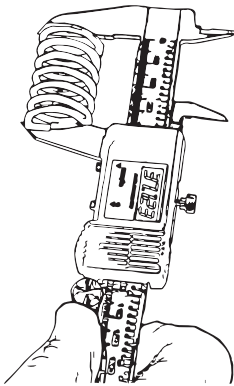


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Figure 7-11. Measure Valve Stem Diameter

Replace valves if stem diameter is less than 0.272 in (6.9 mm), or if total clearance between valve stem and valve guide exceeds 0.0047 in (0.12 mm).

4. See **Figure 7-12**. Check valve springs for free length. Replace valve springs if free length is less than 1.437 in (36.5 mm).



004442

Figure 7-12. Check Valve Springs

Gasket Removal

Verify sealing surface is completely clean and free of old gasket material before installing a new gasket. Residual gasket material or dirt can prevent the new gasket from sealing completely. A poorly sealed gasket may cause the engine to leak oil, lose compression, or overheat.

It is critical to completely remove old gasket material without damaging sealing surfaces. Aluminum is very soft and easy to scratch and gouge. Even a new gasket may not seal completely if the underlying surface is damaged. Therefore, Generac does not recommend using a wire brush or abrasive sanding equipment to remove gaskets. Instead, use a metal gasket scraper specifically designed for this purpose. Scrape gasket at a shallow angle—almost parallel to the sealing surface. Wipe sealing surface with a clean, dry, lint-free rag after removing gasket residue.

If the gasket is difficult to scrape off, spray it with an aerosol gasket removal compound. Allow the compound to soak into the gasket for the length of time recommended by the manufacturer (typically 10–20 minutes). Then use the gasket scraper again.

NOTE: Some gasket removal compounds contain chemicals damaging to plastics. Read and follow the directions on the product label before use.

Assemble Cylinder Head

1. Install new valve stem seals.
 - a. See [Figure 7-13](#). Press seal onto intake valve guide bushing until fully seated.
 - b. Lubricate inner top lip of seal with oil.

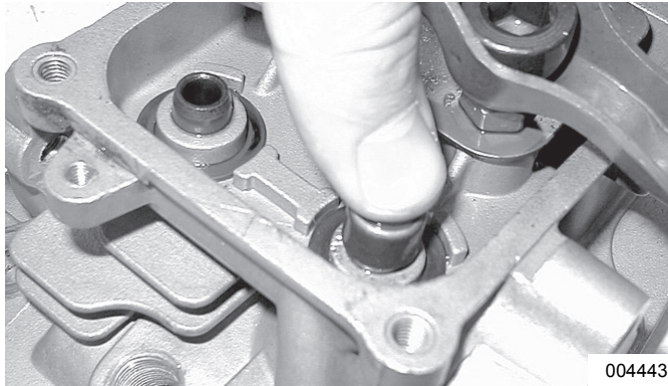


Figure 7-13. Install Valve Stem Seals

2. See [Figure 7-14](#). Install intake (A) and exhaust (B) valves.

NOTE: Lightly coat valve stems with oil before installing valves. Verify lubricant is not on valve face and seat.

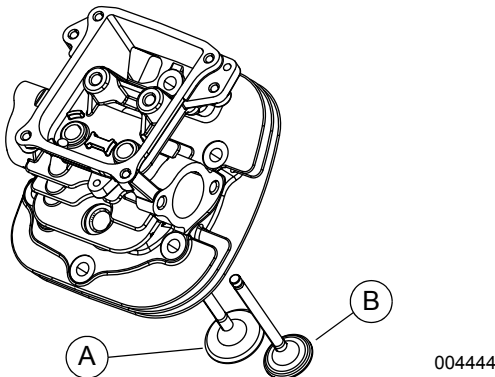


Figure 7-14. Install Valves

3. Place a shop rag or short section of rubber fuel line under valves inside combustion chamber to hold valve in place while compressing spring.
4. Install valve springs and valve spring retainers over valve stems.

5. See [Figure 7-15](#). Compress valve spring and install keepers.

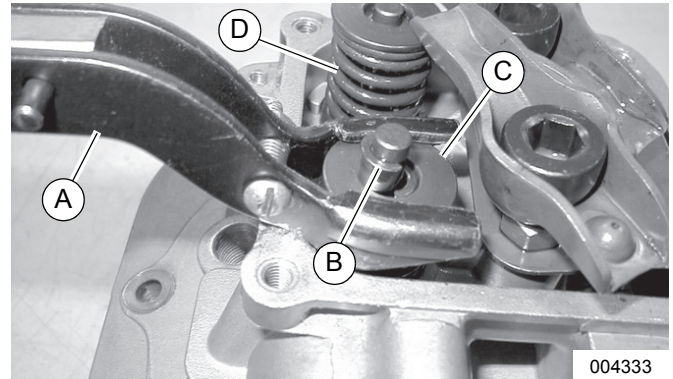


Figure 7-15. Compress Valve Spring; Install Keepers

A	Valve spring compressor	C	Valve spring retainer
B	Valve spring keeper	D	Valve spring

6. Repeat procedure for other valve(s).
7. Set push rod guide plate in place with tabs facing upward, and loosely install rocker arm assemblies (ball stud, rocker arm, and jam nut).
8. Repeat step 7 for other head.

Install Cylinder Head

1. Install cylinder head with new gasket.
2. See [Figure 7-16](#). Tighten head bolts in A-B-C-D-E-F sequence to 22 ft-lb (29.9 Nm) torque.

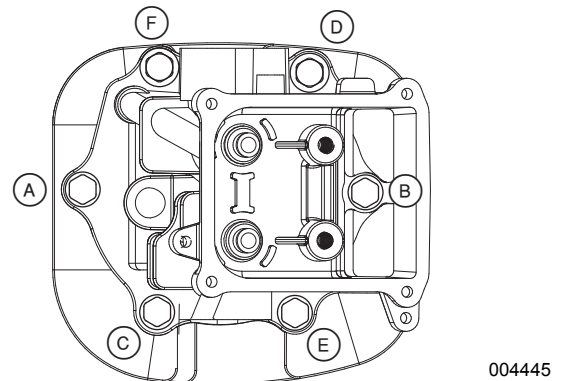


Figure 7-16. Torque Sequence for Cylinder Heads

3. Insert push rods into recess in tappets.

Adjust Valve Clearance

1. Set #1 cylinder at TDC compression stroke.
 - a. See [Figure 7-17](#). Adjust rocker arms and check clearance.

Valve Clearance (cold) Intake and Exhaust	0.003 in (0.076 mm)
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- b. Tighten jam nut to 174 **in-lb** (19.6 Nm) torque.

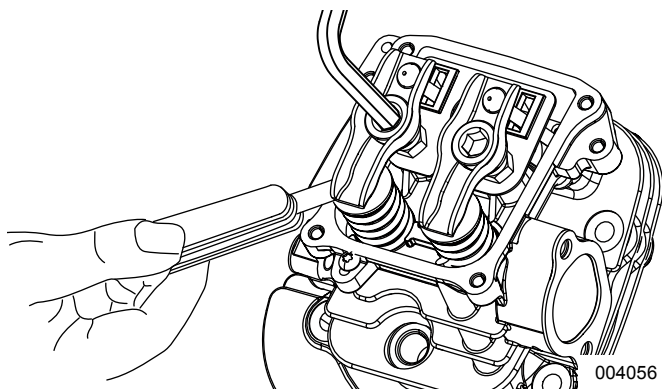


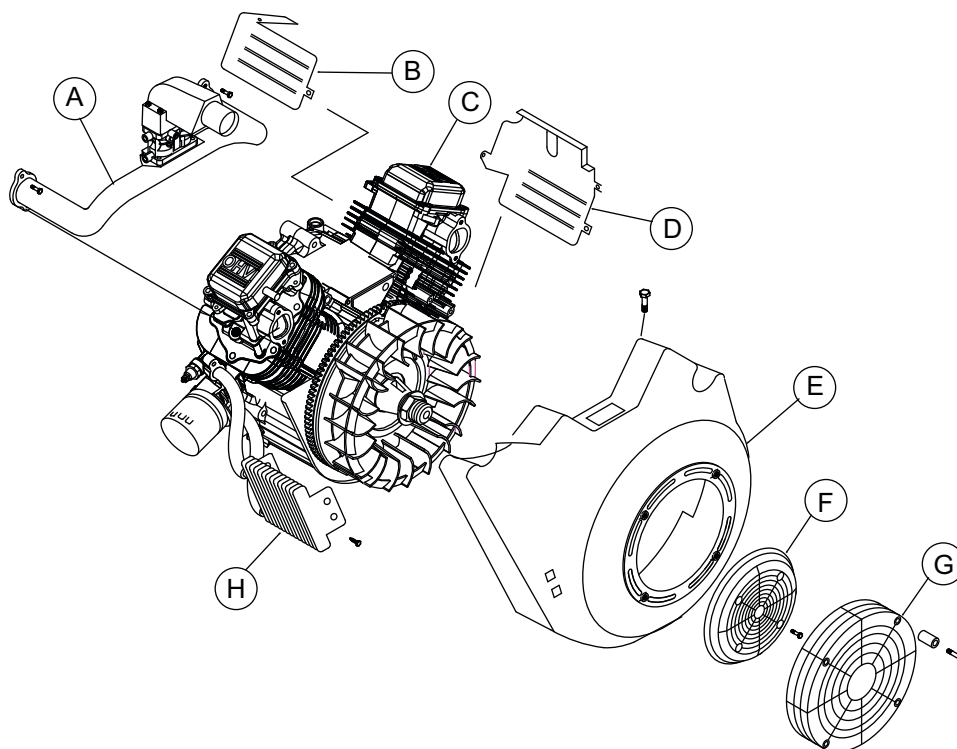
Figure 7-17. Adjust Valve Clearance

2. Set #2 cylinder at TDC compression stroke, and repeat adjustment steps 1a and 1b.
3. Install valve covers with new gaskets. Tighten screws to 60 **in-lb** (6.8 Nm) torque.

Assembly

See [Figure 7-18](#). Proceed as follows to install all parts as shown.

1. Install cylinder wrappers.
 - a. Tighten M5 screws to 25 **in-lb** (2.8 Nm) torque.
 - b. Tighten M6 screws to 40 **in-lb** (4.5 Nm) torque.
2. Install spark plugs. Tighten to 168 **in-lb** (19 Nm) torque.
3. Install exhaust manifold. Tighten screws to 168 **in-lb** (19 Nm) torque.
4. Install blower housing. Tighten screws to 40 **in-lb** (4.5 Nm) torque.
5. Install intake manifold with new gaskets.
 - a. Tighten screws to 168 **in-lb** (19 Nm) torque.
 - b. Assemble governor link to carburetor.
6. Install rotating screen. Tighten screws to 17 **in-lb** (1.9 Nm) torque.
7. Install finger guard.
 - a. If engine is equipped with hex head screws, tighten screws to 40 **in-lb** (4.5 Nm) torque.
 - b. If engine is equipped with finger screws, tighten screws by hand to approximately 12 **in-lb** (1.3 Nm) torque.
8. Assemble air cleaner.
9. Make static and dynamic governor adjustments as outlined in Section 6: [Mechanical Governor](#).



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Figure 7-18. General Assembly

A	Intake manifold assembly	D	Lower wrapper	G	Finger guard
B	Upper wrapper	E	Blower housing	H	Oil cooler
C	Valve cover	F	Rotating screen	—	—

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Section 8: Electric Starter

General Information

See **Figure 8-1**. The starter motor utilizes a solenoid, similar to an automobile starter, to assist in pinion gear engagement. When the starter motor is activated, the pinion gear engages a ring gear attached to the engine flywheel and cranks the engine.

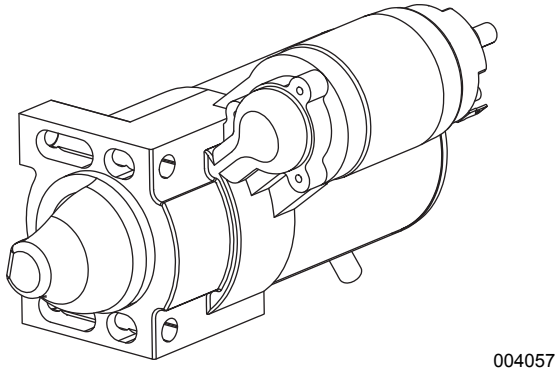


Figure 8-1. Starter Motor

Troubleshooting

NOTE: If a starting problem is encountered, check the engine thoroughly to eliminate it as the cause of starting difficulty. Check engine for free rotation by removing spark plugs and flywheel guard and turning the crankshaft with a socket and ratchet or breaker bar. Verify it rotates freely.

WARNING: Do not rotate engine with electric starter while spark plugs are removed, unless using a compression gauge. Arcing at the spark plug ends may ignite gasoline vapor exiting the spark plug hole.

Engine cranks slowly:

- Additional load affecting performance (see note above).
- Discharged battery.
- Faulty electrical connection (battery circuit).
- Discharged battery (see alternators).
- Dirty or worn starter motor commutator, bearing, weak magnets, etc.
- Worn brushes or weak brush spring.
- Wrong oil viscosity for temperature expected.
- Battery leads too long or wire too small.
- Underpowered battery.

Engine will not crank:

- Faulty safety interlocks.
- Discharged or defective battery.
- Faulty electrical connections.
- Faulty starter motor switch (open circuit).
- Open circuit in starter motor.
- Brushes sticking, etc.
- Faulty solenoid.

Starter motor spins but does not crank engine:

- Sticking pinion gear due to dirt.
- Damaged pinion or ring gear.
- Battery faulty or damaged.
- Incorrect rotation due to reversed motor polarity — all motors rotate counterclockwise viewed from pinion gear.
- Damaged solenoid.

Starter motor spins but will not stop:

- Defective starter switch.
- Solenoid stuck engaged.

Test Equipment

Digital Multimeter (DMM)

See **Figure 8-2**. A digital multimeter (DMM) may be used to read volts, ohms, amperes, and test diodes (rectifiers).

NOTE: The DMM is equipped with two fuses to prevent DMM damage if input limits are exceeded. Check fuses if the DMM displays a reading of 0.00 when testing DC output. Refer to the DMM operators' manual for more information.

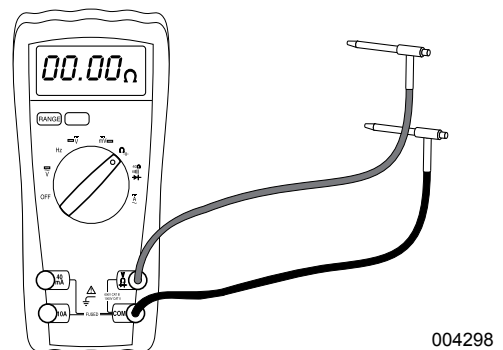


Figure 8-2. Digital Multimeter (DMM)

Test Starter Motor

Test Starter Solenoid

The solenoid is a normally open, electrically activated switch. With the key switch in the "START" position, the switch closes, allowing battery current to flow to the starter motor and crank the engine.

NOTE: Solenoid may be tested while mounted on engine.

NOTE: A jumper test lead is required for this test.

1. Remove positive battery cable from battery. Then remove battery cable from stud terminal on solenoid.
2. Disconnect wire from tab terminal on solenoid.
3. Keyswitch must be in OFF position.
4. Insert RED test lead into $V\Omega\rightarrow$ receptacle on DMM.
5. Insert BLACK test lead into **COM** receptacle on DMM.
6. Rotate DMM selector to $\rightarrow$ position.
7. See [Figure 8-3](#). Attach one DMM test lead to each stud terminal on starter solenoid.

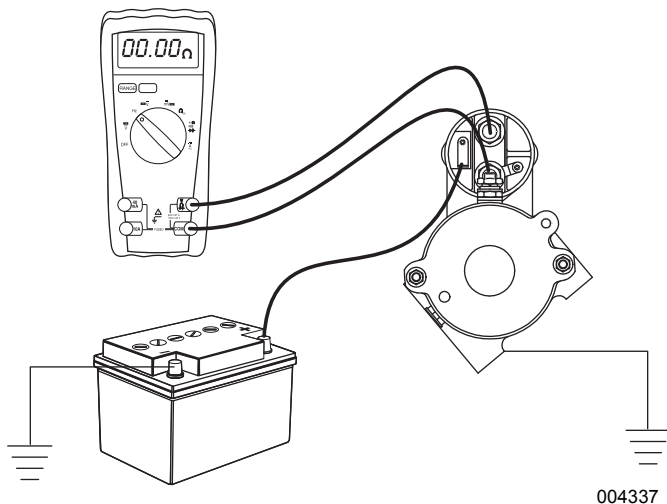


Figure 8-3. Test Starter Solenoid

8. Attach one end of jumper lead to positive terminal on battery.
9. Touch jumper wire to tab terminal on solenoid.
 - a. A "click" should be heard as the solenoid switch closes.
 - b. DMM should make a continuous tone.

Check Starter Voltage Drop

Verify battery is at least 70% charged (12.5 V or higher) before starting this test. Also, verify a correctly sized battery is being used. Check engine manufacturer's Operator's Manual for battery size recommendations.

With engine and all other accessories turned off, measure battery voltage across the battery posts using a DMM set to the 20V scale. Do not measure across battery wire terminal ends. Voltage should read 12.5 V or higher.

Do not crank engine longer than 15 seconds during any of the following tests. Proceed as follows to check starter voltage drop:

1. Prevent engine from starting by either grounding ignition at ground stud on side of engine, or removing spark plug leads. Using a DMM, place (+) positive lead on positive battery post, and (-) negative lead on negative battery post. Crank engine and observe DMM reading.
2. See [Figure 8-4](#). Place DMM (+) positive lead on starter battery terminal stud, and (-) negative lead on starter housing. Crank engine and observe DMM reading.

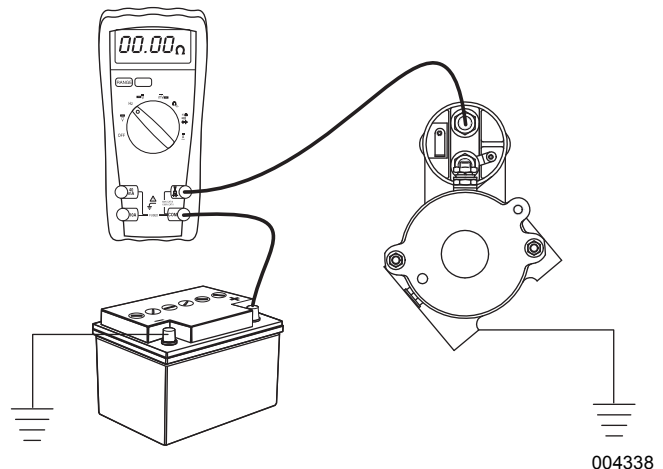


Figure 8-4. Test Starter Voltage Drop

3. Compare the two voltage readings. If readings are the same (or vary within one volt or less), there is no excessive voltage drop on the positive side. If a voltage drop of greater than one volt exists, proceed to step 4.

4. See **Figure 8-4**. Set DMM on the 2V scale. Place DMM (+) positive lead on (+) positive battery post, and DMM (-) negative lead on starter (+) positive terminal stud. With engine still disabled, crank engine and observe DMM reading. There should be less than a 0.2 V drop. If drop is greater than 0.2 V, check each wire and terminal connection. No more than 0.2 volts should be observed at each connection.
5. Check (-) negative side of system. Place DMM (+) positive lead on starter case, and DMM (-) negative lead on (-) negative battery post. Crank engine and observe the reading. Voltage should be less than 0.2 V.
6. If voltage reads greater than 0.2 volts, check all wire connections on the starter (-) negative side. There should be no readings higher than 0.2 volts.

Starter Amp Draw Test

This test requires a special clamp-on tool called an ammeter. Most repair shops should be able to perform this test.

1. Place ammeter clamp on (+) positive cable running from the battery (+) post to the starter post.
2. Ground ignition and crank engine (no longer than 15 seconds). Starter amp draw should be less than 100 amps during cranking.

Excessive draw could be caused by the following:

- **Faulty starter.** To verify, remove starter from engine and bench check prior to replacing. Perform an amp draw test on the bench. Starter should not exceed 62 amps with a minimum pinion speed of 3250 rpm.
- **Engine oil viscosity too heavy** (see engine manufacturer's Operators Manual).
- Some equipment manufacturers may use a **constant drive pump**. Constant drive pump oil may be too heavy, or belt tension may be too tight.
- **Internal engine problem.** Turn the engine over with a socket and ratchet to verify it rotates easily.
- **Battery cable loose** or too small.
- **Battery terminals corroded.**
- **Battery CCA too low.**

Battery Information

A 12 volt battery is required to operate the starter motor on Generac OHVI V-Twin engines. For best starter life and performance, battery rating should be at least 450 cold cranking amps at 0 °F (-18 °C) for gasoline engines and 525 cold cranking amps for LP engines.

WARNING: Wear eye protection when servicing the battery. Avoid skin contact. If contact does occur, flush with cold water and consult a physician.

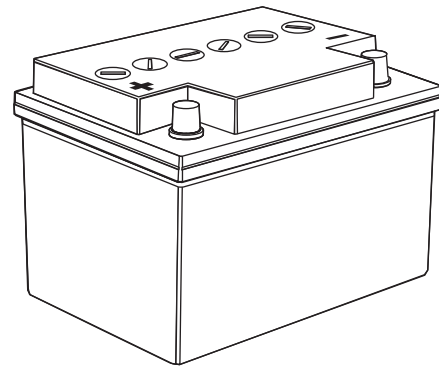


WARNING

Explosion. Batteries emit explosive gases. Always disconnect negative battery cable first to avoid spark. Failure to do so could result in death or serious injury.

(000238)

WARNING: Batteries produce hydrogen, an explosive gas. Do not store or charge a battery near an open flame or devices which utilize a pilot light or can create a spark.



004339

Figure 8-5. Typical Battery

Check Battery

1. Physical check; clean if necessary. Look for:
 - Corrosion
 - Dirt
 - Terminals and clamps—good condition
 - Terminals and clamps—secure connections
2. Bring battery to full charge.

WARNING: DO NOT exceed charge rate of 1/10 ampere for every ampere of battery rating. Consult battery manufacturer for maximum charge recommendations.

- a. Use a "trickle" charge to reduce charge rate.
- b. Fill non-sealed battery cells with distilled water after charging (for batteries which have been in service).

NOTE: Periodically unplug charger if battery becomes hot to the touch or emits excessive gas or electrolyte solution.

- See [Figure 8-6](#). With battery fully charged, check specific gravity readings of each cell with a battery hydrometer. Record readings. All readings should be above 1.250 (compensating for temperature). If specific gravity readings vary by more than 0.50, or if all cells read less than 1.225, replace battery.

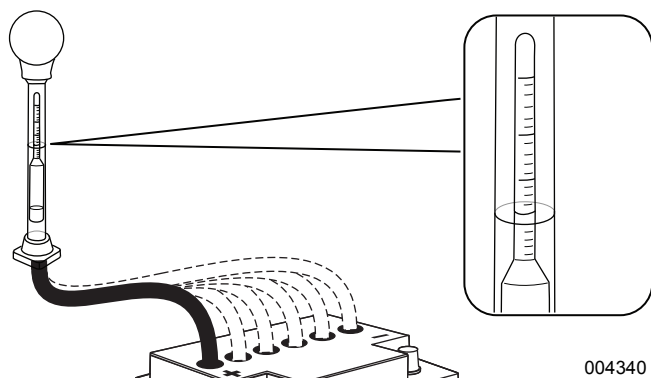


Figure 8-6. Check 12 V Battery Cells

Testing Battery

- Set DMM to read DC Volts.
- Attach RED DMM test lead to positive (+) battery terminal. Attach BLACK DMM test lead to negative (-) battery terminal.
- With ignition switch OFF, press starter button. If ignition switch and starter switch are the same switch, disconnect wires from spark plugs and ground ignition using two ignition testers.
- Turn switch to START. DMM should display 9 volts or more while cranking engine. Replace battery if less than 9 volts is measured.

IMPORTANT NOTE: Do not crank starter motor for more than 15 seconds without allowing starter motor to cool at least two minutes.

Battery Recommendations

Minimum recommended battery size is 450 CCA @ 0 °F (-18 °C) for gasoline engines and 525 CCA @ 0 °F (-18 °C) for LP engines.

Battery Cable Recommendations:

These cable sizes are based on total length of cable from battery positive post to starter, plus ground return to battery negative post.

- #4 AWG — 6 ft (1.8 m) or less
- #2 AWG — 12 ft (3.7 m) or less

Optional Wire Harness

See [Figure 8-7](#) or [Figure 8-8](#). Engines supplied with a wire harness have a four or five terminal Packard connector (A) on the harness. A typical wire harness consists of four colored wires as follows:

- **Brown** wire is used to signal low oil pressure. When low oil pressure is detected, this wire becomes a ground. This is a normally open switch. Some engines use a normally closed switch and will work the opposite as stated above.
- **Yellow** wire is used to ground the ignition. Connect this wire to ground and the ignition will be shut off.
- **White** wire is connected to +12 Volts DC and the carburetor solenoid, which allows fuel to flow. Turn off the +12 V supply, and the fuel solenoid will close, stopping the flow of fuel.
- **Orange** wire (red on some models) is used to start the engine. Supply +12 VDC to this wire and the starter solenoid will engage. Turn off the +12 VDC supply, and the starter will disengage.

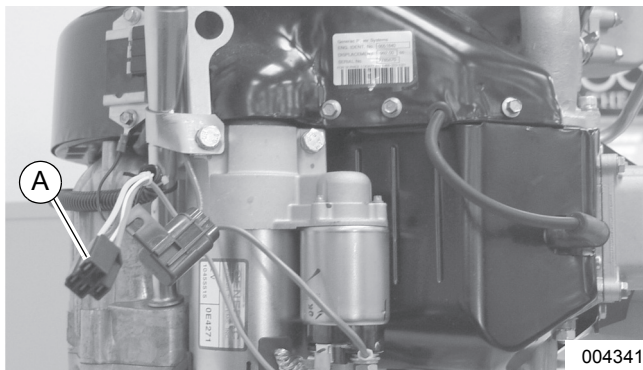


Figure 8-7. Packard Connector—Vertical Shaft

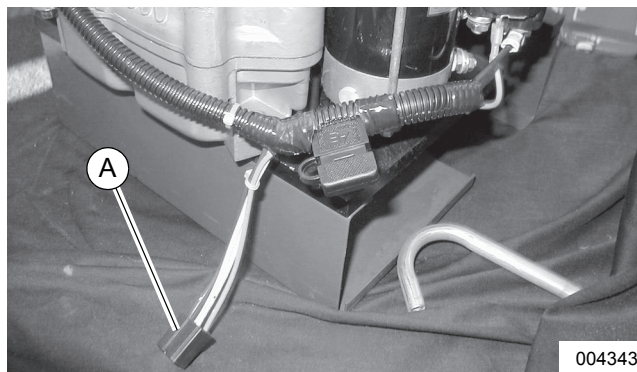


Figure 8-8. Packard Connector—Horizontal Shaft

[Figure 8-9](#) and [Figure 8-10](#) illustrate proper wire harness routing and connections. On horizontal shaft engines, use a cable tie to fasten wiring harness (B) to fuel solenoid. On vertical shaft engines, route wiring harness above fuel solenoid.

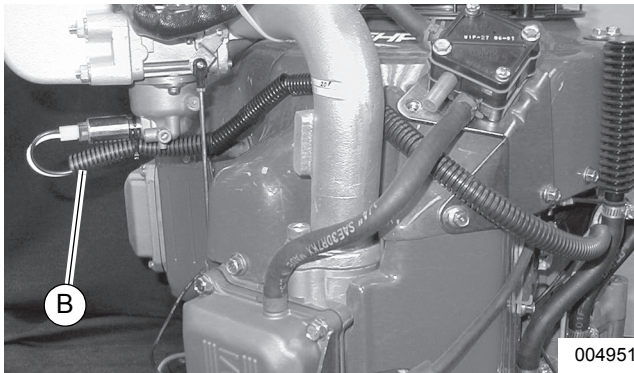


Figure 8-9. Wire Harness Routing—Horizontal Shaft

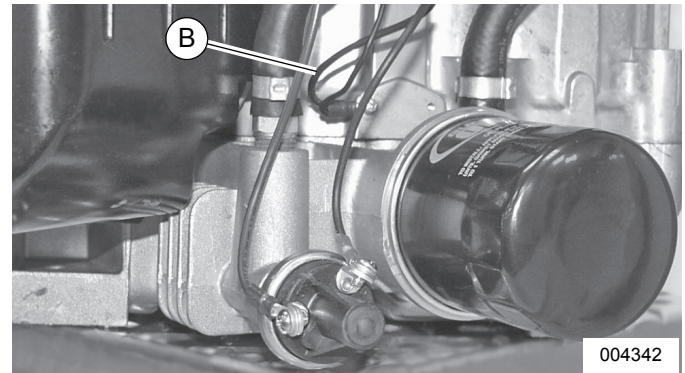


Figure 8-10. Wire Harness Routing—Vertical Shaft

Wire harness connectors vary depending on engine model. Refer to the following chart for wire harness terminal locations.

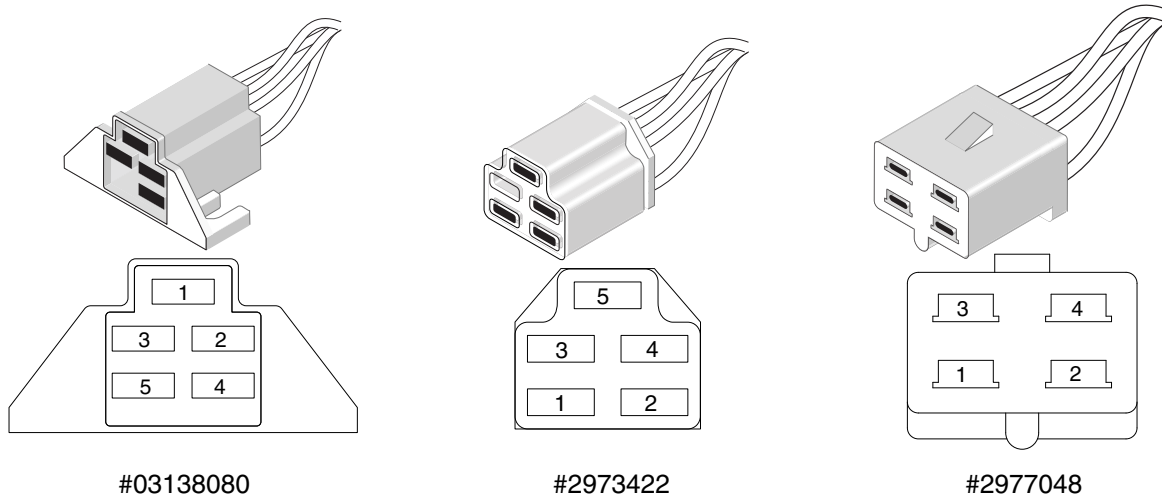
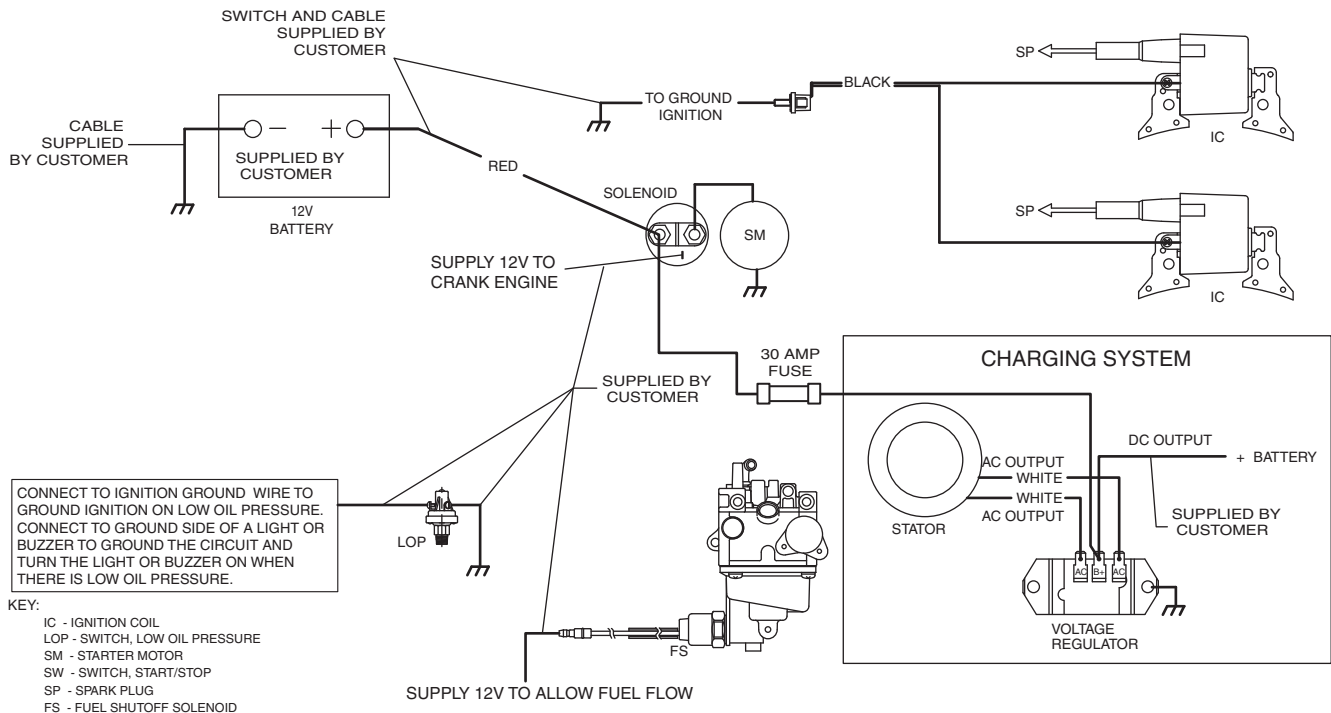


Figure 8-11. Wire Harness Terminal Identification

1	Ignition kill terminal—yellow	3	Carburetor fuel solenoid—white	5	Open
2	Starter solenoid—orange	4	Low oil pressure switch—brown	—	—



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Figure 8-12. Typical 12 Volt Wiring Diagram

Battery Disposal

⚠ WARNING

Environmental Hazard. Always recycle batteries at an official recycling center in accordance with all local laws and regulations. Failure to do so could result in environmental damage, death or serious injury.

(000228)

Do not dispose of waste batteries in household or business trash receptacles. They contain potentially hazardous materials such as metals, chemicals, and plastic requiring proper disposal.

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit the Battery Council International website at: <http://batteryCouncil.org>.

Section 9: Lubrication System

Description

Generac GTH/GTV-760/990 OHVI V-twins use a full pressure lubrication system with an oil filter. A gerotor oil pump draws oil from a screened oil pickup in the sump and pumps oil through the oil filter.

Filtered oil flows through an oil gallery in the sump and is distributed to the main bearings, connecting rod bearings, and camshaft bearings. Engine oil pressure varies with oil viscosity, ambient air temperature differences, operating temperatures and engine load. See [Recommended Oil Type](#).

Oil Pressure - @ 70 °F (21 °C):
15–50 psi (103–345 kPa)

Two pressure relief valves limit the maximum oil pressure in the system to 60 and 40 psi (414–276 kPa) respectively.

Protection Systems

Low Oil Pressure Switch

See [Figure 9-1](#). The engine is usually equipped with a normally open type, low oil pressure switch (A) which closes the circuit between the terminals when oil pressure drops below 8 psi (55 kPa). If one terminal is connected to the ignition ground wire and the other terminal is connected to ground, the engine will shut down on low oil pressure. The switch can also be connected to an alarm or light to indicate low oil pressure, instead of shutting down the engine.

NOTE: Some engine models have a normally closed type switch and work the opposite as above.

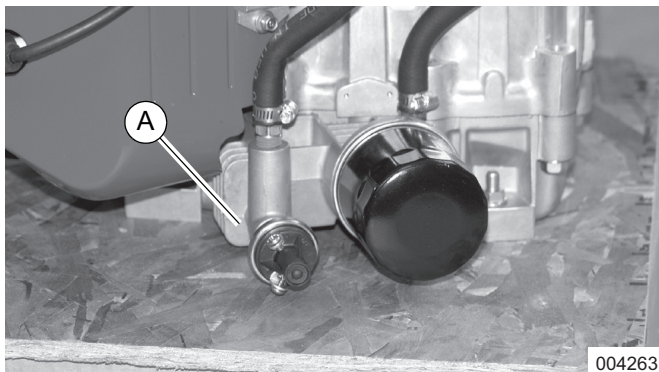


Figure 9-1. Low Oil Pressure Switch

A delay built into the shutdown system on some engine applications allows oil pressure to build during starting. The delay allows the engine to run for about 10 seconds before sensing oil pressure. If the engine is wired for low oil pressure shutdown and the system senses low oil pressure during operation, the engine shuts down. The engine will not run until 8 psi (55 kPa) of oil pressure is reached. If a restart is attempted within five seconds after shutdown, the engine may NOT start. The system needs 10 seconds to reset.

Check Engine Oil Level

Oil capacity of the GTH/GTV-760/990 OHVI engine is approximately 2 qt (1.9L). See [Recommended Oil Type](#) for the proper amount for each model. Proceed as follows to check the engine oil level:

1. See [Figure 9-2](#). Remove dipstick (A) and wipe it dry with a clean cloth.

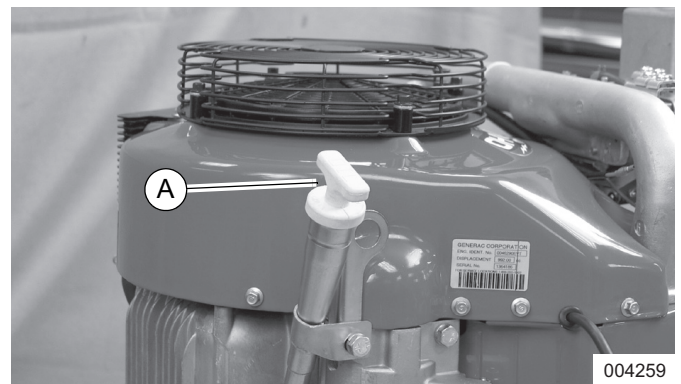


Figure 9-2. Oil Fill/Check Location (Vertical Shaft)

2. Insert dipstick completely, then remove. Oil level should be at the FULL mark. If necessary, add oil until the FULL mark is reached. **Do not fill above the “FULL” mark.**

NOTE: Never operate engine with oil level below the ADD mark on the dipstick. Doing so could damage the engine.

Change Engine Oil And Filter

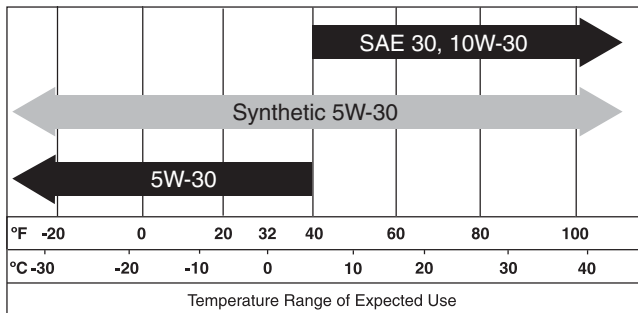
Recommended Oil Type

Correct oil type and weight in the crankcase is extremely important. Check oil before each use and change oil regularly. Incorrect or dirty oil can cause premature engine wear and failure.

Use only high quality detergent oil rated with an API service classification SN or higher.

Recommended oil weights include the following:

- During summer months: SAE 30. An acceptable substitute is SAE 10W-30. After first oil change, synthetic oil is acceptable.
- During winter months: SAE 5W-30 or synthetic 5W-30. DO NOT USE SAE 10W-40. After first oil change, synthetic oil is acceptable.



004058

Figure 9-3. Oil Temperature Range

NOTE: Engine failure may occur if attempting to crank or start the engine without the proper amount of recommended oil.

Oil Change Procedure

See [Specifications](#) for crankcase oil capacities. Use no special additives. Verify the unit is level when filling with oil. DO NOT OVERFILL.

The engine is equipped with an oil filter. Change oil and oil filter after the first eight hours of operation. Change oil and oil filter every 100 hours thereafter. In dirty or dusty operating conditions, or in extremely hot weather, change oil more often.

Change oil while the engine is still warm:

1. See [Figure 9-4](#). Clean area around oil drain plug (A), remove plug, and drain oil completely into a suitable container.

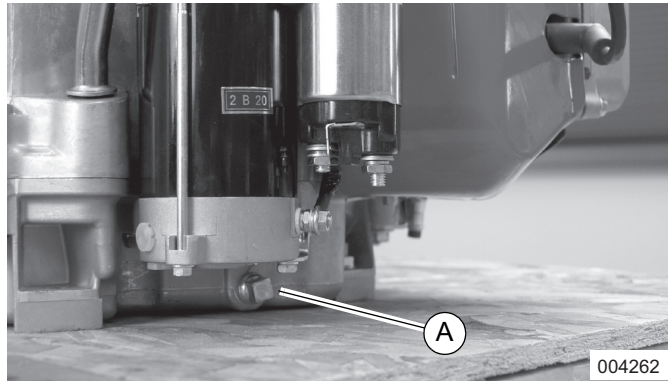


Figure 9-4. Oil Drain Plug

2. After oil is drained, install and tighten oil drain plug.
3. See [Figure 9-5](#). To change oil filter:
 - a. Locate oil filter (B).



Figure 9-5. Remove Oil Filter

- b. Place a suitable container beneath oil filter and rotate filter counterclockwise to remove.
 - c. Coat gasket on new filter with engine oil. Install and rotate new filter clockwise until gasket contacts filter adapter, then tighten an additional 3/4 turn.
4. Remove oil fill cap and insert a clean funnel into oil fill opening. Fill crankcase with recommended oil until oil level is at FULL mark on dipstick.
 5. After crankcase is filled to the proper level, install oil fill cap. Start engine to fill oil filter. Check and adjust oil level as needed.

Oil Cooler

See [Figure 9-6](#). All GTH/GTV-760/990 V-Twin engines are equipped with an oil cooler (A) mounted on the blower housing. Forced air from the flywheel fan flows through the oil cooler fins, dissipating heat from the engine oil.

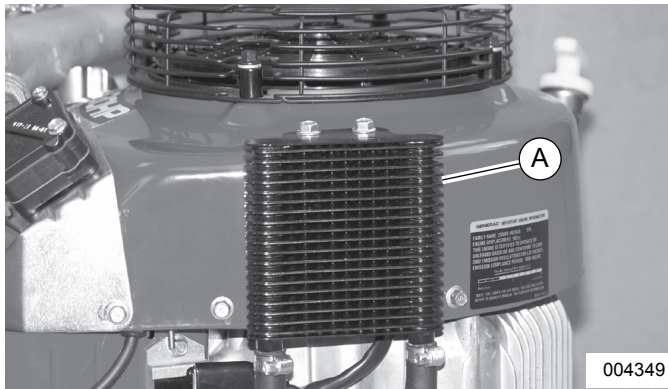


Figure 9-6. Oil Cooler

Clean oil cooler fins every 100 hours, or more often under dusty conditions. Periodically inspect for debris. Clean with compressed air or a soft bristle brush.

Oil Pump

Generac engines are equipped with gerotor style oil pumps classified as a positive displacement pumps. Each revolution displaces a constant amount of oil.

See [Figure 9-7](#) or [Figure 9-8](#). As the gerotor turns, oil is drawn into the oil intake and fills the entire cavity (A) between the inner and the outer rotor. As the gerotor continues to rotate, oil moves from the intake port to the outlet port where the pressurized oil is forced into the entire lubrication system.

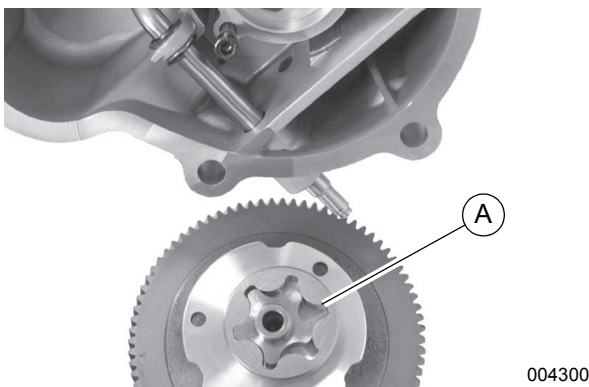


Figure 9-7. Oil Pump, Vertical Engine

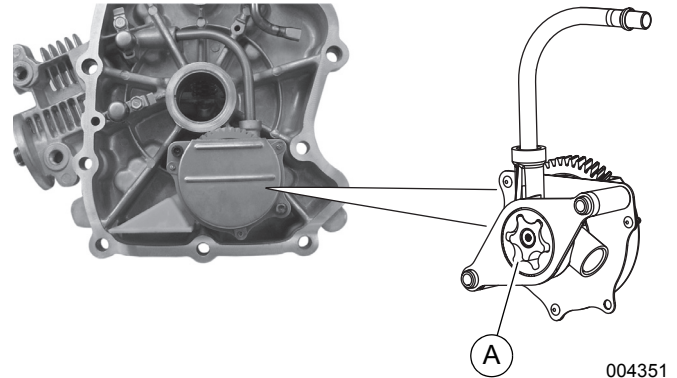


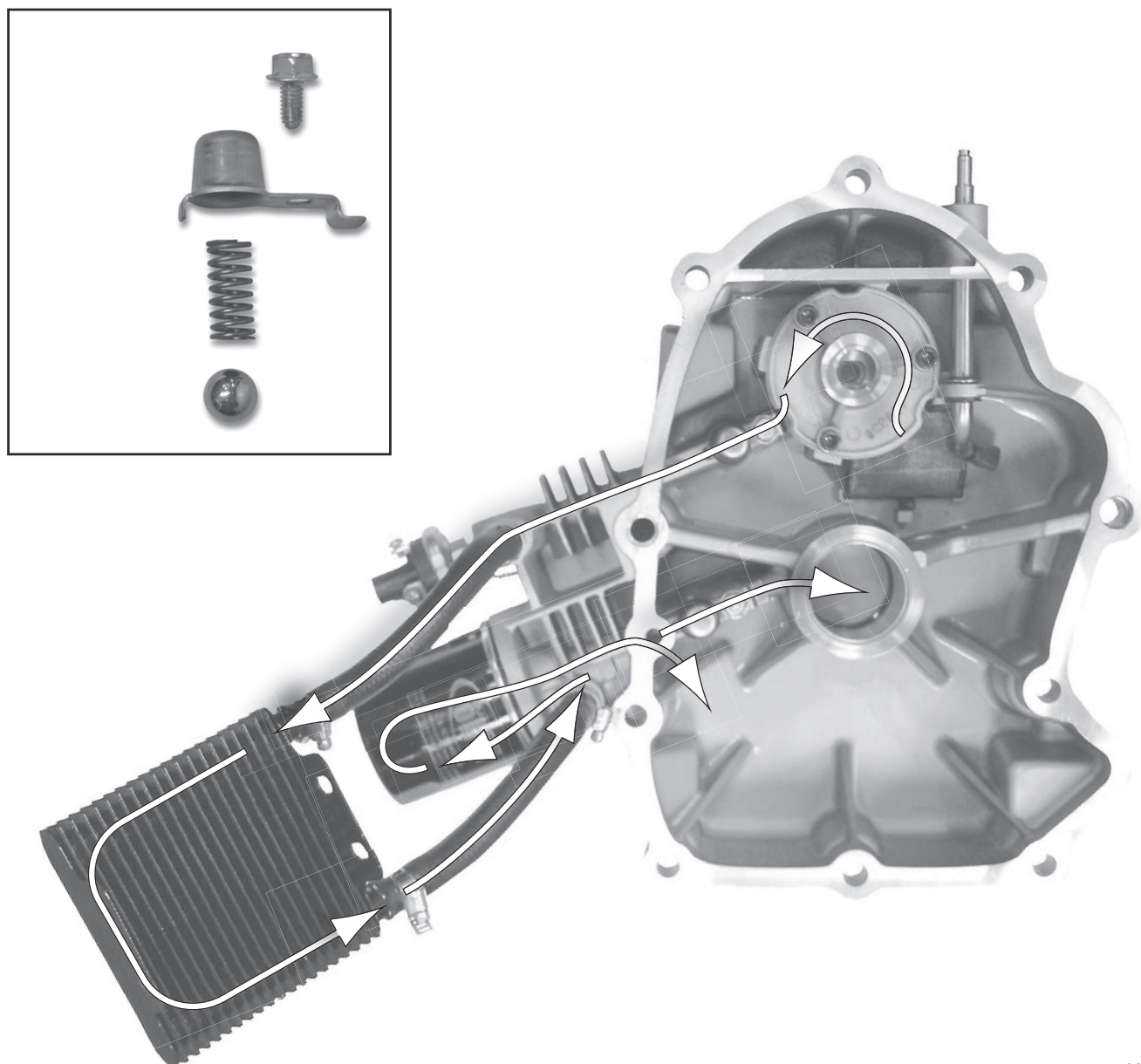
Figure 9-8. Oil Pump, Horizontal Engine

Pressure Relief Valves

A pressure relief valve is necessary to relieve excess pressure build-up in the system when a constant displacement pump is used. The valve is required because the pump supplies more pressurized oil than the system is able to use.

Generac uses two types of relief valves: a “Cold Start” and a “Common” valve. Oil pressure can reach up to 140 psi (965 kPa) when the oil is cold and the engine is started.

[Figure 9-9](#) illustrates oil flow in a typical V-twin engine. To avoid oil filter and oil cooler damage, the Cold Start relief valve is installed in series immediately downstream of the oil pump and set at 60 psi (414 kPa). After the oil goes through the oil cooler and filter, it flows through the Common relief valve. This relief valve is set at 40 psi (276 kPa) and mainly regulates the pressure to the PTO journal and throughout the rest of the oil system



004352

Figure 9-9. Typical V-Twin Engine Oil Flow (Vertical Shaft Engine Shown)

Section 10: Engine Disassembly

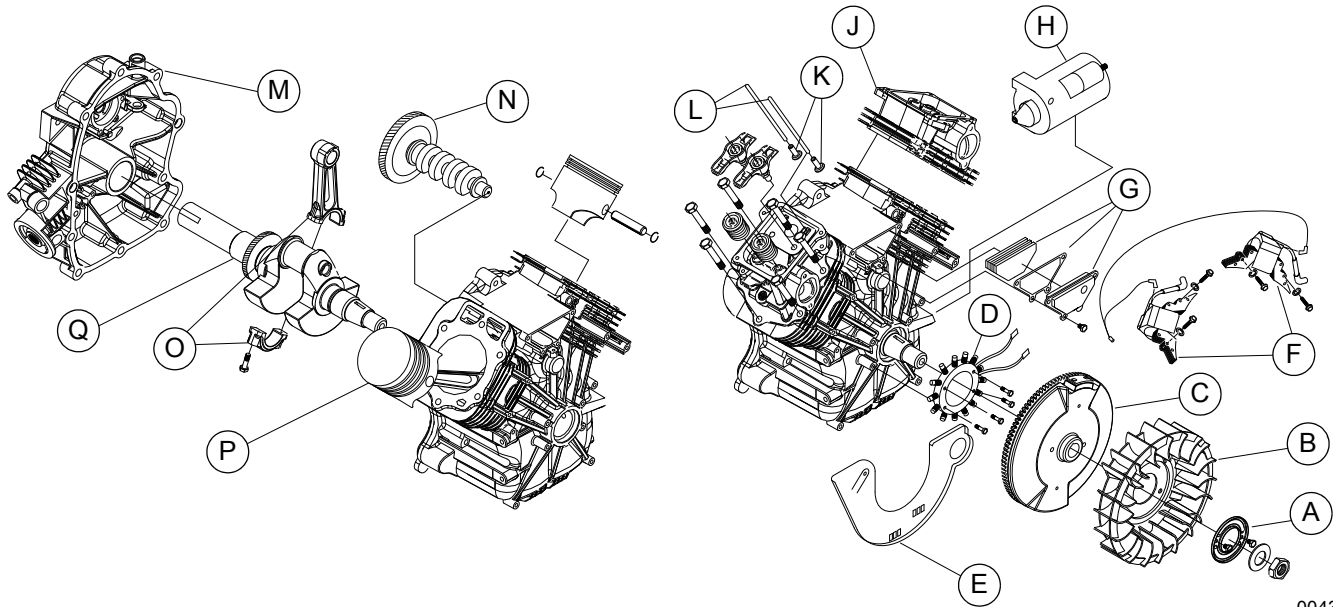
Engine Disassembly

Drain oil, disconnect wire harness, connector, cables and fuel hose. Remove engine from equipment. Remove intake and exhaust systems. Remove engine shroud and all cylinder wrappers.

See [Section 4: Ignition](#). Disconnect stop switch wires at ignition coils and remove ignition coils.

See [Section 5: Carburetion and Fuel System](#). Remove cylinder heads.

1. See [Figure 10-1](#). Remove the following parts:
(C) Flywheel
(G) Breather Assembly
(D) Charging coil
(E) Backing plate
(H) Starter
2. Remove crankcase cover (M) and sump. Discard gasket and O-ring.

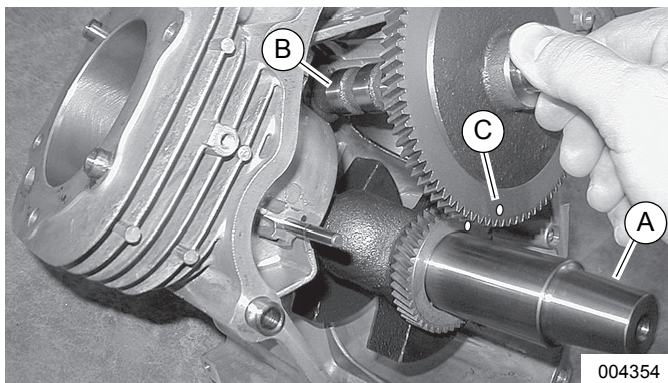


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Figure 10-1. Partial View of Engine Breakdown

A	Fan retainer	E	Backing plate	J	Cylinder head	N	Camshaft
B	Fan	F	Ignition coils	K	Tappets	O	Connecting rod
C	Flywheel	G	Breather assembly	L	Push rods	P	Piston assembly
D	Charging coil	H	Starter	M	Crankcase cover	Q	Crankshaft

3. See [Figure 10-2](#). Rotate crankshaft (A) and camshaft (B) until timing marks (C) align, and remove camshaft.



004354

Figure 10-2. Remove Camshaft

NOTE: If necessary, place engine flywheel side down to prevent tappets from catching on the camshaft.

4. Remove tappets.

NOTE: Remove any carbon or ridge at the top of the cylinder bores to prevent breaking rings when removing piston and connecting rod assemblies.

5. See [Figure 10-3](#). Remove #2 connecting rod cap, and push connecting rod and piston assembly out through top of cylinder.

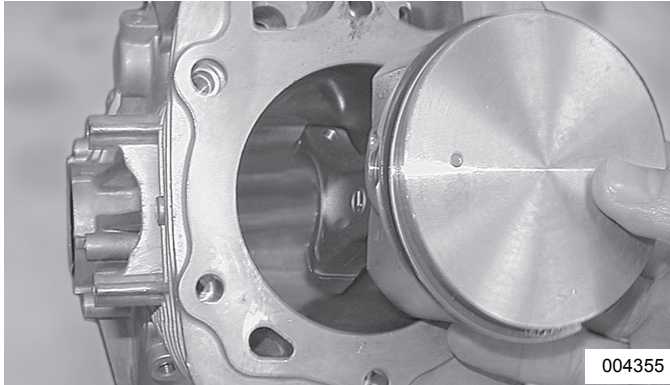


Figure 10-3. Remove Pistons and Connecting Rods

Assemble cap on rod to prevent interchanging.

6. Repeat for cylinder #1.
7. See [Figure 10-4](#). Remove crankshaft.

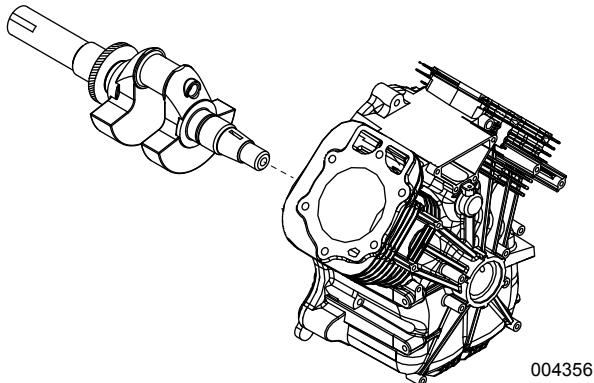


Figure 10-4. Remove Crankshaft

8. Clean all surfaces of gasket material as described in section "Cylinder Head and Valves." under [Gasket Removal](#).
9. Remove oil seals and thoroughly clean components in solvent. Organize components, keeping subassembly parts together.

10. See [Figure 10-5](#) or [Figure 10-6](#). Remove oil pump from crankcase cover/sump.

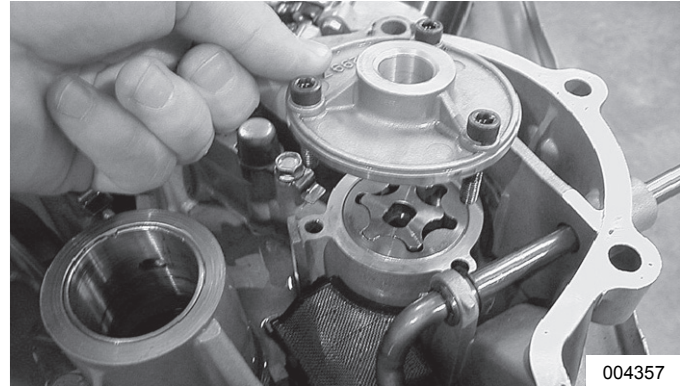


Figure 10-5. Remove Oil Pump (Vertical Shaft Engine)

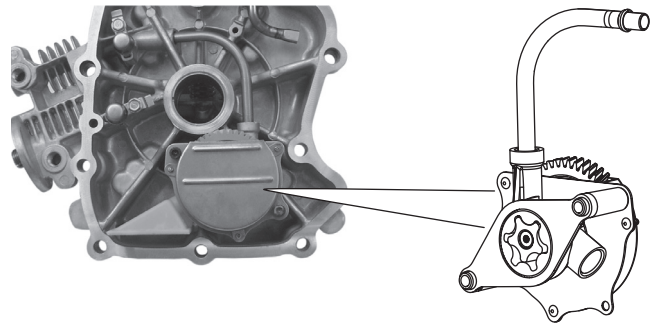


Figure 10-6. Remove Oil Pump (Horizontal Shaft Engine)

Section 11: Cylinder and Crankcase Cover

Check Crankcase

Check crankcase for cracks, stripped threads, or broken fins. Check cylinder bores for damage or scoring.

1. See [Figure 11-1](#). Check cylinder head mounting surface for distortion with a straight edge.

Crankcase must be replaced if mounting surfaces are distorted more than 0.004 in (0.1 mm).

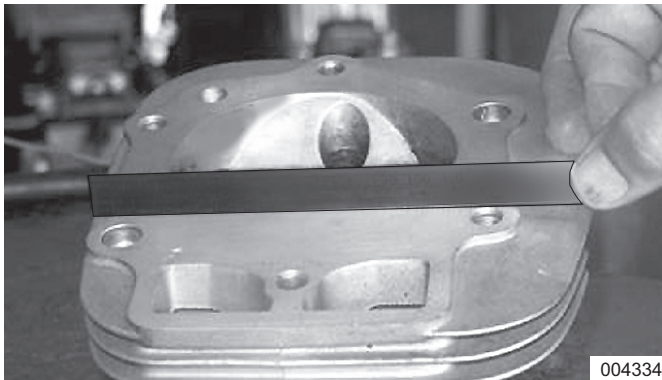


Figure 11-1. Check Cylinder Head Mounting Surface

2. See [Figure 11-2](#). Check each cylinder bore for wear using a cylinder bore gauge or a telescoping gauge and dial caliper.

Standard Bore Diameter: 3.543–3.544 in (90.00–90.025 mm)

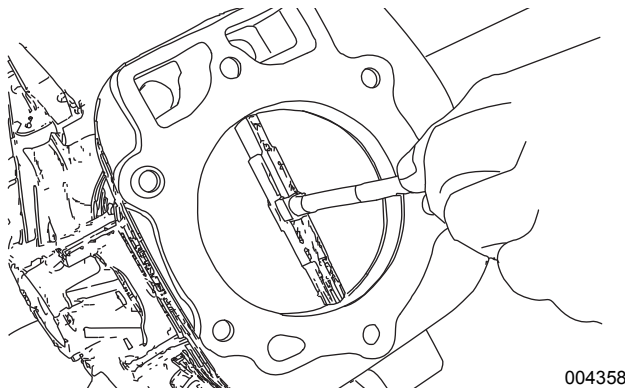
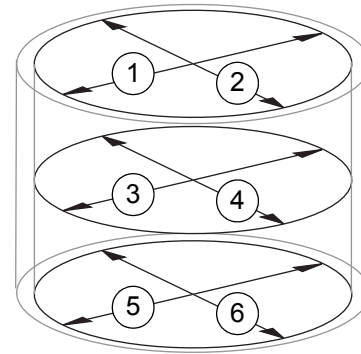


Figure 11-2. Check Cylinder Bore Diameter

See [Figure 11-3](#). Measure cylinder bore in six points at right angles as shown.



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Figure 11-3. Measure Diameter at Six Points

Crankcase must be replaced if either cylinder bore is worn more than 0.003 in (0.075 mm) or more than 0.0015 in (0.035 mm) out of round.

NOTE: If cylinder bores are within specification and show no signs of scoring or other damage, new piston rings may be installed, provided the cylinder bores are reconditioned using a rigid hone with finishing stones to restore the proper cross hatch angle in the cylinder bores. The proper cylinder cross hatch ensures proper lubrication and piston ring break-in.

Refer to [Cylinder Finish \(Cross Hatch\)](#) for correct procedure for cross hatch honing.

Resizing

NOTE: Oversize rings and pistons are NOT available. DO NOT bore or over-hone cylinder.

Cylinder Finish (Cross Hatch)

Recondition cylinder bores to restore cross hatch before installing new piston rings in a cylinder within specification. Avoid honing cylinder bores oversize. Oversize cylinder bores will permanently damage the crankcase.

See **Figure 11-4**. Special honing stones are required when reconditioning a cylinder bore. Honing stones produce the correct cross hatch necessary for proper lubrication and ring seating. Honing is performed as a two step process called plateau honing. These steps must be followed carefully and should be performed by a competent machine shop. The correct cross hatch angle is approximately 45 degrees.

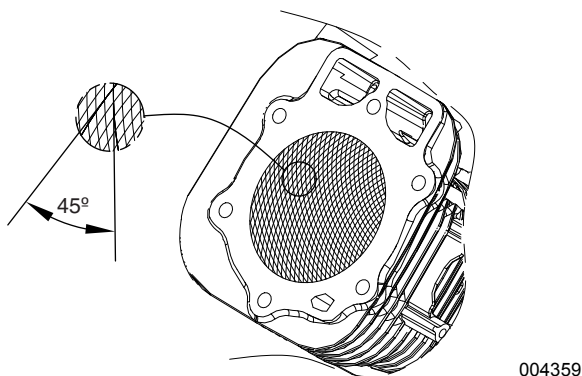


Figure 11-4. Cylinder Cross Hatch

Honing is done with a variable speed half-inch portable drill and a honing fixture. **Figure 11-6** at the end of this topic provides dimensions for making a honing fixture.

Setup

1. See **Figure 11-5**. Use two crankcase cover mounting screws to fasten honing fixture (A) to crankcase.

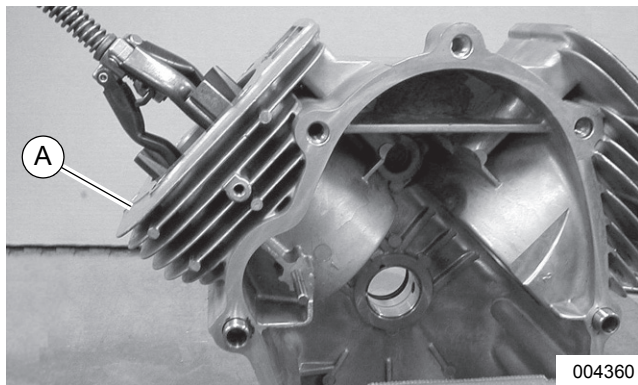


Figure 11-5. Honing Cylinders

2. Cut a wood block and place inside crankcase to prevent hone from extending further than 0.75–1.0 in (19 mm–25 mm) below cylinder bore.
3. Clamp honing fixture and crankcase securely in a vise at a convenient work height.

4. Place hone in middle of cylinder bore. Tighten adjusting knob with fingers until stones fit snugly against cylinder wall. **DO NOT FORCE**. Place hone drive shaft in portable drill chuck and tighten. Verify cylinder and hone are centered and aligned with drill spindle.

Honing Procedure

1. For the first pass, hone each cylinder with a 220 grit diamond metal-bond super-abrasive stone (such as a Sunnen® GMG55). Set drill speed at 240 rpm at 80 strokes per minute. Lubricate cylinder liberally to prevent buildup on honing stones.
2. For the second pass, hone each cylinder with an 800 grit diamond metal-bond super-abrasive stone (such as a Sunnen® RMG807) at 110 rpm drill speed at 80 strokes per minute.

NOTE: Sunnen® lubricating oil (SCC605) is recommended. Automatic transmission fluid is also an acceptable honing oil. Another acceptable honing oil can be made by mixing four parts 30 weight oil with one part kerosene.

Cleaning

Clean each cylinder and crankcase thoroughly after honing.

1. Wash cylinder and crankcase carefully in a commercial solvent or kerosene.
2. Thoroughly wash cylinder and crankcase using a stiff brush with soap and hot water. Rinse thoroughly with hot running water.
3. Repeat washing and rinsing until all traces of honing grit are gone.

IMPORTANT NOTE: Honing grit is highly abrasive and will cause rapid wear to all internal components of the engine unless it is completely removed.

4. When cylinder and crankcase have been thoroughly cleaned, use a clean white rag or paper towel and wipe the cylinder bore. Any remaining honing grit will be visible as a gray residue. If any honing grit is evident, wash and rinse entire cylinder and crankcase, and check again. The cylinder is properly cleaned when there is no trace of honing grit on the rag. Then, apply oil to the cylinder bore to prevent rusting.

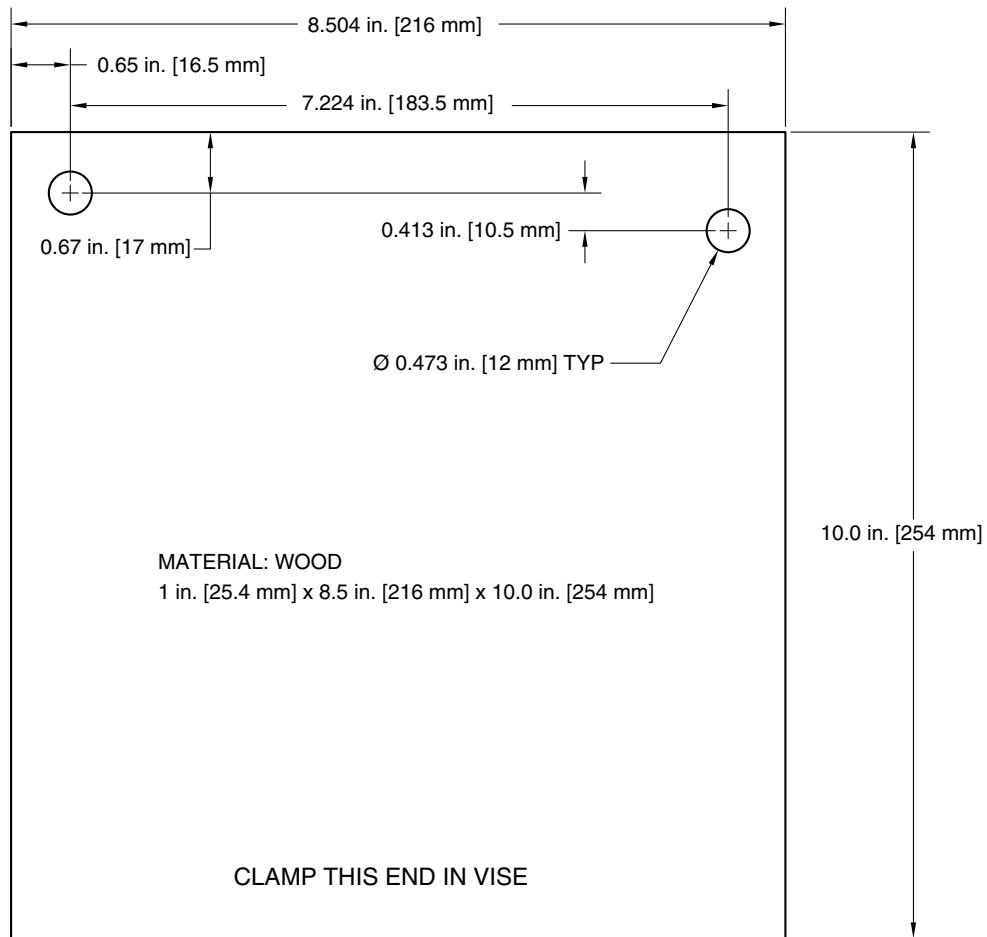


Figure 11-6. Honing Fixture

Bearings

Check Magneto (Flywheel End) Bearing

See [Figure 11-7](#). Check magneto bearing (A) for damage. Damaged bearings cannot be replaced. If undamaged, check for wear using a telescoping gauge and caliper. Measure at several locations. Crankcase must be replaced if measured diameter is larger than 1.506 in (38.25 mm).

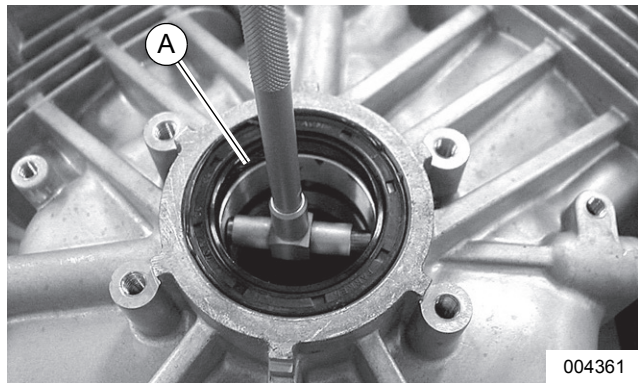


Figure 11-7. Check Magneto Bearing

Crankshaft diameter out of specification may also make it necessary to replace the crankcase. See [Specifications](#) for crankshaft measurements.

Check PTO Bearing

See [Figure 11-8](#). Sump cover must be replaced if PTO bearing (B) is damaged or measures larger than 1.663 in (42.25 mm) diameter.

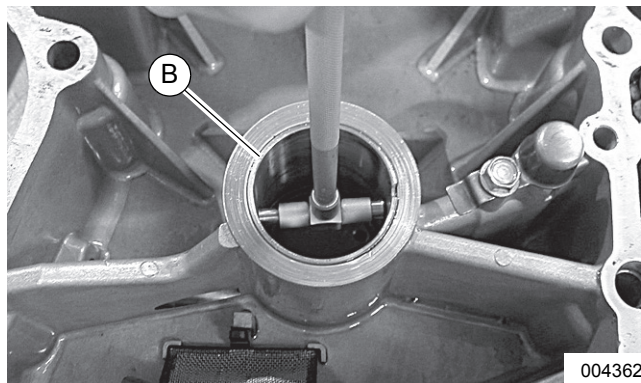


Figure 11-8. Check PTO Bearing

Crankshaft diameter out of specification may also make it necessary to replace sump cover. See [Specifications](#) for crankshaft measurements.

Install PTO Oil Seal

Install a new PTO oil seal and press it in until it is 0.0625 in (1.5 mm) below the mounting surface.

Check Camshaft Bearings

See [Figure 11-9](#) and [Figure 11-10](#). Use a telescoping gauge and caliper to check camshaft bearing diameters. Crankcase or crankcase cover must be replaced if camshaft bearings are worn or diameters are out of specification. (For vertical shaft engines, the oil pump cover must be replaced).

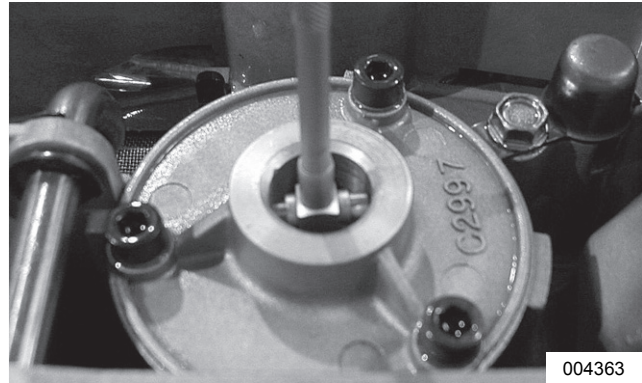


Figure 11-9. Check PTO Side Camshaft Bearing Diameter

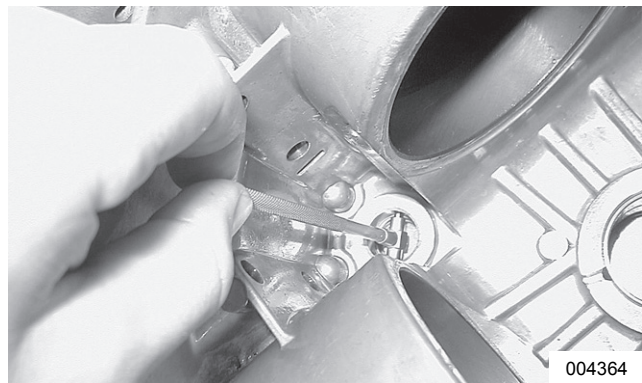


Figure 11-10. Check Magneto Side Camshaft Bearing Diameter

Oil Seals

Always install new oil seals whenever engine is disassembled for major servicing. Lubricate sealing edge of oil seal with clean engine oil before assembly.

Check Crankshaft

See [Figure 11-11](#). Inspect crankshaft threads, keyways, and timing gear for damage or wear. Replace crankshaft if threads, keyways, or timing gear are damaged or worn. Check journals for scoring. Replace crankshaft if journals are scored.

Check journals for wear.

See crankshaft tolerances in [Specifications](#).

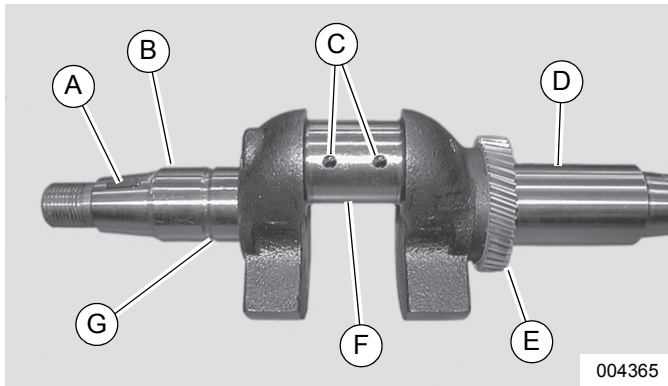


Figure 11-11. Check Crankshaft

A	Flywheel key	E	Timing gear
B	Magneto journal	F	Crankpin
C	Oil galleries	G	Oil gallery
D	PTO journal	—	—

Check Camshaft

See [Figure 11-11](#). Inspect gear teeth, lobes and journals for wear and nicks. Check journals and lobes for scoring and wear. Replace camshaft if measurements are outside range provided in [Specifications](#).

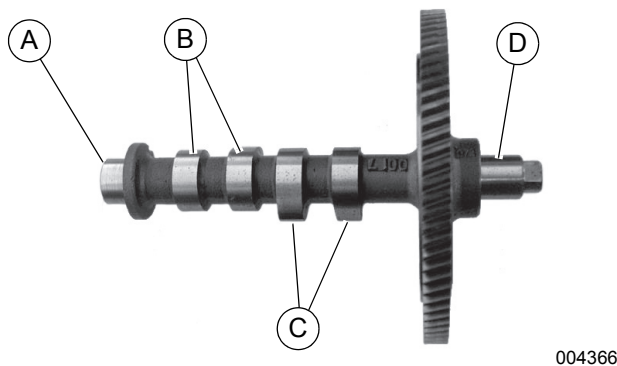


Figure 11-12. Check Camshaft

A	Magneto journal	C	Exhaust lobes
B	Intake lobes	D	PTO journal

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Section 12: Piston, Rings, and Connecting Rod

Overview

Install new piston rings whenever the engine is disassembled for major servicing or overhaul, provided cylinder bores are within specification.

See Section 7: **Cylinder Head and Valves**. Measure cylinder bore diameters before checking pistons and rings. Crankcase must be replaced if cylinder bore diameters are out of tolerance.

Cylinder bore must be replaced if it is more than 0.003 in (0.075 mm) oversize, or 0.0015 in (0.035 mm) out of round.

Disassemble Piston and Connecting Rod

Clean all carbon from cylinder bore (to prevent ring breakage) before attempting to remove piston and connecting rod.:

1. See **Figure 12-1**. Remove piston compression rings using a ring expander similar to the one shown. Remove oil ring.

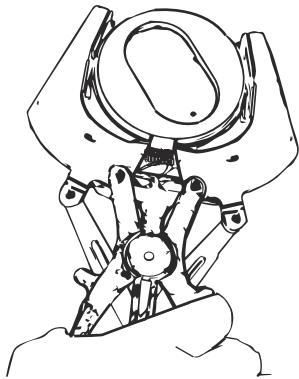
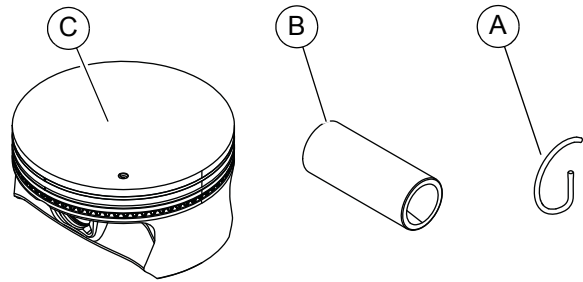


Figure 12-1. Remove Rings

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2. See **Figure 12-2**. Remove two piston pin locks (A) from each piston pin (B). Remove piston (C) from connecting rod.



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Figure 12-2. Remove Piston Pin Locks

NOTE: Piston pin is a slip fit in piston and connecting rod.

NOTE: Keep pistons and connecting rods together as an assembly. They are not interchangeable.

Check Piston and Rings

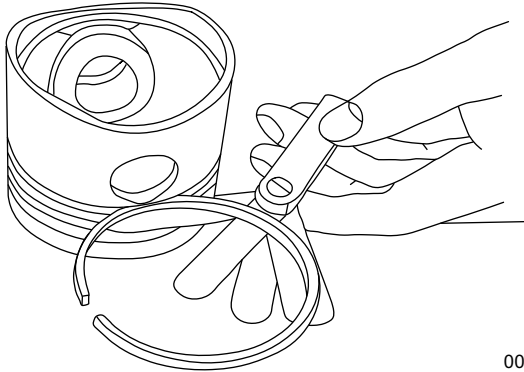
General Rules

Always comply with the following general rules pertaining to piston rings:

- Replace piston rings in sets.
- Use a ring expander to remove or install piston rings, or breakage could result. Do not spread the rings too far apart or they might break.
- Use a ring compressor when installing piston in cylinder.
- When installing NEW rings, deglaze the cylinder walls with a commercially available deglazing tool.

If the cylinder will not be replaced and the piston shows no signs of scoring, carefully remove carbon from piston ring grooves.

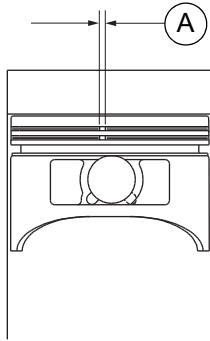
1. See **Figure 12-3**. Check side clearance of ring grooves using new rings. Piston must be replaced if compression ring groove allows insertion of a 0.004 in (0.10 mm) feeler gauge next to ring, or oil ring groove allows insertion of a 0.008 in (0.20 mm) feeler gauge next to ring.



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Figure 12-3. Check Ring Groove Side Clearance

See **Figure 12-4**. Check ring end gap (A).

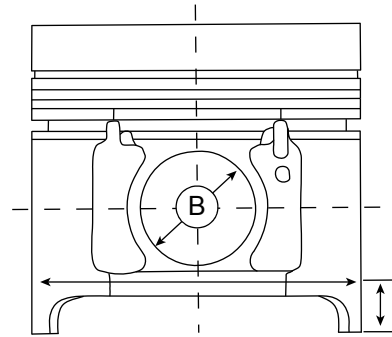


002126

Figure 12-4. Check Ring End Gap

- a. Insert a new ring approximately 1.0 in (25 mm) into cylinder.
- b. If gap is less than measurement indicated in **Specifications**, remove some material from ring ends with a fine file to achieve minimum gap.

2. See **Figure 12-3**. Check piston pin bore diameter (B). Replace piston if bore diameter exceeds 0.7886 in (20.03 mm), or if it is 0.0005 in (0.01 mm) out of round.

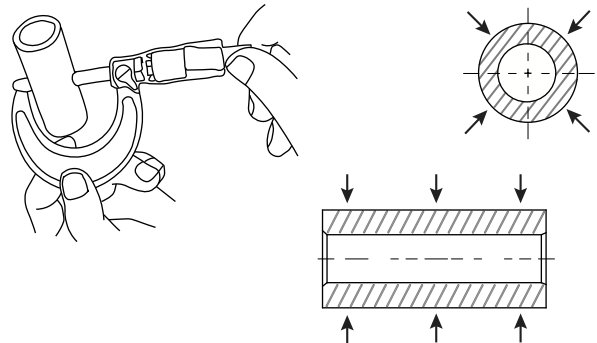


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Figure 12-5. Check Piston Pin Bore

Checking Piston Pin And Connecting Rod

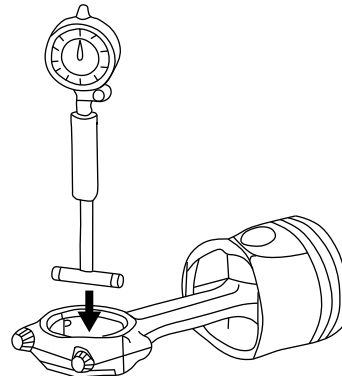
1. See **Figure 12-3**. Check piston pin diameter. Replace piston pin if diameter is less than 0.7862 in (19.97 mm) or out of round by 0.0005 in (0.01 mm) or more.



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Figure 12-6. Check Piston Pin

2. See **Figure 12-7**. Check connecting rod bearings.



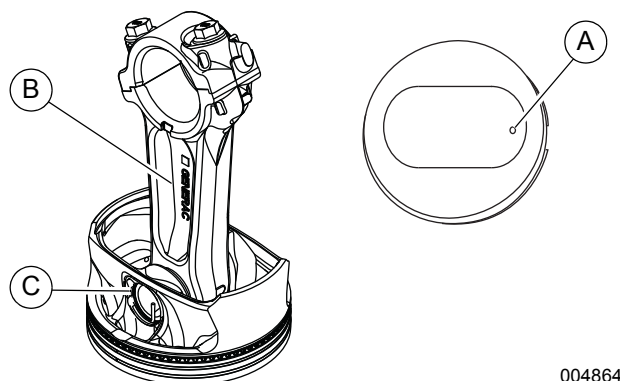
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Figure 12-7. Check Rod Bearings

NOTE: Connecting rod must be replaced if piston pin or crankpin bearing ends are scored or worn.

Assemble Pistons And Connecting Rods

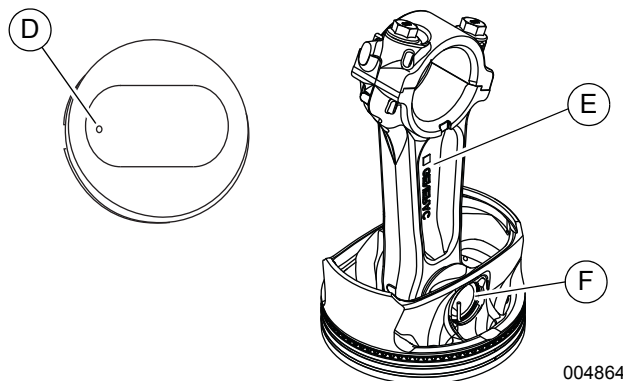
1. See [Figure 12-8](#). Lubricate parts with engine oil and assemble #1 piston and connecting rod.



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Figure 12-8. Assemble #1 Rod and Piston

- a. Notch or casting mark (A) on the piston top must face towards flywheel.
 - b. Embossed number “1” (B) on connecting rod must face PTO side (opposite notch or casting mark on piston).
 - c. Install piston pin locks (C) with needle nose pliers.
2. See [Figure 12-8](#). Lubricate parts with engine oil and assemble #2 piston and connecting rod.



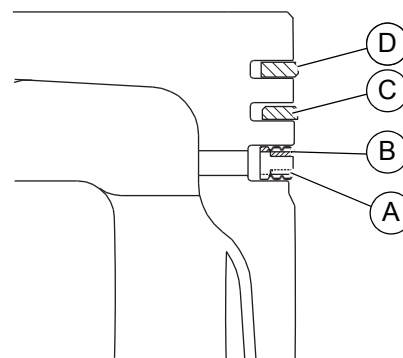
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Figure 12-9. Assemble #2 Rod and Piston

- a. Notch or casting mark (D) on the piston top must face towards flywheel.
- b. Embossed number “2” (E) on connecting rod must face PTO side (away from the flywheel).
- c. Install piston pin locks (F) with needle nose pliers.

Install Piston Rings On Piston

See [Figure 12-10](#). Install piston rings, using ring expander when installing center and top compression rings.



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Figure 12-10. Piston Ring Installation

1. Install oil ring expander.
 - a. Install lower oil scraper ring (A).
 - b. Install upper oil scraper ring (B).
2. Install center compression ring (C) with dimple up.
3. Install top compression ring (D) with dimple up.

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Section 13: Engine Assembly

Install Crankshaft

See [Figure 13-1](#). Lubricate magneto bearing and inside edge of oil seal with engine oil, and install crankshaft.

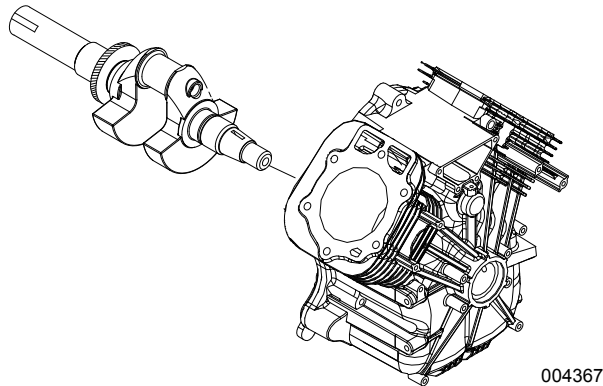


Figure 13-1. Install Crankshaft

Install Piston And Connecting Rod

NOTE: Install #1 piston and connecting rod first.

1. Lubricate piston rings, piston skirt, and ring compressor with oil.
 - a. Rotate the rings so ring end gaps are 90 degrees from each other.
 - b. See [Figure 13-2](#). Place piston inside ring compressor (A) and set upside down on bench with projections on compressor facing up.



Figure 13-2. Compressing Rings

- c. Tighten ring compressor until rings are fully compressed.
 - d. Remove connecting rod cap.
2. Lubricate cylinder bores and crankpin and rotate crankshaft until it is at bottom of stroke.
3. See [Figure 13-3](#). Install #1 piston with notch or casting mark (B) towards flywheel side.

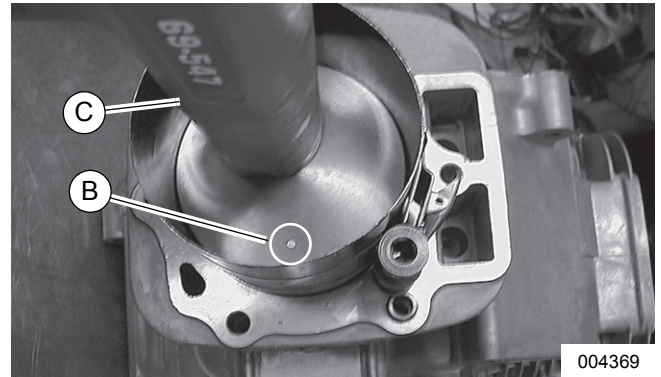


Figure 13-3. Install Piston and Connecting Rod

4. Push piston down by hand or with a wood handle (C) until connecting rod is seated on crankpin.
5. See [Figure 13-2](#). Assemble connecting rod cap (D) to rod with match marks aligned.

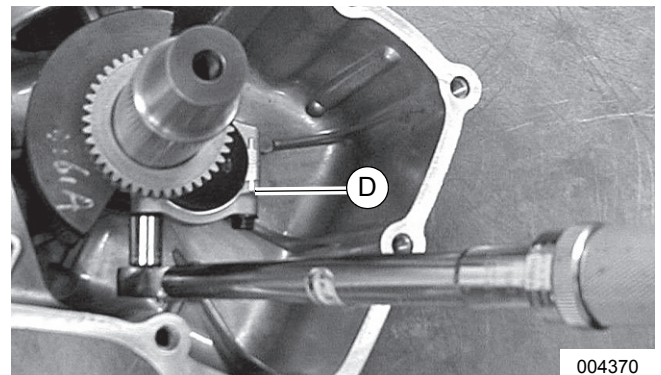


Figure 13-4. Tighten Connecting Rods (#2 rod shown)

6. Tighten screws to 216 in-lb (24.4 Nm) torque.
7. Rotate crankshaft two revolutions to check for binding. Rod should also be free to move sideways on crankpin.
8. Repeat steps 1–5 for #2 cylinder.

NOTE: The number 1 on #1 connecting rod and the number 2 on #2 connecting rod must be facing PTO side.

IMPORTANT NOTE: Failure to use a torque wrench can result in loose connecting rod screws (causing breakage) or tight connecting rod screws (causing scoring).

Install Camshaft

1. Lubricate tappets, camshaft journals, and lobes with engine oil.
2. Install tappets.
3. See [Figure 13-5](#). Align timing marks (A) on camshaft and crankshaft gear, and install camshaft.

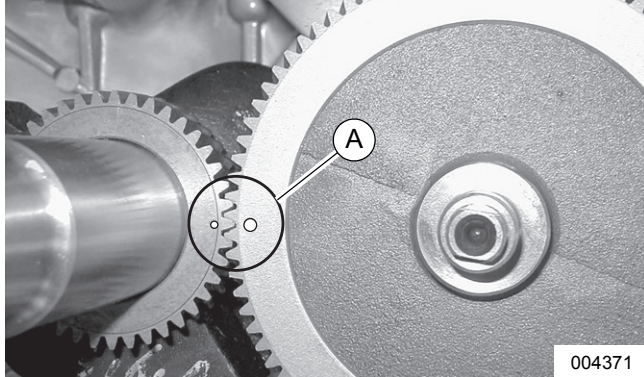


Figure 13-5. Installing Camshaft

4. Assemble governor spool to governor shaft.
 - a. Verify spool engages flyweights. Refer to [Mechanical Governor](#).
 - b. Clean all old gasket material from crankcase and cover. Refer to [Gasket Removal](#) in section "Cylinder Head and Valves" for proper procedure.
 - c. See [Figure 13-6](#). Install new oil passage o-ring (B) in crankcase.

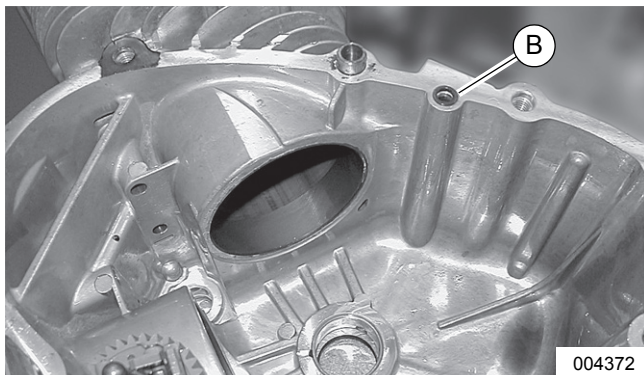


Figure 13-6. Crankcase Oil Passage O-Ring

Install Oil Pump

1. See [Figure 13-7](#). Lubricate inner and outer gerotor (A, B) and set in place.

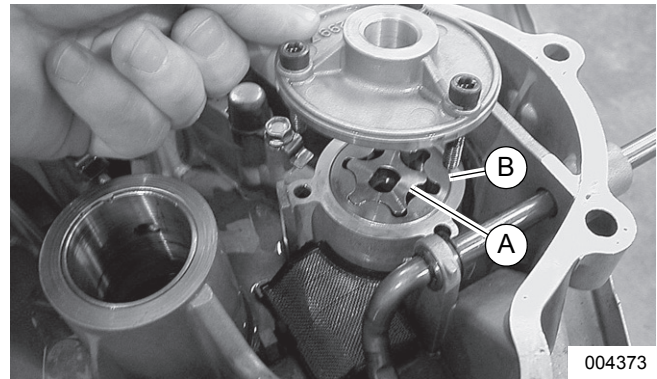


Figure 13-7. Install Oil Pump

2. Install oil pump cover.
3. Tighten screws to 108 in-lb (12.2 Nm) torque.

Install Crankcase Cover

1. Lubricate PTO seal and cam gear bearing.
2. See [Figure 13-8](#). Rotate governor shaft counterclockwise so paddle (C) rests against oil pump pick-up.

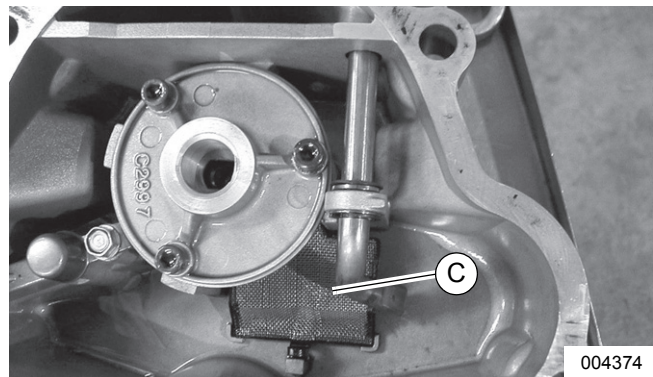


Figure 13-8. Rotating Governor Shaft

3. Rotate inner gerotor to align with camshaft drive.
4. Install crankcase cover with new gasket.

NOTE: Cover will not fully seat on assembly if oil pump drive is not aligned. Rotating crankshaft may align oil pump drive.

5. Install governor support bracket.
 - a. Loosen hex head crankcase bolt if necessary.
 - b. See **Figure 13-9**. Tighten screws in sequence shown to 35 ft-lb (47.5 Nm) torque.

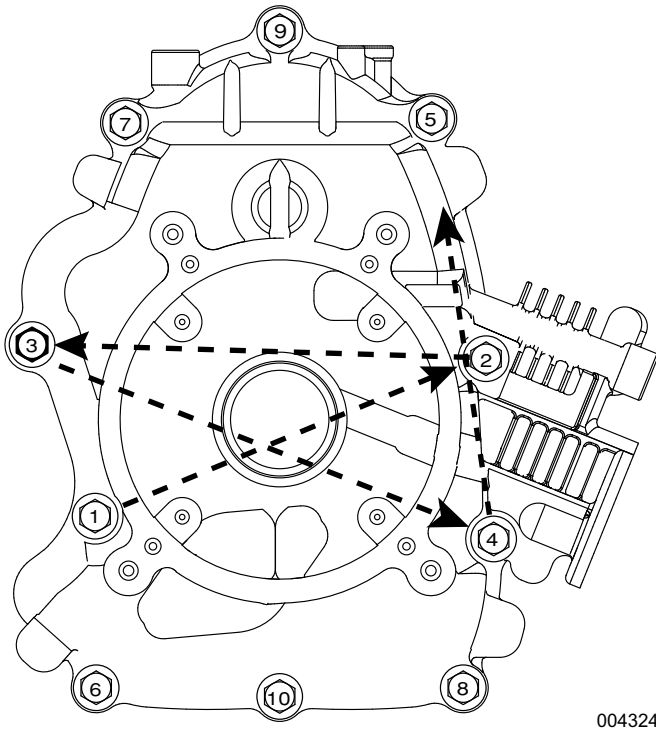


Figure 13-9. Crankcase Cover Torque Sequence

6. Check crankshaft end play. Assembly may be faulty if end play is less than 0.002 in (0.05 mm).

Install Alternator And Ignition Coils

1. See **Figure 13-10**. Install alternator. Tighten screws to 40 in-lb (4.75 Nm) torque.

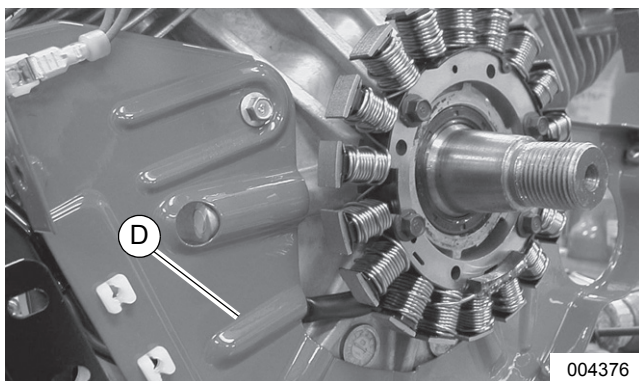


Figure 13-10. Install Alternator

2. Install back plate. Tighten screws to 40 in-lb (4.75 Nm) torque.

Important Note: Route alternator wires through relief (D) in back plate. **DO NOT** pinch wires.

3. See **Figure 13-11**. Assemble ignition coils on engine. Mounting holes in coil are slotted. Push coil as far back as possible and tighten one screw to hold coil in place.

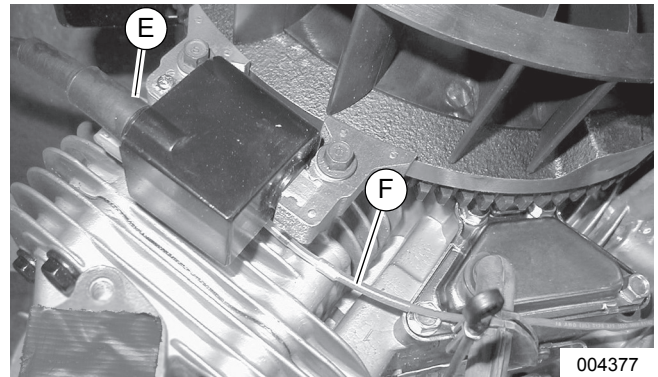


Figure 13-11. Ignition Coils

4. Repeat for second coil.

NOTE: Spark plug lead (E) must be at the top.

5. Install ground wire (F) on tab terminal on ignition coils.

IMPORTANT NOTE: Verify wires are routed over coil mounting posts and under breather tube.

6. Install starter motor. Tighten screws to 228 in-lb (27 Nm) torque.

Install Breather

1. Insert baffle if equipped.
2. Insert breather material.
3. Install breather assembly and gasket.
4. Tighten bolts to 40 in-lb (4.75 Nm) torque.

Install Flywheel

Important: Clean flywheel and crankshaft taper removing all oil, dirt, or grease.

- 1. Insert flywheel key into crankshaft.
- 2. See [Figure 13-12](#). Assemble flywheel (A) on crankshaft.

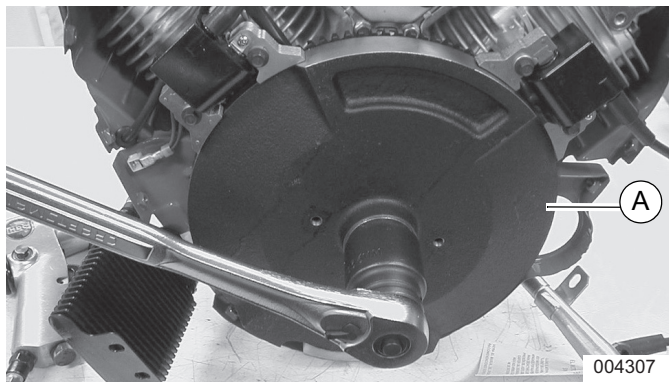


Figure 13-12. Tighten Flywheel Nut

- 3. Install washer (cupped side down) and flywheel nut.
- 4. Tighten flywheel nut to 150 ft-lb (214 Nm) torque.

Adjust Ignition Coil Air Gap

- 1. Rotate flywheel until magnet is under coil laminations.
- 2. See [Figure 13-12](#). Place a 0.008–0.012 in (0.20–0.30 mm) thick, non-magnetic gauge (A) between magnet and coil laminations.

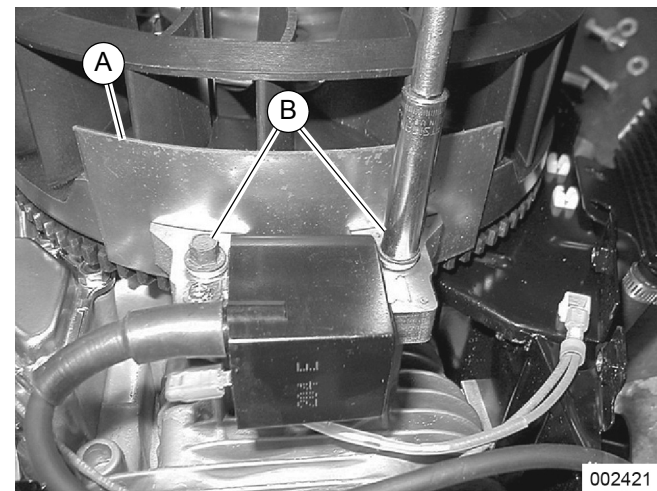


Figure 13-13. Adjust Coil Air Gap

- 3. Loosen mounting screws (B) until magnet pulls coil against thickness gauge. Tighten screws to 40 in-lb (4.75 Nm) torque.
- 4. Rotate flywheel to remove thickness gauge.
- 5. Repeat steps 1 through 4 for second coil.

Install Cylinder Heads

- 1. Install cylinder head with new gasket.
- 2. Lubricate threads of head bolts with one drop of oil.
- 3. See [Figure 13-14](#). Tighten head bolts in sequence shown to 22 ft-lb (29.9 Nm) torque.
- 4. Insert push rods into recess in tappets.

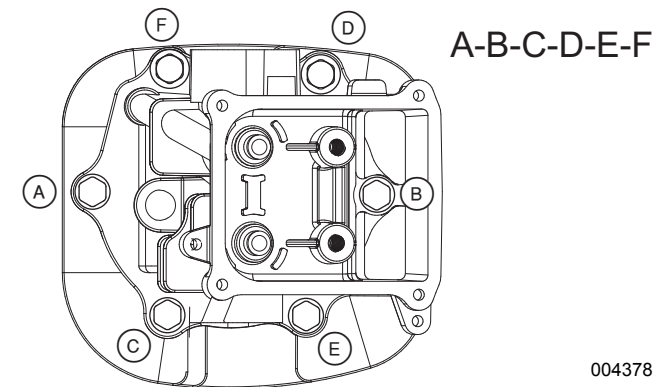


Figure 13-14. Cylinder Head Torque

Install Rocker Arms

- 1. Lubricate rocker arms and ball studs with clean engine oil.
- 2. See [Figure 13-15](#). Assemble ball studs (A), rocker arms (B), jam nuts (C), and guide plates (D) on cylinder head.

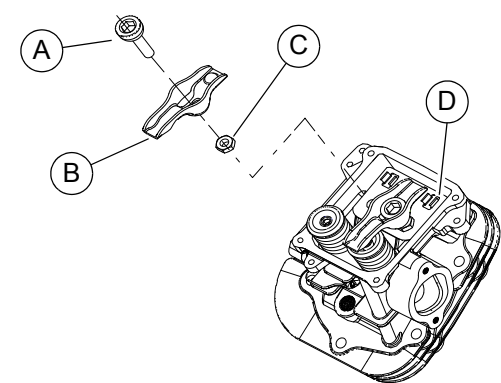


Figure 13-15. Install Rocker Arms

A	Ball stud	C	Jam nut
B	Rocker arm	D	Guide plate

NOTE: Verify push rods are properly seated in tappets and rocker arms.

Adjust Valve Clearance

1. Set #1 cylinder at TDC (Top Dead Center), compression stroke.
2. See [Figure 13-16](#). Adjust rocker arms to specified clearance.

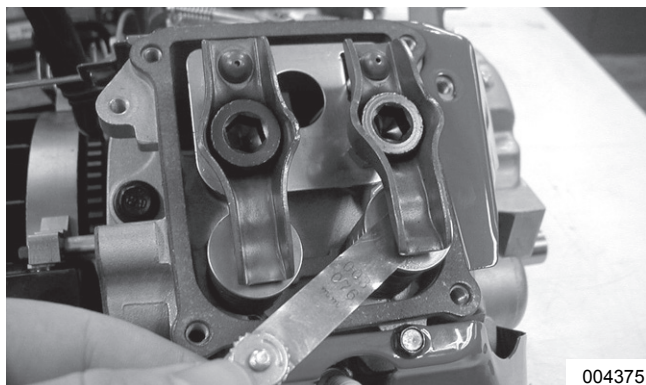


Figure 13-16. Adjust Valve Clearance

**Valve Clearance (cold) INTAKE and EXHAUST:
0.003 in, +/- 0.001 in (0.076 mm, +/- 0.02 mm)**

3. Tighten ball stud jam nuts to 174 **in-lb** (19.68 Nm) torque.
4. Repeat for #2 cylinder.
5. See [Figure 13-17](#). Install valve covers with new gaskets. Tighten bolts to 60 **in-lb** (6.8 Nm) torque.

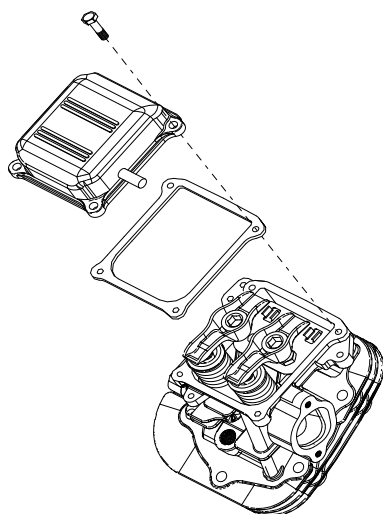


Figure 13-17. Install Valve Covers and Gaskets

General Assembly

- 1. See **Figure 13-18**. Install upper and lower wrappers (F, G).
 - Tighten M5 screws to 25 **in-lb** (2.8 Nm) torque.
 - Tighten M6 screws to 40 **in-lb** (4.5 Nm) torque.
- 2. See **Figure 13-19**. Connect ignition ground wire to ignition ground terminal in backing plate.
- 3. Connect all remaining wires to their proper locations.

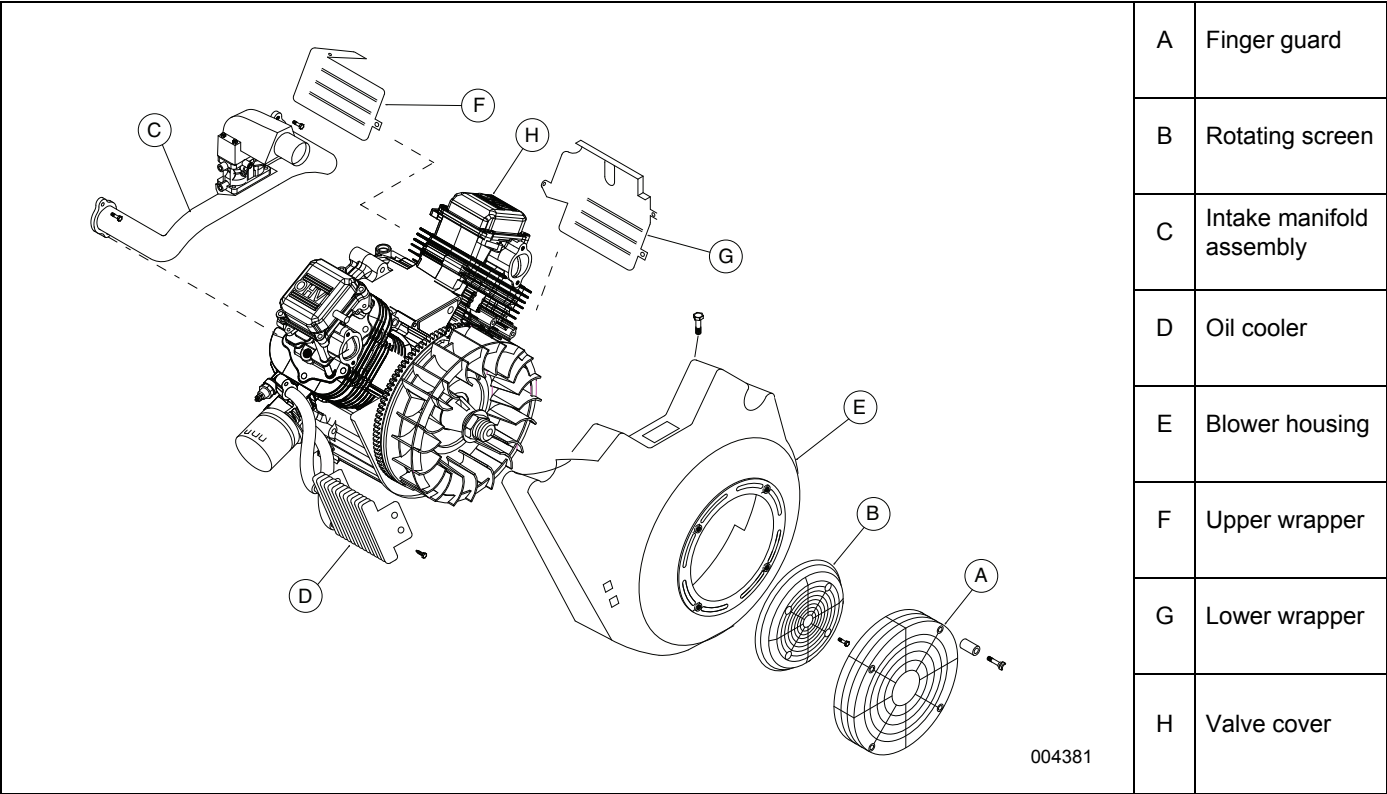


Figure 13-18. Exploded View

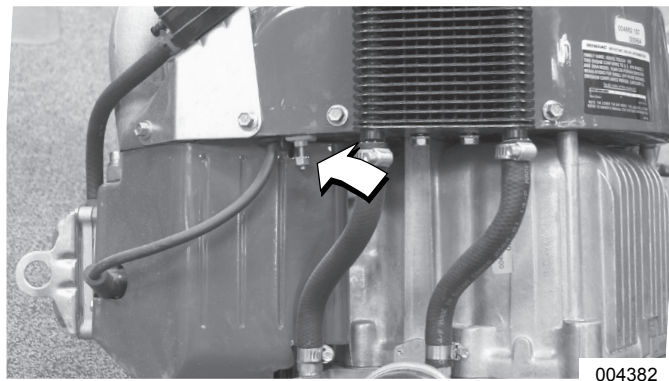


Figure 13-19. Ground Terminal

4. Assemble governor lever to governor arm. DO NOT tighten at this time.
5. Install speed adjust assembly (refer to **Mechanical Governor**).
 - Tighten screws to 53 **in-lb** (6.0 Nm) torque.
6. Install exhaust.
 - Tighten screws to 168 **in-lb** (19 Nm) torque.
7. Install fan and retaining ring.
 - Tighten screws to 192 **in-lb** (21.7 Nm) torque.
8. Install blower housing.
 - Tighten screws to 40 **in-lb** (4.5 Nm) torque.
9. Install rotating screen.
 - Tighten screws to 17 **in-lb** (1.9 Nm) torque.
10. Install intake manifold assembly.
 - Tighten bolts to 168 **in-lb** (19 Nm) torque.
11. Install finger guard.
 - Tighten screws by hand to approximately 12 **in-lb** (1.3 Nm) torque.
12. Connect governor linkage and speed control linkage (refer to **Mechanical Governor**).
13. Install spark plugs.
 - Tighten spark plugs to 168 **in-lb** (19 Nm) torque.

Adjust Governor

Note: Refer to **Static Governor Adjustment** in section **Mechanical Governor**.

IMPORTANT NOTE: Complete static governor adjustment **BEFORE STARTING OR RUNNING ENGINE!** Failure to make the static adjustment first could result in engine overspeed. Overspeed may result in engine damage, property damage, or personal injury.

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Section 14: Troubleshooting

Problem Solving

Introduction

Problems affecting engine operation may be classified as one of, or a combination of two or more of, the following:

- Engine will not start.
- Engine starts with difficulty.
- Engine lacks power.
- Engine vibrates.
- Engine overheats.
- Engine burns oil.

In many cases, the cause will be readily apparent. If this is not the case, check the fuel system, ignition system, and engine compression. These checks can be performed quickly and are the best methods of finding the cause of a failure. They may also predict future failures.

Check Fuel System

Verify fuel tank has been properly filled with correct fuel. If engine is equipped with a fuel shutoff valve, confirm valve is open. Verify fuel is flowing through fuel line and to carburetor. Adjust carburetor as needed. Verify choke closes all the way. If engine does not start, remove and inspect spark plug.

If spark plug is WET, look for the following:

- Overchoking.
- Dirty air cleaner.
- Excessively rich fuel-air mixture.
- Water in fuel.
- Fuel leaking past carburetor float needle.

If spark plug is DRY, look for the following:

- Carburetor mounting gasket leaks.
- Dirt or gum in carburetor or fuel line.
- Carburetor float needle stuck closed.
- Inoperative fuel pump (if equipped).
- Clogged fuel filter (if equipped).

Check Ignition System

See testing procedures in section [Ignition](#).

Check Compression

To check engine compression:

1. Remove spark plug.
2. Insert an automotive type compression gauge into the spark plug hole.
3. Crank engine until there is no further increase in pressure. The highest reading obtained is engine compression pressure.

If compression is poor, look for one or more of the following causes:

- Loose cylinder head bolts.
- Failed cylinder head gasket.
- Burned valves or valve seats.
- Insufficient valve clearance.
- Warped cylinder head.
- Warped valve stem.
- Worn or broken piston ring(s).
- Worn or damaged cylinder bore.
- Broken connecting rod.
- Worn valve seats or valves.
- Worn valve guides.

PROBLEM	POSSIBLE CAUSES	REMEDY
Engine will not start or starts with difficulty.	Empty fuel tank.	Fill fuel tank.
	Low oil level.	Replenish oil as necessary.
	Dirty air cleaner.	Clean or replace air cleaner element.
	Obstructed fuel line.	Clean fuel line or replace.
	Obstructed fuel tank cap vent hole.	Open cap vent hole.
	Fouled spark plug.	Replace spark plug.
	Incorrect spark plug.	Replace with correct spark plug.
	Loose or defective ignition wiring.	Check wiring, repair or replace.
	No ignition spark.	Check ignition system, replace faulty part(s).
	Incorrect valve clearance.	Reset valve clearance.
	Poor compression.	Check for worn or scored cylinder.
Engine knocks.	Carbon in combustion chamber.	Clean carbon from head and piston.
	Loose flywheel.	Check flywheel key and keyway; replace parts if necessary. Tighten flywheel nut.
	Worn cylinder.	Replace crankcase.
	Loose or worn connecting rod.	Replace connecting rod. Check crankpin diameter to see if crankshaft is faulty.
	Incorrect valve clearance.	Reset to correct clearance.
	Overloaded engine.	Reduce excessive load.
Engine misses under load.	Incorrect spark plug.	Install correct spark plug.
	Fouled spark plug.	Replace spark plug.
	Cracked porcelain on spark plug.	Replace spark plug.
	Incorrect spark plug gap.	Regap spark plug.
	Incorrect valve clearance.	Reset valve clearance.
	Weak valve springs.	Replace weak valve spring(s).
Engine lacks power.	Governor not set correctly.	Adjust governor.
	Incorrect spark plug.	Install correct spark plug.
	Incorrect valve clearance.	Reset valve clearance.
	Worn piston rings.	Replace piston rings.
	Low oil level.	Replenish oil to proper level.
	Obstructed air cleaner.	Clean or replace air cleaner.
	Worn or burned valves or valve seats.	Grind valves and valve seats.

PROBLEM	POSSIBLE CAUSES	REMEDY
Engine overheats.	Obstructed air flow.	Remove obstructions.
	Clogged cooling fins.	Clean cooling fins.
	Carbon buildup in combustion chamber.	Remove cylinder head and clean carbon.
	Overloaded engine	Reduce excessive loading.
	Lack of lubrication.	Replenish oil to proper level.
Engine vibrates excessively.	Loose engine mounting bolts.	Tighten mounting bolts.
	Bent crankshaft.	Replace crankshaft.
	Problem in equipment connected to engine.	Check equipment driven by engine.
Engine surges or runs unevenly.	Plugged vent hole in fuel tank cap.	Open vent hole in tank cap.
	Sticking or binding governor parts.	Clean governor parts and replace as necessary.
	Binding carburetor throttle linkage or throttle shaft.	Clean, lubricate, adjust or repair sticking parts.
	Intermittent ignition spark.	Test ignition as per Ignition section; repair or replace faulty parts as necessary.
	Improper carburetor adjustment.	Adjust carburetor.
	Dirty carburetor.	Clean carburetor.
Engine uses excessive amounts of oil.	Engine speed too fast.	Adjust engine speed to specifications.
	Oil level too high.	Drain excess oil from crankcase.
	Loose oil filler cap, or damaged oil filler cap gasket.	Replace cap gasket and tighten.
	Damaged or dirty breather.	Clean or replace breather as necessary.
	Clogged drain holes in breather.	Unclog drain holes in breather.
	Leaking gaskets or seals.	Replace gaskets or seals.
	Excessively worn valve guides.	Replace cylinder head.
	Worn cylinder walls.	Replace crankcase.
	Worn piston rings and/or grooves.	Install new rings and/or piston.
	Obstructed oil passages.	Clean out oil passages.

PROBLEM	POSSIBLE CAUSES	REMEDY
9. Crankshaft oil seal leaks.	Dirty or faulty crankcase breather.	Clean or replace breather.
	Hardened or worn oil seal.	Replace oil seal.
	Worn crankshaft seal contact surface.	Check crankshaft size and replace if worn.
	Worn crankshaft sleeve bearing.	Replace sleeve bearing.
	Seal not seated squarely in bore.	Replace with new seal.
	New seal damaged during installation.	Replace with new seal.
	Bent crankshaft.	Check crankshaft for straightness; replace if necessary.
10. Breather is passing oil.	Engine speed too fast.	Adjust engine speed to specifications.
	Oil level too high.	Drain excess oil from crankcase.
	Loose oil filler cap, or damaged oil filler cap gasket.	Replace cap gasket and tighten cap.
	Damaged or dirty breather.	Clean or replace breather as necessary.
	Clogged drain holes in breather.	Unclog drain holes in breather.
	Aligned piston ring end gaps.	Rotate ring end gaps, stagger 90°.
	Rings not properly seated.	Check for worn or "out of round" cylinder. Install new rings. Break in new rings with engine under varying loads.

Section 15: Specifications

Model	GTH & GTV 760	GTH & GTV 990	GTH & GTV 1000
Bore	3.54 in (90 mm)	3.54 in (90 mm)	3.54 in (90 mm)
Stroke	2.36 in (60 mm)	3.07 in (78 mm)	3.09 in (78.58 mm)
Displacement	763 cc	992 cc	999 cc

Model	GTV 760/990/1000	GTH 760/990/1000
Oil Capacity With Oil Filter Change	2.4 qt (2.3 L)	2.3 qt (2.2 L)
Oil Capacity Without Oil Filter Change	2.1 qt (2.0 L)	2.0 qt (1.9L)

Common Specifications	GTH & GTV 760/990/1000
Ignition Coil Air Gap	0.008–0.012 in (0.20–0.30 mm)
Crankshaft End Play	0.002–0.015 in (0.05–0.40 mm)
Spark Plug Gap	0.040 in (1.01 mm)
Valve Clearance (Cold) –Intake –Exhaust	0.002–0.004 in (0.05–0.10 mm) 0.002–0.004 in (0.05–0.10 mm)
Normal Compression –GTH/GTV 760 –GTH/GTV 990/1000	130–160 psi (896–1103 kPa) 160–190 psi (1103–1310 kPa)

Torque Specifications	GTH & GTV 760/990/1000
Alternator (To Crankcase) Screws	40 in-lb (4.75 Nm)
Ignition Coil Screws	40 in-lb (4.75 Nm)
Carburetor Adaptor Mounting Bolts	48 in-lb (5.4 Nm)
Connecting Rod Bolts	216 in-lb (24.4 Nm)
Crankcase Cover Bolts	35 ft-lb (47.5 Nm)
Cylinder Head Bolts	22 ft-lb (29.9 Nm)
Exhaust Manifold Bolts	168 in-lb (19 Nm)
Flywheel Nut	150 ft-lb (214 Nm)
Flywheel Fan Retainer Bolts	182 in-lb (21.7 Nm)
Governor Lever (Clinching Screw)	100 in-lb (11.3 Nm)
Intake Manifold Screws	168 in-lb (19 Nm)
Oil Pump Cover	108 in-lb (12.2 Nm)
Rocker Arm Jam Nut	168 in-lb (19 Nm)
Spark Plug	168 in-lb (19 Nm)
Starter Motor Bolts	228 in-lb (27 Nm)
Rocker Cover Screws	60 in-lb (6.8 Nm)
Blower Housing Screws	40 in-lb (4.75 Nm)

Standard and Reject Dimensions

Description	Standard Dimension	Reject Dimension
CYLINDER		
Bore	3.543–3.544 in (90.00–90.025 mm)	3.547 in (90.10 mm)
Out of Round		0.0015 in (0.035 mm)
Main Bearing, Mag End	1.498–1.5 in (38.044–38.099 mm)	1.506 in (38.25 mm)
Cam Bearing	0.787–0.789 in (20.00–20.03 mm)	0.790 in (20.06 mm)
CYLINDER HEAD		
Valve Guide	0.2758–0.2764 in (7.005–7.020 mm)	0.278 in (7.06 mm)
Valve Stem	0.2734–0.2748 in (6.945–6.98 mm)	0.272 in (6.9 mm)
Valve Spring Free Length	1.49–1.43 in (38.0–36.5 mm)	Less than: 1.437 in (36.5 mm)
CRANKCASE COVER		
Main Bearing, PTO End	1.6553–1.6574 in (42.044–42.099 mm)	1.663 in (42.25 mm)
Cam Bearing	0.7087–0.7096 in (18.0–18.025 mm)	0.711 in (18.06 mm)

CRANKSHAFT		
Crankpin GT1000	1.614–1.577 in (41.00–40.08 mm)	1.576 in (40.05 mm)
Crankpin GT760/990	1.535–1.536 in (38.99–39.01 mm)	1.534 in (38.96 mm)
Magneto Journal	1.496–1.4965 in (38.00–38.012 mm)	1.490 in (37.85 mm)
PTO Journal	1.6535–1.654 in (42.00–42.012 mm)	1.648 in (41.85 mm)
CAMSHAFT		
Magneto Journal		
PTO Journal	0.7072–0.708 in (17.964–17.982 mm)	0.705 in (17.93 mm)
Lobes	1.2299–1.2393 in (31.239–31.479 mm)	1.221 in (31.02 mm)
CONNECTING ROD		
Crankpin Bearing GT1000	1.6165–1.6169 in (41.06–41.07 mm)	1.6177 in (41.09 mm)
Crankpin Bearing GT760/990	1.5378–1.5382 in (39.06–39.07 mm)	1.539 in (39.09 mm)
Piston Pin Bearing	0.7882–0.7886 in (20.02–20.03)	0.7894 in (20.05 mm)
PISTON PIN		
Diameter	0.7868–0.7872 in (19.984–19.995 mm)	0.7862 in (19.97 mm)
Out of Round		0.0005 in (0.01 mm)
PISTON PIN BEARING (PISTON)		
Bore	0.7874–0.7882 in (20.00–20.02 mm)	0.7886 in (20.03 mm)
Out of Round		0.0005 in (0.01 mm)
PISTON RINGS		
End Gap – Top and Center	0.0098–0.0197 in (0.25–0.5 mm)	0.0295 in (0.75 mm)
End Gap – Oil	0.015–0.045 in (0.38–1.15 mm)	0.059 in (1.5 mm)
Ring Side Clearance – Top and Center	0.0016–0.0035 in (0.04–0.09 mm)	0.004 in (0.10 mm)
Ring Side Clearance – Oil	0.0005–0.007 in (0.012–0.18 mm)	0.008 in (0.20 mm)

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Generac Power Systems, Inc.
S45 W29290 Hwy. 59
Waukesha, WI 53189
1-888-GENERAC (1-888-436-3722)
www.generac.com